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COHERENT ALGEBRAIC SHEAVES

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Introduction

It is well known that cohomological methods, and sheaf theory in particular, play an increasingly important role, not only in the theory of functions of several complex variables (cf. [5]), but also in classical algebraic geometry (it will suffice to cite the recent work of Kodaira and Spencer on the Riemann–Roch theorem). The algebraic character of these methods suggested that they could also be applied to abstract algebraic geometry; the purpose of the present paper is to show that this is indeed the case.

The contents of the various chapters are as follows:

Chapter I is devoted to the general theory of sheaves. It contains proofs of the results from that theory which are used in the other two chapters. The various algebraic operations that may be performed on sheaves are described in §1; we have followed Cartan's account ([2], [5]) fairly closely. Section 2 studies coherent sheaves of modules; these sheaves generalize coherent analytic sheaves (cf. [3], [5]) and enjoy entirely analogous properties. In §3 the cohomology groups of a space X with coefficients in a sheaf $\mathfrak{F}$ are defined. In the later applications, X is an algebraic variety equipped with the Zariski topology; it is therefore not a Hausdorff topological space, and the methods used by Leray [10] or Cartan [3] (based on "partitions of unity" or "fine" sheaves) do not apply to it. We have therefore had to return to the Čech procedure and define the cohomology groups $H^q(X, \mathfrak{F})$ by passage to the direct limit over increasingly fine open coverings. A further difficulty, related to the non-Hausdorffness of X, arises in the "cohomology exact sequence" (cf. nos. 24 and 25): we have been able to establish this exact sequence only in certain special cases, which are nevertheless sufficient for the applications we had in view (cf. nos. 24 and 47).

Chapter II begins with a definition of algebraic varieties analogous to Weil's ([17], Chap. VII), but also encompassing reducible varieties. We note in this connection that, contrary to Weil's usage, we do not reserve the term "variety" for irreducible varieties. We define the structure of an algebraic variety by specifying a topology (the Zariski topology) and a subsheaf of the sheaf of germs of functions (the sheaf of local rings). A coherent algebraic sheaf on an algebraic variety V is simply a coherent sheaf of $\mathcal{O}_V$-modules, where $\mathcal{O}_V$ denotes the sheaf of local rings of V; we give various examples in §2. Section 3 is devoted to affine varieties. The results obtained are entirely analogous to those concerning Stein varieties (cf. [3], [5]): if $\mathfrak{F}$ is a coherent algebraic sheaf on the affine variety V, then $H^q(V, \mathfrak{F}) = 0$ for every $q > 0$, and $\mathfrak{F}_x$ is generated by $H^0(V, \mathfrak{F})$ for every $x \in V$. Moreover (§4), $\mathfrak{F}$ is uniquely determined by $H^0(V, \mathfrak{F})$, regarded as a module over the coordinate ring of V.

Chapter III, concerning projective varieties, contains the essential results of this paper. We begin by establishing a correspondence between coherent algebraic sheaves $\mathfrak{F}$ on projective space $X = \mathbf{P}_r(K)$ and graded S-modules satisfying condition (TF) of no. 56 (where S denotes the polynomial algebra $K[t_0, \ldots, t_r]$). This correspondence is bijective if two S-modules which differ only in their homogeneous components of sufficiently low degree are identified (for the precise statements, see nos. 57, 59, and 65). From that point on, every question concerning $\mathfrak{F}$ can be translated into a question concerning the associated S-module M. Thus, in §3 we give a procedure for determining the $H^q(X, \mathfrak{F})$ algebraically from M. This enables us, in particular, to study the properties of $H^q(X, \mathfrak{F}(n))$ as $n \to +\infty$ (for the definition of $\mathfrak{F}(n)$, see no. 54); the resulting statements are given in nos. 65 and 66. In §4 we relate the groups $H^q(X, \mathfrak{F})$ to the functors

Ext_S^q introduced by Cartan and Eilenberg [6]. This allows us, in §5, to study the behavior of $H^q(X, \mathfrak{F}(n))$ as $n \to -\infty$ and to give a homological characterization of varieties that are "k-fold of the first kind." Section 6 presents some elementary properties of the Euler–Poincaré characteristic of a projective variety with coefficients in a coherent algebraic sheaf.

Elsewhere we shall show how the general results of the present paper can be applied to various particular problems, notably to extend the "duality theorem" of [15] to the abstract case, as well as part of Kodaira and Spencer's results on the Riemann–Roch theorem; in these applications the theorems of nos. 66, 75, and 76 play an essential role. We shall also show that, when the base field is the field of complex numbers, the theory of coherent algebraic sheaves is essentially identical to that of coherent analytic sheaves (cf. [4]).

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Chapter I. Sheaves

§1. Operations on sheaves

1. Definition of a sheaf. Let X be a topological space. A *sheaf of abelian groups* on X (or simply a *sheaf*) consists of:

(a) *A function $x \mapsto \mathfrak{F}_x$ assigning to each $x \in X$ an abelian group $\mathfrak{F}_x$,*

(b) *A topology on the set $\mathfrak{F}$, the disjoint union of the sets $\mathfrak{F}_x$.*

If f is an element of $\mathfrak{F}_x$, we put $\pi(f) = x$; the map π is called the *projection* of $\mathfrak{F}$ onto X. The subset of $\mathfrak{F} \times \mathfrak{F}$ formed by the pairs (f, g) such that $\pi(f) = \pi(g)$ will be denoted by $\mathfrak{F} + \mathfrak{F}$.

With these definitions in place, we can state the two axioms to which the data (a) and (b) are subject:

(I) *For every $f \in \mathfrak{F}$ there exist a neighborhood V of f and a neighborhood U of $\pi(f)$ such that the restriction of π to V is a homeomorphism of V onto U.*

(In other words, π must be a local homeomorphism.)

(II) *The map $f \mapsto -f$ is continuous from $\mathfrak{F}$ to $\mathfrak{F}$, and the map $(f, g) \mapsto f + g$ is continuous from $\mathfrak{F} + \mathfrak{F}$ to $\mathfrak{F}$.*

Observe that, even if X is Hausdorff (which we have not assumed), $\mathfrak{F}$ need not be Hausdorff, as is shown by the example of the sheaf of germs of functions (cf. no. 3).

Example of a sheaf. Let G be an abelian group and put $\mathfrak{F}_x = G$ for every $x \in X$. The set $\mathfrak{F}$ may be identified with the product $X \times G$; if it is given the product topology of the topology on X and the discrete topology on G, one obtains a sheaf, called the *constant sheaf* isomorphic to G, and often identified with G.

2. Sections of a sheaf. Let $\mathfrak{F}$ be a sheaf on the space X, and let U be a subset of X. A *section* of $\mathfrak{F}$ over U is a continuous map $s : U \to \mathfrak{F}$ such that $\pi \circ s$ is the identity map of U. Thus $s(x) \in \mathfrak{F}_x$ for every $x \in U$. The set of sections of $\mathfrak{F}$ over U will be denoted by $\Gamma(U, \mathfrak{F})$; axiom (II) implies that $\Gamma(U, \mathfrak{F})$ is an abelian group. If $U \subset V$ and s is a section over V, the restriction of s to U is a section over U; this gives a homomorphism $\rho_U^V : \Gamma(V, \mathfrak{F}) \to \Gamma(U, \mathfrak{F})$.

If U is open in X, then $s(U)$ is open in $\mathfrak{F}$; and, as U ranges over a basis of open subsets of X, the sets $s(U)$ range over a basis of open subsets of $\mathfrak{F}$. This is simply another way of expressing axiom (I).

Let us also note a consequence of axiom (I): for every $f \in \mathfrak{F}_x$, there exists a section s over a neighborhood of x such that $s(x) = f$, and any two sections with this property coincide in a neighborhood of x. In other words, $\mathfrak{F}_x$ is the direct limit of the $\Gamma(U, \mathfrak{F})$ over the directed set of neighborhoods of x.

3. Construction of sheaves. Suppose that, for every open set $U \subset X$, an abelian group $\mathfrak{F}_U$ is given and, for every pair of open sets $U \subset V$, a homomorphism $\varphi_U^V : \mathfrak{F}_V \to \mathfrak{F}_U$ is given, in such a way that the transitivity condition $\varphi_U^V \circ \varphi_V^W = \varphi_U^W$ holds whenever $U \subset V \subset W$.

The collection $(\mathfrak{F}_U, \varphi_U^V)$ then defines a sheaf $\mathfrak{F}$ as follows:

(a) Put $\mathfrak{F}_x = \varinjlim \mathfrak{F}_U$, the direct limit over the directed set of open neighborhoods U of x. Thus, if x belongs to the open set U, there is a canonical homomorphism $\varphi_x^U : \mathfrak{F}_U \to \mathfrak{F}_x$.

(b) If $t \in \mathfrak{F}_U$, let $[t, U]$ denote the set of the elements $\varphi_x^U(t)$ as x ranges over U. Thus $[t, U] \subset \mathfrak{F}$, and we give $\mathfrak{F}$ the topology generated by the sets $[t, U]$. Consequently, an element $f \in \mathfrak{F}_x$ has as a neighborhood basis in $\mathfrak{F}$ the sets $[t, U]$ for which $x \in U$ and $\varphi_x^U(t) = f$.

It follows immediately that the data (a) and (b) satisfy axioms (I) and (II), that is, that $\mathfrak{F}$ is indeed a sheaf. We shall call it the sheaf *defined by the system* $(\mathfrak{F}_U, \varphi_U^V)$.

If $t \in \mathfrak{F}_U$, the map $x \mapsto \varphi_x^U(t)$ is a section of $\mathfrak{F}$ over U; hence there is a canonical homomorphism $\iota : \mathfrak{F}_U \to \Gamma(U, \mathfrak{F})$.

Proposition 1. *The homomorphism $\iota : \mathfrak{F}_U \to \Gamma(U, \mathfrak{F})$ is injective*[1] *if and only if the following condition holds:*

If $t \in \mathfrak{F}_U$ and there exists an open covering $\{U_i\}$ of U such that $\varphi_{U_i}^U(t) = 0$ for every i, then $t = 0$.

If $t \in \mathfrak{F}_U$ satisfies the preceding condition, then

$$\varphi_x^U(t) = \varphi_x^{U_i} \circ \varphi_{U_i}^U(t) = 0 \qquad \text{if } x \in U_i,$$

which means that $\iota(t) = 0$. Conversely, suppose that $\iota(t) = 0$ for some $t \in \mathfrak{F}_U$. Since $\varphi_x^U(t) = 0$ for $x \in U$, the definition of the direct limit gives an open neighborhood $U(x)$ of x such that $\varphi_{U(x)}^U(t) = 0$. The sets $U(x)$ then form an open covering of U satisfying the condition in the statement.

Proposition 2. *Let U be an open subset of X, and suppose that $\iota : \mathfrak{F}_V \to \Gamma(V, \mathfrak{F})$ is injective for every open set $V \subset U$. The homomorphism $\iota : \mathfrak{F}_U \to \Gamma(U, \mathfrak{F})$ is surjective*[1] *(hence bijective) if and only if the following condition holds:*

[1]Recall (cf. [1]) that a map $f : E \to E'$ is called *injective* if $f(e_1) = f(e_2)$ implies $e_1 = e_2$, *surjective* if $f(E) = E'$, and *bijective* if it is both injective and surjective. An injective (respectively surjective, bijective) map is called an injection (respectively a surjection, a bijection).

For every open covering $\{U_i\}$ of U and every family $\{t_i\}$, where $t_i \in \mathfrak{F}_{U_i}$ and

$$\varphi^{U_i}_{U_i \cap U_j}(t_i) = \varphi^{U_j}_{U_i \cap U_j}(t_j)$$

for every pair (i, j), there exists $t \in \mathfrak{F}_U$ such that $\varphi^U_{U_i}(t) = t_i$ for every i.

The condition is necessary. Each t_i defines a section $s_i = \iota(t_i)$ over U_i, and $s_i = s_j$ over $U_i \cap U_j$. Hence there exists a section s over U which agrees with s_i on U_i for every i. If $\iota : \mathfrak{F}_U \to \Gamma(U, \mathfrak{F})$ is surjective, there exists $t \in \mathfrak{F}_U$ such that $\iota(t) = s$. Put $t'_i = \varphi^U_{U_i}(t)$. The section defined by t'_i over U_i is precisely s_i; hence $\iota(t_i - t'_i) = 0$, which implies $t_i = t'_i$ because ι is assumed to be injective.

The condition is sufficient. If s is a section of $\mathfrak{F}$ over U, there exist an open covering $\{U_i\}$ of U and elements $t_i \in \mathfrak{F}_{U_i}$ such that $\iota(t_i)$ is the restriction of s to U_i. It follows that $\varphi^{U_i}_{U_i \cap U_j}(t_i)$ and $\varphi^{U_j}_{U_i \cap U_j}(t_j)$ define the same section over $U_i \cap U_j$ and are therefore equal, by the hypothesis on ι. If $t \in \mathfrak{F}_U$ satisfies $\varphi^U_{U_i}(t) = t_i$, then $\iota(t)$ agrees with s on every U_i, and hence on U. $\square$

Proposition 3. *If $\mathfrak{F}$ is a sheaf of abelian groups on X, the sheaf defined by the system $(\Gamma(U, \mathfrak{F}), \rho^V_U)$ is canonically isomorphic to $\mathfrak{F}$.*

This follows immediately from the properties of sections stated in no. 2.

Proposition 3 shows that every sheaf can be defined by a suitable system $(\mathfrak{F}_U, \varphi^V_U)$. Note that different systems may define the *same* sheaf $\mathfrak{F}$. Nevertheless, if the systems $(\mathfrak{F}_U, \varphi^V_U)$ are required to satisfy the conditions of Propositions 1 and 2, there is, up to isomorphism, only one possible system: the one formed by $(\Gamma(U, \mathfrak{F}), \rho^V_U)$.

Example. Let G be an abelian group, and take $\mathfrak{F}_U$ to be the set of functions on U with values in G. Define $\varphi^V_U : \mathfrak{F}_V \to \mathfrak{F}_U$ by restricting a function. This gives a system $(\mathfrak{F}_U, \rho^V_U)$ and hence a sheaf $\mathfrak{F}$, called the *sheaf of germs of functions* with values in G. It is immediate that the system $(\mathfrak{F}_U, \varphi^V_U)$ satisfies the conditions of Propositions 1 and 2; consequently, the sections of $\mathfrak{F}$ over an open set U may be identified with the elements of $\mathfrak{F}_U$.

4. Gluing of sheaves.

Let $\mathfrak{F}$ be a sheaf on X, and let U be a subset of X. The set $\pi^{-1}(U) \subset \mathfrak{F}$, endowed with the topology induced from that of $\mathfrak{F}$, forms a sheaf on U, called the sheaf *induced* by $\mathfrak{F}$ on U and denoted by $\mathfrak{F}(U)$ (or simply $\mathfrak{F}$ when no confusion can arise).

We shall now see that, conversely, a sheaf on X can be defined from sheaves on open sets which cover X.

Proposition 4. *Let $\mathfrak{U} = \{U_i\}_{i \in I}$ be an open covering of X, and, for each $i \in I$, let $\mathfrak{F}_i$ be a sheaf on U_i. For every pair (i, j), let θ_{ij} be an isomorphism from $\mathfrak{F}_j(U_i \cap U_j)$ onto $\mathfrak{F}_i(U_i \cap U_j)$. Suppose that $\theta_{ij} \circ \theta_{jk} = \theta_{ik}$ at every point of $U_i \cap U_j \cap U_k$, for every triple (i, j, k).*

Then there exist a sheaf $\mathfrak{F}$ and, for each $i \in I$, an isomorphism η_i from $\mathfrak{F}(U_i)$ onto $\mathfrak{F}_i$ such that $\theta_{ij} = \eta_i \circ \eta_j^{-1}$ at every point of $U_i \cap U_j$. Moreover, $\mathfrak{F}$ and the η_i are determined up to isomorphism by the preceding condition.

The uniqueness of $\{\mathfrak{F}, \eta_i\}$ is clear. To prove existence, one could define $\mathfrak{F}$ as the quotient space of the disjoint union of the $\mathfrak{F}_i$. We shall instead use the procedure of no. 3. If U is open in X, let $\mathfrak{F}_U$ be the group whose elements are the families $\{s_k\}_{k \in I}$ with $s_k \in \Gamma(U \cap U_k, \mathfrak{F}_k)$ and $s_k = \theta_{kj}(s_j)$ on $U \cap U_j \cap U_k$. If $U \subset V$, define φ^V_U in the evident way. The sheaf defined by the system $(\mathfrak{F}_U, \varphi^V_U)$ is the required sheaf $\mathfrak{F}$. Furthermore, if $U \subset U_i$, the map which sends a family $\{s_k\} \in \mathfrak{F}_U$ to the element $s_i \in \Gamma(U, \mathfrak{F}_i)$[2] is an isomorphism from $\mathfrak{F}_U$ onto $\Gamma(U, \mathfrak{F}_i)$ by the transitivity condition. This gives an isomorphism $\eta_i : \mathfrak{F}(U_i) \to \mathfrak{F}_i$ which plainly satisfies the required condition.

The sheaf $\mathfrak{F}$ is said to be obtained by *gluing* the sheaves $\mathfrak{F}_i$ by means of the isomorphisms θ_{ij}.

5. Extension and restriction of a sheaf.

[2]The printed French has $\Gamma(U_i, \mathfrak{F}_i)$ here. Since the family is defined over the fixed open set $U \subset U_i$, and the codomain stated immediately afterward is $\Gamma(U, \mathfrak{F}_i)$, the subscript U_i is a source typo.

Let X be a topological space, let Y be a *closed* subspace of X, and let $\mathfrak{F}$ be a sheaf on X. We shall say that $\mathfrak{F}$ *is concentrated on* Y, or *is zero outside* Y, if $\mathfrak{F}_x = 0$ for every $x \in X - Y$.

PROPOSITION 5. *If the sheaf $\mathfrak{F}$ is concentrated on Y, the homomorphism*

$$\rho_Y^X : \Gamma(X, \mathfrak{F}) \longrightarrow \Gamma(Y, \mathfrak{F}(Y))$$

is bijective.

If a section of $\mathfrak{F}$ over X is zero over Y, it is zero everywhere, since $\mathfrak{F}_x = 0$ when $x \notin Y$. This shows that ρ_Y^X is injective. Conversely, let s be a section of $\mathfrak{F}(Y)$ over Y, and extend s to X by setting $s(x) = 0$ when $x \notin Y$. The map $x \mapsto s(x)$ is plainly continuous on $X - Y$. On the other hand, if $x \in Y$, there exists a section s' of $\mathfrak{F}$ over a neighborhood U of x such that $s'(x) = s(x)$. Since s is continuous on Y by hypothesis, there is a neighborhood V of x, contained in U, such that $s'(y) = s(y)$ for every $y \in V \cap Y$. Since $\mathfrak{F}_y = 0$ when $y \notin Y$, we also have $s'(y) = s(y)$ for $y \in V - (V \cap Y)$. Thus s and s' agree on V, proving that s is continuous near Y, and hence everywhere. It follows that ρ_Y^X is surjective, completing the proof.

We shall now show that the sheaf $\mathfrak{F}(Y)$ determines the sheaf $\mathfrak{F}$ unambiguously.

PROPOSITION 6. *Let Y be a closed subspace of a space X, and let $\mathfrak{G}$ be a sheaf on Y. Put $\mathfrak{F}_x = \mathfrak{G}_x$ if $x \in Y$ and $\mathfrak{F}_x = 0$ if $x \notin Y$, and let $\mathfrak{F}$ be the disjoint union of the $\mathfrak{F}_x$. Then $\mathfrak{F}$ can be endowed with a unique sheaf structure on X such that $\mathfrak{F}(Y) = \mathfrak{G}$.*

Let U be open in X. If s is a section of $\mathfrak{G}$ over $U \cap Y$, extend s by zero on $U - (U \cap Y)$. As s ranges over $\Gamma(U \cap Y, \mathfrak{G})$, this gives a group $\mathfrak{F}_U$ of maps from U to $\mathfrak{F}$. Proposition 5 shows that, if $\mathfrak{F}$ is endowed with a sheaf structure such that $\mathfrak{F}(Y) = \mathfrak{G}$, then $\mathfrak{F}_U = \Gamma(U, \mathfrak{F})$, proving the uniqueness of the structure in question. Existence follows from the procedure of no. 3, applied to the $\mathfrak{F}_U$ and to the restriction homomorphisms $\varphi_U^V : \mathfrak{F}_V \to \mathfrak{F}_U$.[3]

The sheaf $\mathfrak{F}$ is said to be obtained by *extending the sheaf* $\mathfrak{G}$ *by zero outside* Y; it is denoted by $\mathfrak{G}^Y$, or simply by $\mathfrak{G}$ if no confusion can result.

6. Sheaves of rings and sheaves of modules.

The notion of a sheaf defined in no. 1 is that of a sheaf of *abelian groups*. Clearly, analogous definitions exist for every algebraic structure (one could even define "sheaves of sets," in which $\mathcal{F}_x$ has no algebraic structure and only axiom (I) is postulated). In what follows, we shall mainly encounter sheaves of *rings* and sheaves of *modules*.

A sheaf of rings $\mathcal{A}$ is a sheaf of abelian groups $\mathcal{A}_x$, $x \in X$, in which each $\mathcal{A}_x$ is endowed with a ring structure such that the map $(f, g) \mapsto f \cdot g$ is continuous from $\mathcal{A} + \mathcal{A}$ to $\mathcal{A}$ (with the notation of no. 1). We shall always assume that each $\mathcal{A}_x$ has a unit element which depends continuously on x.

If $\mathcal{A}$ is a sheaf of rings satisfying the preceding condition, then $\Gamma(U, \mathcal{A})$ is a ring with unit, and $\rho_U^V : \Gamma(V, \mathcal{A}) \to \Gamma(U, \mathcal{A})$ is a unital homomorphism when $U \subset V$. Conversely, suppose that rings $\mathcal{A}_U$ with unit and unital homomorphisms $\varphi_U^V : \mathcal{A}_V \to \mathcal{A}_U$ are given, with $\varphi_U^V \circ \varphi_V^W = \varphi_U^W$. Then the sheaf $\mathcal{A}$ defined by the system $(\mathcal{A}_U, \varphi_U^V)$ is a sheaf of rings. For example, if G is a ring with unit, the sheaf of germs of functions with values in G (defined in no. 3) is a sheaf of rings.

Let $\mathcal{A}$ be a sheaf of rings. A sheaf $\mathcal{F}$ is called a *sheaf of $\mathcal{A}$-modules* if each $\mathcal{F}_x$ is endowed with the structure of a unital $\mathcal{A}_x$-module (on the left, to fix ideas), varying "continuously" with x in the following sense: if $\mathcal{A} + \mathcal{F}$ is the subset of $\mathcal{A} \times \mathcal{F}$ formed by the pairs (a, f) such that $\pi(a) = \pi(f)$, then the map $(a, f) \mapsto a \cdot f$ is continuous from $\mathcal{A} + \mathcal{F}$ to $\mathcal{F}$.

If $\mathcal{F}$ is a sheaf of $\mathcal{A}$-modules, then $\Gamma(U, \mathcal{F})$ is a unital module over $\Gamma(U, \mathcal{A})$. Conversely, suppose that $\mathcal{A}$ is defined by the system $(\mathcal{A}_U, \varphi_U^V)$ as above, and let $\mathcal{F}$ be the sheaf defined by a system $(\mathcal{F}_U, \psi_U^V)$, where each $\mathcal{F}_U$ is a unital $\mathcal{A}_U$-module and

$$\psi_U^V(a \cdot f) = \varphi_U^V(a) \cdot \psi_U^V(f)$$

[3] The printed French reverses the source and target here, writing $\mathfrak{F}_U \to \mathfrak{F}_V$. The maps are restrictions for $U \subset V$, so their direction is necessarily $\mathfrak{F}_V \to \mathfrak{F}_U$, as in no. 3.

for $a \in \mathcal{A}_V$ and $f \in \mathcal{F}_V$. Then $\mathcal{F}$ is a sheaf of $\mathcal{A}$-modules.

Every sheaf of abelian groups may be regarded as a sheaf of **Z**-modules, where **Z** denotes the constant sheaf isomorphic to the ring of integers. This will allow us, in what follows, to restrict our attention to sheaves of modules.

7. Subsheaf and quotient sheaf.

Let $\mathcal{A}$ be a sheaf of rings and $\mathcal{F}$ a sheaf of $\mathcal{A}$-modules. For every $x \in X$, let $\mathcal{G}_x$ be a subset of $\mathcal{F}_x$. We say that $\mathcal{G} = \bigcup \mathcal{G}_x$ is a *subsheaf* of $\mathcal{F}$ if:

(a) *$\mathcal{G}_x$ is an $\mathcal{A}_x$-submodule of $\mathcal{F}_x$ for every $x \in X$;*

(b) *$\mathcal{G}$ is an open subset of $\mathcal{F}$.*

Condition (b) may equivalently be expressed as follows:

> (b′) *If x is a point of X, and if s is a section of $\mathcal{F}$ over a neighborhood of x such that $s(x) \in \mathcal{G}_x$, then $s(y) \in \mathcal{G}_y$ for every y sufficiently near x.*

It is clear that, if these conditions hold, $\mathcal{G}$ is a sheaf of $\mathcal{A}$-modules.

Let $\mathcal{G}$ be a subsheaf of $\mathcal{F}$, and put $\mathcal{H}_x = \mathcal{F}_x/\mathcal{G}_x$ for every $x \in X$. Endow $\mathcal{H} = \bigcup \mathcal{H}_x$ with the quotient topology induced by that of $\mathcal{F}$. One readily sees that this makes $\mathcal{H}$ into a sheaf of $\mathcal{A}$-modules, called the *quotient sheaf* of $\mathcal{F}$ by $\mathcal{G}$ and denoted by $\mathcal{F}/\mathcal{G}$. It may also be defined by the procedure of no. 3: if U is open in X, put

$$\mathcal{H}_U = \Gamma(U, \mathcal{F})/\Gamma(U, \mathcal{G}),$$

where φ_U^V is the homomorphism induced on the quotient by

$$\rho_U^V : \Gamma(V, \mathcal{F}) \longrightarrow \Gamma(U, \mathcal{F}).$$

The sheaf defined by the system $(\mathcal{H}_U, \varphi_U^V)$ is precisely $\mathcal{H}$.

Either definition of $\mathcal{H}$ shows that, if s is a section of $\mathcal{H}$ near x, there exists a section t of $\mathcal{F}$ near x such that the class of $t(y)$ modulo $\mathcal{G}_y$ is equal to $s(y)$ for every y sufficiently near x. Of course, this is no longer true globally in general: if U is open in X, one has only the exact sequence

$$0 \longrightarrow \Gamma(U, \mathcal{G}) \longrightarrow \Gamma(U, \mathcal{F}) \longrightarrow \Gamma(U, \mathcal{H}),$$

the homomorphism $\Gamma(U, \mathcal{F}) \to \Gamma(U, \mathcal{H})$ not being surjective in general (cf. no. 24).

8. Homomorphisms.

Let $\mathcal{A}$ be a sheaf of rings, and let $\mathcal{F}$ and $\mathcal{G}$ be two sheaves of $\mathcal{A}$-modules. An $\mathcal{A}$-*homomorphism* (also called an $\mathcal{A}$-linear homomorphism, or simply a homomorphism) from $\mathcal{F}$ to $\mathcal{G}$ consists, for every $x \in X$, of an $\mathcal{A}_x$-homomorphism $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$, in such a way that the map $\varphi : \mathcal{F} \to \mathcal{G}$ defined by the φ_x is continuous. Equivalently, if s is a section of $\mathcal{F}$ over U, then $x \mapsto \varphi_x(s(x))$ is a section of $\mathcal{G}$ over U (which we denote by $\varphi(s)$ or $\varphi \circ s$). For example, if $\mathcal{G}$ is a subsheaf of $\mathcal{F}$, the injection $\mathcal{G} \to \mathcal{F}$ and the projection $\mathcal{F} \to \mathcal{F}/\mathcal{G}$ are homomorphisms.

Proposition 7. *Let φ be a homomorphism from $\mathcal{F}$ to $\mathcal{G}$. For every $x \in X$, let $\mathfrak{N}_x$ be the kernel of φ_x, and let $\mathfrak{I}_x$ be the image of φ_x. Then $\mathfrak{N} = \bigcup \mathfrak{N}_x$ is a subsheaf of $\mathcal{F}$, $\mathfrak{I} = \bigcup \mathfrak{I}_x$ is a subsheaf of $\mathcal{G}$, and φ induces an isomorphism from $\mathcal{F}/\mathfrak{N}$ onto $\mathfrak{I}$.*

Since φ_x is an $\mathcal{A}_x$-homomorphism, $\mathfrak{N}_x$ and $\mathfrak{I}_x$ are submodules of $\mathcal{F}_x$ and $\mathcal{G}_x$, respectively,[4] and φ_x induces an isomorphism from $\mathcal{F}_x/\mathfrak{N}_x$ onto $\mathfrak{I}_x$. Furthermore, if s is a local section of $\mathcal{F}$ such that $s(x) \in \mathfrak{N}_x$, then $(\varphi \circ s)(x) = 0$, whence $(\varphi \circ s)(y) = 0$ for y sufficiently near x, and therefore $s(y) \in \mathfrak{N}_y$. This shows that $\mathfrak{N}$ is a subsheaf of $\mathcal{F}$. If t is a local section of $\mathcal{G}$ such that

[4]The printed French says "submodules of $\mathcal{F}$ and $\mathcal{G}$," omitting the stalk subscripts. The objects $\mathfrak{N}_x$ and $\mathfrak{I}_x$ are $\mathcal{A}_x$-modules, and the same sentence immediately uses $\mathcal{F}_x/\mathfrak{N}_x$, so $\mathcal{F}_x$ and $\mathcal{G}_x$ are required.

$t(x) \in \mathfrak{I}_x$, there exists a local section s of $\mathcal{F}$ such that $(\varphi \circ s)(x) = t(x)$; hence $\varphi \circ s = t$ near x. This shows that $\mathfrak{I}$ is a subsheaf of $\mathcal{G}$, isomorphic to $\mathcal{F}/\mathfrak{N}$.

The sheaf $\mathfrak{N}$ is called the *kernel* of φ and is denoted by $\mathrm{Ker}(\varphi)$; the sheaf $\mathfrak{I}$ is called the *image* of φ and is denoted by $\mathrm{Im}(\varphi)$; the sheaf $\mathcal{G}/\mathfrak{I}$ is called the *cokernel* of φ and is denoted by $\mathrm{Coker}(\varphi)$. A homomorphism φ is said to be *injective*, or one-to-one, if every φ_x is injective, which is equivalent to $\mathrm{Ker}(\varphi) = 0$; it is said to be *surjective* if every φ_x is surjective, which is equivalent to $\mathrm{Coker}(\varphi) = 0$; it is said to be *bijective* if it is both injective and surjective. In this case Proposition 7 shows that it is an isomorphism from $\mathcal{F}$ onto $\mathcal{G}$, and φ^{-1} is also a homomorphism. All definitions concerning homomorphisms of modules carry over in the same way to sheaves of modules. For example, a sequence of homomorphisms is said to be *exact* if the image of each homomorphism coincides with the kernel of the following homomorphism. If $\varphi : \mathcal{F} \to \mathcal{G}$ is a homomorphism, the sequences

$$0 \longrightarrow \mathrm{Ker}(\varphi) \longrightarrow \mathcal{F} \longrightarrow \mathrm{Im}(\varphi) \longrightarrow 0$$

and

$$0 \longrightarrow \mathrm{Im}(\varphi) \longrightarrow \mathcal{G} \longrightarrow \mathrm{Coker}(\varphi) \longrightarrow 0$$

are exact.

If φ is a homomorphism from $\mathcal{F}$ to $\mathcal{G}$, the map $s \mapsto \varphi \circ s$ is a $\Gamma(U, \mathcal{A})$-homomorphism from $\Gamma(U, \mathcal{F})$ to $\Gamma(U, \mathcal{G})$. Conversely, suppose that $\mathcal{A}$, $\mathcal{F}$, and $\mathcal{G}$ are defined, as in no. 6, by systems $(\mathcal{A}_U, \varphi_U^V)$, $(\mathcal{F}_U, \psi_U^V)$, and $(\mathcal{G}_U, \chi_U^V)$. Suppose also that, for every open set $U \subset X$, an $\mathcal{A}_U$-homomorphism $\varphi_U : \mathcal{F}_U \to \mathcal{G}_U$ is given such that

$$\chi_U^V \circ \varphi_V = \varphi_U \circ \psi_U^V.$$

Passing to the direct limit, the φ_U define a homomorphism $\varphi : \mathcal{F} \to \mathcal{G}$.

9. Direct sum of two sheaves.

Let $\mathcal{A}$ be a sheaf of rings, and let $\mathcal{F}$ and $\mathcal{G}$ be two sheaves of $\mathcal{A}$-modules. For every $x \in X$, form the module $\mathcal{F}_x + \mathcal{G}_x$, the *direct sum* of $\mathcal{F}_x$ and $\mathcal{G}_x$; an element of $\mathcal{F}_x + \mathcal{G}_x$ is a pair (f, g), with $f \in \mathcal{F}_x$ and $g \in \mathcal{G}_x$. Let $\mathcal{H}$ be the disjoint union of the $\mathcal{F}_x + \mathcal{G}_x$ as x ranges over X. We may identify $\mathcal{H}$ with the subset of $\mathcal{F} \times \mathcal{G}$ formed by the pairs (f, g) such that $\pi(f) = \pi(g)$. If $\mathcal{H}$ is endowed with the topology induced by that of $\mathcal{F} \times \mathcal{G}$, it is immediately verified that $\mathcal{H}$ is a sheaf of $\mathcal{A}$-modules. It is called the *direct sum* of $\mathcal{F}$ and $\mathcal{G}$ and is denoted by $\mathcal{F} + \mathcal{G}$. Every section of $\mathcal{F} + \mathcal{G}$ over an open set $U \subset X$ has the form $x \mapsto (s(x), t(x))$, where s and t are sections of $\mathcal{F}$ and $\mathcal{G}$ over U. In other words, $\Gamma(U, \mathcal{F} + \mathcal{G})$ is isomorphic to the direct sum $\Gamma(U, \mathcal{F}) + \Gamma(U, \mathcal{G})$.

The definition of the direct sum extends inductively to a finite number of sheaves of $\mathcal{A}$-modules. In particular, the direct-sum sheaf of p sheaves isomorphic to a given sheaf $\mathcal{F}$ will be denoted by $\mathcal{F}^p$.

10. Tensor product of two sheaves.

Let $\mathcal{A}$ be a sheaf of rings, let $\mathcal{F}$ be a sheaf of right $\mathcal{A}$-modules, and let $\mathcal{G}$ be a sheaf of left $\mathcal{A}$-modules. For every $x \in X$, put $\mathcal{H}_x = \mathcal{F}_x \otimes \mathcal{G}_x$, where the tensor product is taken over the ring $\mathcal{A}_x$ (cf., for example, [6], Chap. II, §2), and let $\mathcal{H}$ be the disjoint union of the $\mathcal{H}_x$.

Proposition 8. *There exists a unique sheaf structure on the set $\mathcal{H}$ such that, if s and t are sections of $\mathcal{F}$ and $\mathcal{G}$ over an open set U, the map*

$$x \longmapsto s(x) \otimes t(x) \in \mathcal{H}_x$$

is a section of $\mathcal{H}$ over U.

The sheaf $\mathcal{H}$ so defined is called the tensor product (over $\mathcal{A}$) of $\mathcal{F}$ and $\mathcal{G}$ and is denoted by $\mathcal{F} \otimes_{\mathcal{A}} \mathcal{G}$. If the $\mathcal{A}_x$ are commutative, it is a sheaf of $\mathcal{A}$-modules.

Suppose that $\mathcal{H}$ is endowed with a sheaf structure satisfying the stated condition, and let s_i and t_i be sections of $\mathcal{F}$ and $\mathcal{G}$ over an open set $U \subset X$. Then the map $x \mapsto \sum_i s_i(x) \otimes t_i(x)$ is a section of $\mathcal{H}$ over U. Every $h \in \mathcal{H}_x$ can be written as $h = \sum_i f_i \otimes g_i$, with $f_i \in \mathcal{F}_x$ and $g_i \in \mathcal{G}_x$, and hence also as $\sum_i s_i(x) \otimes t_i(x)$, where the s_i and t_i are sections defined in a neighborhood U of x. It follows that every section of $\mathcal{H}$ must locally be equal to a section of the preceding form, proving the uniqueness of the sheaf structure on $\mathcal{H}$.

We now prove existence. We may suppose that $\mathcal{A}$, $\mathcal{F}$, and $\mathcal{G}$ are defined, as in no. 6, by systems $(\mathcal{A}_U, \varphi_U^V)$, $(\mathcal{F}_U, \psi_U^V)$, and $(\mathcal{G}_U, \chi_U^V)$. Put $\mathcal{H}_U = \mathcal{F}_U \otimes \mathcal{G}_U$, where the tensor product is taken over $\mathcal{A}_U$. Passing to the tensor product, the homomorphisms ψ_U^V and χ_U^V define a homomorphism $\eta_U^V : \mathcal{H}_V \to \mathcal{H}_U$. Moreover,

$$\varinjlim_{x\in U} \mathcal{H}_U = \left(\varinjlim_{x\in U} \mathcal{F}_U\right) \otimes \left(\varinjlim_{x\in U} \mathcal{G}_U\right) = \mathcal{H}_x,$$

where the tensor product is taken over $\mathcal{A}_x$ (commutation of the tensor product with direct limits; cf., for example, [6], Chap. VI, Exer. 18). The sheaf defined by the system $(\mathcal{H}_U, \eta_U^V)$ may therefore be identified with $\mathcal{H}$, which is thereby endowed with a sheaf structure plainly satisfying the required condition. Finally, if the $\mathcal{A}_x$ are commutative, we may suppose that the $\mathcal{A}_U$ are commutative as well (it is enough to take $\mathcal{A}_U = \Gamma(U, \mathcal{A})$); hence $\mathcal{H}_U$ is an $\mathcal{A}_U$-module, and $\mathcal{H}$ is a sheaf of $\mathcal{A}$-modules.

Now let φ be an $\mathcal{A}$-homomorphism from $\mathcal{F}$ to $\mathcal{F}'$, and let ψ be an $\mathcal{A}$-homomorphism from $\mathcal{G}$ to $\mathcal{G}'$. Then $\varphi_x \otimes \psi_x$ is a homomorphism (of abelian groups in general, and of $\mathcal{A}_x$-modules if $\mathcal{A}_x$ is commutative), and the definition of $\mathcal{F} \otimes_{\mathcal{A}} \mathcal{G}$ shows that the collection of the $\varphi_x \otimes \psi_x$ is a homomorphism from $\mathcal{F} \otimes_{\mathcal{A}} \mathcal{G}$ to $\mathcal{F}' \otimes_{\mathcal{A}} \mathcal{G}'$. This homomorphism is denoted by $\varphi \otimes \psi$; if ψ is the identity map, we write φ instead of $\varphi \otimes 1$.

All the usual properties of the tensor product of two modules carry over directly to the tensor product of two sheaves of modules. For example, every exact sequence

$$\mathcal{F} \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F}'' \longrightarrow 0$$

gives rise to an exact sequence

$$\mathcal{F} \otimes_{\mathcal{A}} \mathcal{G} \longrightarrow \mathcal{F}' \otimes_{\mathcal{A}} \mathcal{G} \longrightarrow \mathcal{F}'' \otimes_{\mathcal{A}} \mathcal{G} \longrightarrow 0.$$

There are canonical isomorphisms

$$\mathcal{F} \otimes_{\mathcal{A}} (\mathcal{G}_1 + \mathcal{G}_2) \simeq \mathcal{F} \otimes_{\mathcal{A}} \mathcal{G}_1 + \mathcal{F} \otimes_{\mathcal{A}} \mathcal{G}_2, \qquad \mathcal{F} \otimes_{\mathcal{A}} \mathcal{A} \simeq \mathcal{F},$$

and, supposing the $\mathcal{A}_x$ commutative to simplify the notation,

$$\mathcal{F} \otimes_{\mathcal{A}} \mathcal{G} \simeq \mathcal{G} \otimes_{\mathcal{A}} \mathcal{F}, \qquad \mathcal{F} \otimes_{\mathcal{A}} (\mathcal{G} \otimes_{\mathcal{A}} \mathcal{K}) \simeq (\mathcal{F} \otimes_{\mathcal{A}} \mathcal{G}) \otimes_{\mathcal{A}} \mathcal{K}.$$

11. Sheaf of germs of homomorphisms from one sheaf to another.

Let $\mathcal{A}$ be a sheaf of rings, and let $\mathcal{F}$ and $\mathcal{G}$ be two sheaves of $\mathcal{A}$-modules. If U is an open subset of X, let $\mathcal{H}_U$ be the group of homomorphisms from $\mathcal{F}(U)$ to $\mathcal{G}(U)$ (we shall also say "a homomorphism from $\mathcal{F}$ to $\mathcal{G}$ over U" in place of "a homomorphism from $\mathcal{F}(U)$ to $\mathcal{G}(U)$"). Restriction of a homomorphism defines $\varphi_U^V : \mathcal{H}_V \to \mathcal{H}_U$. The sheaf defined by the system $(\mathcal{H}_U, \varphi_U^V)$ is called the *sheaf of germs of homomorphisms* from $\mathcal{F}$ to $\mathcal{G}$ and is denoted by $\mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G})$. If the $\mathcal{A}_x$ are commutative, $\mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G})$ is a sheaf of $\mathcal{A}$-modules.

An element of $\mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G})_x$, being a germ of a homomorphism from $\mathcal{F}$ to $\mathcal{G}$ in a neighborhood of x, unambiguously defines an $\mathcal{A}_x$-homomorphism from $\mathcal{F}_x$ to $\mathcal{G}_x$; hence there is a canonical homomorphism

$$\rho : \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G})_x \longrightarrow \mathrm{Hom}_{\mathcal{A}_x}(\mathcal{F}_x, \mathcal{G}_x).$$

In contrast to the operations studied thus far, however, ρ is not in general a bijection; in no. 14 we shall give a sufficient condition for it to be one.

If $\varphi : \mathcal{F}' \to \mathcal{F}$ and $\psi : \mathcal{G} \to \mathcal{G}'$ are homomorphisms, there is an evident homomorphism

$$\mathrm{Hom}_{\mathcal{A}}(\varphi, \psi) : \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}', \mathcal{G}').$$

Every exact sequence

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{G}' \longrightarrow \mathcal{G}''$$

gives rise to an exact sequence

$$0 \longrightarrow \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G}') \longrightarrow \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G}'').$$

There are also canonical isomorphisms

$$\mathrm{Hom}_{\mathcal{A}}(\mathcal{A}, \mathcal{G}) \simeq \mathcal{G},$$

$$\mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G}_1 + \mathcal{G}_2) \simeq \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G}_1) + \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G}_2),$$

and

$$\mathrm{Hom}_{\mathcal{A}}(\mathcal{F}_1 + \mathcal{F}_2, \mathcal{G}) \simeq \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}_1, \mathcal{G}) + \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}_2, \mathcal{G}).$$

§2. Coherent sheaves of modules

In this section, X denotes a topological space and $\mathcal{A}$ a sheaf of rings on X. We suppose that all the $\mathcal{A}_x$, $x \in X$, are commutative and possess an identity element varying continuously with x. All sheaves considered through no. 16 are sheaves of $\mathcal{A}$-modules, and all homomorphisms are $\mathcal{A}$-homomorphisms.

12. Definitions.

Let $\mathcal{F}$ be a sheaf of $\mathcal{A}$-modules, and let $s_1, \ldots, s_p$ be sections of $\mathcal{F}$ over an open set $U \subset X$. The rule sending every family $f_1, \ldots, f_p$ of elements of $\mathcal{A}_x$ to the element $\sum_{i=1}^{p} f_i\, s_i(x)$ of $\mathcal{F}_x$ gives a homomorphism $\varphi : \mathcal{A}^p \to \mathcal{F}$ defined over U (more precisely, in the notation of no. 4, φ is a homomorphism from $\mathcal{A}^p(U)$ to $\mathcal{F}(U)$). The kernel $\mathcal{R}(s_1, \ldots, s_p)$ of φ is a subsheaf of $\mathcal{A}^p$, called the *sheaf of relations* among the s_i; the image of φ is the subsheaf of $\mathcal{F}$ generated by the s_i. Conversely, every homomorphism $\varphi : \mathcal{A}^p \to \mathcal{F}$ defines sections $s_1, \ldots, s_p$ of $\mathcal{F}$ by the formulas

$$s_1(x) = \varphi_x(1, 0, \ldots, 0), \qquad \ldots, \qquad s_p(x) = \varphi_x(0, \ldots, 0, 1).$$

Definition 1. *A sheaf of $\mathcal{A}$-modules $\mathcal{F}$ is said to be of finite type if it is locally generated by finitely many of its sections.*

In other words, for every point $x \in X$ there must exist an open set U containing x and finitely many sections $s_1, \ldots, s_p$ of $\mathcal{F}$ over U such that every element of $\mathcal{F}_y$, $y \in U$, is a linear combination of the $s_i(y)$ with coefficients in $\mathcal{A}_y$. By the preceding discussion, this is equivalent to saying that the restriction of $\mathcal{F}$ to U is isomorphic to a quotient sheaf of some $\mathcal{A}^p$.

Proposition 1. *Let $\mathcal{F}$ be a sheaf of finite type. If $s_1, \ldots, s_p$ are sections of $\mathcal{F}$ defined over a neighborhood of a point $x \in X$ and generating $\mathcal{F}_x$, then they generate $\mathcal{F}_y$ for every y sufficiently close to x.*

Since $\mathcal{F}$ is of finite type, there are finitely many sections $t_1, \ldots, t_q$ of $\mathcal{F}$ near x which generate $\mathcal{F}_y$ for y sufficiently close to x. Since the $s_j(x)$ generate $\mathcal{F}_x$, there exist sections f_{ij} of $\mathcal{A}$ near x such that $t_i(x) = \sum_{j=1}^{p} f_{ij}(x) s_j(x)$. It follows that, for y sufficiently close to x,

$$t_i(y) = \sum_{j=1}^{p} f_{ij}(y) s_j(y),$$

and hence the $s_j(y)$ generate $\mathcal{F}_y$, q.e.d.

Definition 2. *A sheaf of $\mathcal{A}$-modules $\mathcal{F}$ is said to be coherent if:*

(a) *$\mathcal{F}$ is of finite type;*

(b) *if $s_1, \ldots, s_p$ are sections of $\mathcal{F}$ over an open set $U \subset X$, the sheaf of relations among the s_i is of finite type (on the open set U).*

Notice the *local* character of Definitions 1 and 2.

PROPOSITION 2. *Locally, every coherent sheaf is isomorphic to the cokernel of a homomorphism $\varphi : \mathcal{A}^q \to \mathcal{A}^p$.*

This follows immediately from the definitions and from the remarks preceding Definition 1.

PROPOSITION 3. *Every finite-type subsheaf of a coherent sheaf is coherent.*

Indeed, if a sheaf $\mathcal{F}$ satisfies condition (b) of Definition 2, then every subsheaf of $\mathcal{F}$ plainly satisfies it as well.

13. Main properties of coherent sheaves.

THEOREM 1. *Let*

$$0 \longrightarrow \mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H} \longrightarrow 0$$

be an exact sequence of homomorphisms. If two of the three sheaves $\mathcal{F}$, $\mathcal{G}$, and $\mathcal{H}$ are coherent, then so is the third.

Suppose first that $\mathcal{G}$ and $\mathcal{H}$ are coherent. Locally there is then a surjective homomorphism $\gamma : \mathcal{A}^p \to \mathcal{G}$. Let $\mathcal{S}$ be the kernel of $\beta \circ \gamma$. Since $\mathcal{H}$ is coherent, $\mathcal{S}$ is of finite type by condition (b); hence $\gamma(\mathcal{S})$ is of finite type and therefore coherent by Proposition 3. Since α is an isomorphism from $\mathcal{F}$ onto $\gamma(\mathcal{S})$, it follows that $\mathcal{F}$ is coherent.

Next suppose that $\mathcal{F}$ and $\mathcal{G}$ are coherent. Since $\mathcal{G}$ is of finite type, so is $\mathcal{H}$, and it remains to show that $\mathcal{H}$ satisfies condition (b) of Definition 2. Let $s_1, \ldots, s_p$ be finitely many sections of $\mathcal{H}$ near a point $x \in X$. Since the question is local, we may suppose that there are sections $s'_1, \ldots, s'_p$ of $\mathcal{G}$ such that $s_i = \beta(s'_i)$. Let also $n_1, \ldots, n_q$ be finitely many sections of $\mathcal{F}$ near x which generate $\mathcal{F}_y$ for y sufficiently close to x. A family $f_1, \ldots, f_p$ of elements of $\mathcal{A}_y$ belongs to $\mathcal{R}(s_1, \ldots, s_p)_y$ if and only if there exist $g_1, \ldots, g_q \in \mathcal{A}_y$ such that

$$\sum_{i=1}^{p} f_i s'_i = \sum_{j=1}^{q} g_j \alpha(n_j) \qquad \text{at } y.$$

The sheaf of relations among the s'_i and the $\alpha(n_j)$ is of finite type because $\mathcal{G}$ is coherent. Its image under the canonical projection $\mathcal{A}^{p+q} \to \mathcal{A}^p$ is $\mathcal{R}(s_1, \ldots, s_p)$, which is therefore of finite type. This proves that $\mathcal{H}$ is coherent.

Finally, suppose that $\mathcal{F}$ and $\mathcal{H}$ are coherent. Since the question is local, we may suppose that $\mathcal{F}$ is generated by finitely many sections $n_1, \ldots, n_q$ and $\mathcal{H}$ by finitely many sections $s_1, \ldots, s_p$; we may also suppose that there are sections s'_i of $\mathcal{G}$ such that $s_i = \beta(s'_i)$. The sections s'_i and $\alpha(n_j)$ then plainly generate $\mathcal{G}$, so $\mathcal{G}$ is of finite type.

Now let $t_1, \ldots, t_r$ be finitely many sections of $\mathcal{G}$ near a point x. Since $\mathcal{H}$ is coherent, there are sections f^i_j of $\mathcal{A}^r$ ($1 \le i \le r$, $1 \le j \le s$), defined near x, which generate the sheaf of relations among the $\beta(t_i)$. Put

$$u_j = \sum_{i=1}^{r} f^i_j t_i.$$

Since $\sum_{i=1}^{r} f^i_j \beta(t_i) = 0$, the u_j lie in $\alpha(\mathcal{F})$. Because $\mathcal{F}$ is coherent, the sheaf of relations among the u_j is generated near x by finitely many sections, say g^j_k ($1 \le j \le s$, $1 \le k \le t$). We claim that the families

$$\left(\sum_{j=1}^{s} g^j_k f^i_j \right)_{1 \le i \le r}$$

generate $\mathcal{R}(t_1, \ldots, t_r)$ near x. Indeed, suppose that $\sum_{i=1}^{r} f_i t_i = 0$ at y, with $f_i \in \mathcal{A}_y$. Then $\sum_{i=1}^{r} f_i \beta(t_i) = 0$, so there are $g_j \in \mathcal{A}_y$ such that $f_i = \sum_{j=1}^{s} g_j f^i_j$. Substitution into $\sum_i f_i t_i = 0$

gives $\sum_{j=1}^{s} g_j u_j = 0$. Thus the family (g_j) is a linear combination of the families (g_k^j), proving the claim. It follows that $\mathcal{G}$ satisfies condition (b), and the proof is complete.

COROLLARY. *The direct sum of a finite family of coherent sheaves is coherent.*

THEOREM 2. *Let φ be a homomorphism from a coherent sheaf $\mathcal{F}$ to a coherent sheaf $\mathcal{G}$. Then the kernel, cokernel, and image of φ are coherent sheaves.*

Since $\mathcal{F}$ is coherent, $\mathrm{Im}(\varphi)$ is of finite type and hence coherent by Proposition 3. Applying Theorem 1 to the exact sequences

$$0 \longrightarrow \mathrm{Ker}(\varphi) \longrightarrow \mathcal{F} \longrightarrow \mathrm{Im}(\varphi) \longrightarrow 0$$

and

$$0 \longrightarrow \mathrm{Im}(\varphi) \longrightarrow \mathcal{G} \longrightarrow \mathrm{Coker}(\varphi) \longrightarrow 0$$

shows that $\mathrm{Ker}(\varphi)$ and $\mathrm{Coker}(\varphi)$ are coherent.

COROLLARY. *Let $\mathcal{F}$ and $\mathcal{G}$ be two coherent subsheaves of a coherent sheaf $\mathcal{H}$. Then $\mathcal{F} + \mathcal{G}$ and $\mathcal{F} \cap \mathcal{G}$ are coherent.*

For $\mathcal{F} + \mathcal{G}$ this follows from Proposition 3; as for $\mathcal{F} \cap \mathcal{G}$, it is the kernel of the homomorphism $\mathcal{F} \to \mathcal{H}/\mathcal{G}$.

14. Operations on coherent sheaves.

We have just seen that the direct sum of finitely many coherent sheaves is coherent. We shall now prove analogous results for the functors $\otimes$ and Hom.

PROPOSITION 4. *If $\mathcal{F}$ and $\mathcal{G}$ are two coherent sheaves, then $\mathcal{F} \otimes_{\mathcal{A}} \mathcal{G}$ is coherent.*

By Proposition 2, $\mathcal{F}$ is locally isomorphic to the cokernel of a homomorphism $\varphi : \mathcal{A}^q \to \mathcal{A}^p$. Thus $\mathcal{F} \otimes_{\mathcal{A}} \mathcal{G}$ is locally isomorphic to the cokernel of

$$\varphi : \mathcal{A}^q \otimes_{\mathcal{A}} \mathcal{G} \longrightarrow \mathcal{A}^p \otimes_{\mathcal{A}} \mathcal{G}.$$

But $\mathcal{A}^q \otimes_{\mathcal{A}} \mathcal{G}$ and $\mathcal{A}^p \otimes_{\mathcal{A}} \mathcal{G}$ are respectively isomorphic to $\mathcal{G}^q$ and $\mathcal{G}^p$, which are coherent by the corollary to Theorem 1. Hence $\mathcal{F} \otimes_{\mathcal{A}} \mathcal{G}$ is coherent by Theorem 2.

PROPOSITION 5. *Let $\mathcal{F}$ and $\mathcal{G}$ be two sheaves, with $\mathcal{F}$ coherent. For every $x \in X$, the stalk $\mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G})_x$ is isomorphic to $\mathrm{Hom}_{\mathcal{A}_x}(\mathcal{F}_x, \mathcal{G}_x)$.*

More precisely, we shall show that the homomorphism

$$\rho : \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G})_x \longrightarrow \mathrm{Hom}_{\mathcal{A}_x}(\mathcal{F}_x, \mathcal{G}_x),$$

defined in no. 11, is bijective. First let $\psi : \mathcal{F} \to \mathcal{G}$ be a homomorphism defined near x which vanishes on $\mathcal{F}_x$. Since $\mathcal{F}$ is of finite type, it follows at once that ψ vanishes in a neighborhood of x; thus ρ is injective.

To prove that ρ is surjective, let $\varphi : \mathcal{F}_x \to \mathcal{G}_x$ be an $\mathcal{A}_x$-homomorphism. We must find a homomorphism $\psi : \mathcal{F} \to \mathcal{G}$ defined near x such that $\psi_x = \varphi$. Let $m_1, \ldots, m_p$ be finitely many sections of $\mathcal{F}$ near x which generate $\mathcal{F}_y$ for every y sufficiently close to x, and let f_j^i ($1 \leq i \leq p$, $1 \leq j \leq q$) be sections of $\mathcal{A}^p$ generating $\mathcal{R}(m_1, \ldots, m_p)$ near x. There are local sections $n_1, \ldots, n_p$ of $\mathcal{G}$ such that $n_i(x) = \varphi(m_i(x))$. Put

$$p_j = \sum_{i=1}^{p} f_j^i n_i, \qquad 1 \leq j \leq q.$$

The p_j are local sections of $\mathcal{G}$ which vanish at x, and hence at every point of some neighborhood U of x. It follows that, for $y \in U$, a relation $\sum_i f_i m_i(y) = 0$ with $f_i \in \mathcal{A}_y$ implies $\sum_i f_i n_i(y) = 0$. Consequently, for every element $m = \sum_i f_i m_i(y) \in \mathcal{F}_y$, we may set

$$\psi_y(m) = \sum_{i=1}^{p} f_i n_i(y) \in \mathcal{G}_y,$$

and this formula defines $\psi_y(m)$ unambiguously. The collection of the ψ_y, $y \in U$, is a homomorphism $\psi : \mathcal{F} \to \mathcal{G}$ over U with $\psi_x = \varphi$. This completes the proof.

PROPOSITION 6. *If $\mathcal{F}$ and $\mathcal{G}$ are coherent sheaves, then* $\mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G})$ *is coherent.*

The question being local, Proposition 2 allows us to suppose that there is an exact sequence

$$\mathcal{A}^q \longrightarrow \mathcal{A}^p \longrightarrow \mathcal{F} \longrightarrow 0.$$

It follows from the preceding proposition that the sequence

$$0 \longrightarrow \mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathrm{Hom}_{\mathcal{A}}(\mathcal{A}^p, \mathcal{G}) \longrightarrow \mathrm{Hom}_{\mathcal{A}}(\mathcal{A}^q, \mathcal{G})$$

is exact. Now $\mathrm{Hom}_{\mathcal{A}}(\mathcal{A}^p, \mathcal{G})$ is isomorphic to $\mathcal{G}^p$ and hence is coherent, and the same holds for $\mathrm{Hom}_{\mathcal{A}}(\mathcal{A}^q, \mathcal{G})$. Theorem 2 therefore shows that $\mathrm{Hom}_{\mathcal{A}}(\mathcal{F}, \mathcal{G})$ is coherent.

15. Coherent sheaves of rings.

The sheaf of rings $\mathcal{A}$ may be regarded as a sheaf of $\mathcal{A}$-modules; if this sheaf of $\mathcal{A}$-modules is coherent, we shall say that $\mathcal{A}$ is a *coherent sheaf of rings*. Since $\mathcal{A}$ is plainly of finite type, this means that $\mathcal{A}$ satisfies condition (b) of Definition 2.[5] In other words:

DEFINITION 3. *The sheaf $\mathcal{A}$ is a coherent sheaf of rings if the sheaf of relations among any finite number of sections of $\mathcal{A}$ over an open set U is a sheaf of finite type on U.*

EXAMPLES. (1) If X is a complex analytic variety, the sheaf of germs of holomorphic functions on X is a coherent sheaf of rings, by a theorem of K. Oka (cf. [3], Exposé XV, or [5], §5).

(2) If X is an algebraic variety, its sheaf of local rings is a coherent sheaf of rings (cf. no. 37, Proposition 1).

When $\mathcal{A}$ is a coherent sheaf of rings, the following results hold.

PROPOSITION 7. *A sheaf of $\mathcal{A}$-modules is coherent if and only if it is locally isomorphic to the cokernel of a homomorphism $\varphi : \mathcal{A}^q \to \mathcal{A}^p$.*

Necessity is precisely Proposition 2; sufficiency follows because $\mathcal{A}^p$ and $\mathcal{A}^q$ are coherent, together with Theorem 2.

PROPOSITION 8. *A subsheaf of $\mathcal{A}^p$ is coherent if and only if it is of finite type.*

This is a special case of Proposition 3.

COROLLARY. *The sheaf of relations among a finite number of sections of a coherent sheaf is coherent.*

Indeed, this sheaf is of finite type by the very definition of coherent sheaves.

PROPOSITION 9. *Let $\mathcal{F}$ be a coherent sheaf of $\mathcal{A}$-modules. For every $x \in X$, let $\mathcal{J}_x$ be the ideal of $\mathcal{A}_x$ consisting of the elements $a \in \mathcal{A}_x$ such that $a \cdot f = 0$ for every $f \in \mathcal{F}_x$. The $\mathcal{J}_x$ form a coherent sheaf of ideals (called the annihilator of $\mathcal{F}$).*

Indeed, $\mathcal{J}_x$ is the kernel of the homomorphism

$$\mathcal{A}_x \longrightarrow \mathrm{Hom}_{\mathcal{A}_x}(\mathcal{F}_x, \mathcal{F}_x);$$

one then applies Propositions 5 and 6 and Theorem 2.

More generally, the *conductor* $\mathcal{F} : \mathcal{G}$ of a coherent sheaf $\mathcal{G}$ into a coherent subsheaf $\mathcal{F}$ is a coherent sheaf of ideals (it is the annihilator of $\mathcal{G}/\mathcal{F}$).

16. Change of rings.

The notions of a sheaf of finite type and of a coherent sheaf are relative to a specified sheaf of rings $\mathcal{A}$. When several sheaves of rings are under consideration, we shall say "of finite type over $\mathcal{A}$" and "$\mathcal{A}$-coherent" to specify that the sheaves in question are sheaves of $\mathcal{A}$-modules.

THEOREM 3. *Let $\mathcal{A}$ be a coherent sheaf of rings and let $\mathcal{J}$ be a coherent sheaf of ideals of $\mathcal{A}$. Let $\mathcal{F}$ be a sheaf of $\mathcal{A}/\mathcal{J}$-modules. Then $\mathcal{F}$ is $\mathcal{A}/\mathcal{J}$-coherent if and only if it is $\mathcal{A}$-coherent. In particular, $\mathcal{A}/\mathcal{J}$ is a coherent sheaf of rings.*

It is clear that "of finite type over $\mathcal{A}$" is equivalent to "of finite type over $\mathcal{A}/\mathcal{J}$." Moreover, if $\mathcal{F}$ is $\mathcal{A}$-coherent and $s_1, \ldots, s_p$ are sections of $\mathcal{F}$ over an open set U, the sheaf of relations among

[5] The French source prints "Proposition 2" here; condition (b) is the condition in Definition 2.

the s_i with coefficients in $\mathcal{A}$ is of finite type over $\mathcal{A}$. It follows immediately that the sheaf of relations among the s_i with coefficients in $\mathcal{A}/\mathcal{J}$ is of finite type over $\mathcal{A}/\mathcal{J}$, since it is the image of the former under the canonical map

$$\mathcal{A}^p \longrightarrow (\mathcal{A}/\mathcal{J})^p.$$

Thus $\mathcal{F}$ is $\mathcal{A}/\mathcal{J}$-coherent. In particular, since $\mathcal{A}/\mathcal{J}$ is $\mathcal{A}$-coherent, it is also $\mathcal{A}/\mathcal{J}$-coherent; in other words, $\mathcal{A}/\mathcal{J}$ is a coherent sheaf of rings. Conversely, if $\mathcal{F}$ is $\mathcal{A}/\mathcal{J}$-coherent, it is locally isomorphic to the cokernel of a homomorphism

$$\varphi : (\mathcal{A}/\mathcal{J})^q \longrightarrow (\mathcal{A}/\mathcal{J})^p,$$

and since $\mathcal{A}/\mathcal{J}$ is $\mathcal{A}$-coherent, Theorem 2 shows that $\mathcal{F}$ is $\mathcal{A}$-coherent.

17. Extension and restriction of a coherent sheaf.

Let Y be a closed subspace of the space X. If $\mathcal{G}$ is a sheaf on Y, we denote by $\mathcal{G}^X$ the sheaf obtained by extending $\mathcal{G}$ by zero outside Y; it is a sheaf on X (cf. no. 5). If $\mathcal{A}$ is a sheaf of rings on Y, then $\mathcal{A}^X$ is a sheaf of rings on X, and if $\mathcal{F}$ is a sheaf of $\mathcal{A}$-modules, then $\mathcal{F}^X$ is a sheaf of $\mathcal{A}^X$-modules.

Proposition 10. *The sheaf $\mathcal{F}$ is of finite type over $\mathcal{A}$ if and only if $\mathcal{F}^X$ is of finite type over $\mathcal{A}^X$.*

Let U be an open set of X, and put $V = U \cap Y$. Every homomorphism $\varphi : \mathcal{A}^p \to \mathcal{F}$ over V defines a homomorphism $\varphi^X : (\mathcal{A}^X)^p \to \mathcal{F}^X$ over U, and conversely; φ is surjective if and only if φ^X is surjective. The proposition follows immediately.

The same argument proves:

Proposition 11. *The sheaf $\mathcal{F}$ is $\mathcal{A}$-coherent if and only if $\mathcal{F}^X$ is $\mathcal{A}^X$-coherent.*

Taking $\mathcal{F} = \mathcal{A}$ gives:

Corollary. *The sheaf $\mathcal{A}$ is a coherent sheaf of rings if and only if $\mathcal{A}^X$ is a coherent sheaf of rings.*

§3. Cohomology of a space with coefficients in a sheaf

In this section, X denotes a topological space, whether Hausdorff or not. By a *covering* of X we shall always mean an open covering.

18. Cochains of a covering.

Let $\mathfrak{U} = \{U_i\}_{i\in I}$ be a covering of X. If $s = (i_0, \ldots, i_p)$ is a finite sequence of elements of I, put

$$U_s = U_{i_0 \cdots i_p} = U_{i_0} \cap \cdots \cap U_{i_p}.$$

Let $\mathcal{F}$ be a sheaf of abelian groups on X. If p is an integer ≥ 0, a *p-cochain of $\mathfrak{U}$ with values in $\mathcal{F}$* is a function f which assigns to every sequence $s = (i_0, \ldots, i_p)$ of $p+1$ elements of I a section $f_s = f_{i_0\cdots i_p}$ of $\mathcal{F}$ over $U_{i_0\cdots i_p}$. The p-cochains form an abelian group, denoted $C^p(\mathfrak{U}, \mathcal{F})$; it is the product group

$$\prod_s \Gamma(U_s, \mathcal{F}),$$

where the product ranges over all sequences s of $p+1$ elements of I. The family of the $C^p(\mathfrak{U}, \mathcal{F})$, $p = 0, 1, \ldots$, is denoted $C(\mathfrak{U}, \mathcal{F})$. A p-cochain is also called a cochain of degree p.

A p-cochain f is said to be *alternating* if:

(a) $f_{i_0\cdots i_p} = 0$ whenever two of the indices $i_0, \ldots, i_p$ are equal;

(b) $f_{i_{\sigma 0}\cdots i_{\sigma p}} = \varepsilon_\sigma f_{i_0\cdots i_p}$ if σ is a permutation of $\{0, \ldots, p\}$, where ε_σ denotes the sign of σ.

The alternating p-cochains form a subgroup $C'^p(\mathfrak{U}, \mathcal{F})$ of $C^p(\mathfrak{U}, \mathcal{F})$; the family of the $C'^p(\mathfrak{U}, \mathcal{F})$ is denoted $C'(\mathfrak{U}, \mathcal{F})$.

19. Simplicial operations.

Let $S(I)$ be the simplex whose vertex set is I. An (ordered) simplex of $S(I)$ is a sequence $s = (i_0, \ldots, i_p)$ of elements of I; p is called the dimension of s. Let

$$K(I) = \sum_{p=0}^{\infty} K_p(I)$$

be the complex defined by $S(I)$: by definition, $K_p(I)$ is the free group with the set of simplices of dimension p in $S(I)$ as basis.

If s is a simplex of $S(I)$, we denote its set of vertices by $|s|$.

A map $h : K_p(I) \to K_q(I)$ is called a *simplicial endomorphism* if

(i) h is a homomorphism;

(ii) for every simplex s of dimension p in $S(I)$,

$$h(s) = \sum_{s'} c_s^{s'} s', \qquad c_s^{s'} \in \mathbf{Z},$$

where the sum ranges over the simplices s' of dimension q such that $|s'| \subset |s|$.

Let h be a simplicial endomorphism and let $f \in C^q(\mathfrak{U}, \mathcal{F})$ be a cochain of degree q. For every simplex s of dimension p, put

$$({}^t hf)_s = \sum_{s'} c_s^{s'} \rho_s^{s'}(f_{s'}),$$

where $\rho_s^{s'}$ denotes the restriction homomorphism

$$\Gamma(U_{s'}, \mathcal{F}) \longrightarrow \Gamma(U_s, \mathcal{F}),$$

which is defined because $|s'| \subset |s|$. The map $s \mapsto ({}^t hf)_s$ is a p-cochain, denoted ${}^t hf$. The map $f \mapsto {}^t hf$ is a homomorphism

$${}^t h : C^q(\mathfrak{U}, \mathcal{F}) \longrightarrow C^p(\mathfrak{U}, \mathcal{F}),$$

and one immediately verifies the formulas

$${}^t(h_1 + h_2) = {}^t h_1 + {}^t h_2, \qquad {}^t(h_1 \circ h_2) = {}^t h_2 \circ {}^t h_1, \qquad {}^t 1 = 1.$$

Note. In practice, the restriction homomorphism $\rho_s^{s'}$ is often left unwritten.

20. The cochain complexes.

Apply the preceding construction to the simplicial endomorphism

$$\partial : K_{p+1}(I) \longrightarrow K_p(I)$$

defined by the usual formula

$$\partial(i_0, \ldots, i_{p+1}) = \sum_{j=0}^{p+1} (-1)^j (i_0, \ldots, \widehat{\imath_j}, \ldots, i_{p+1}),$$

where, as usual, the hat means that the symbol beneath it is to be omitted.

We thereby obtain a homomorphism ${}^t\partial : C^p(\mathfrak{U}, \mathcal{F}) \to C^{p+1}(\mathfrak{U}, \mathcal{F})$, which we denote by d; thus, by definition,

$$(df)_{i_0 \cdots i_{p+1}} = \sum_{j=0}^{p+1} (-1)^j \rho_j (f_{i_0 \cdots \widehat{\imath_j} \cdots i_{p+1}}),$$

where ρ_j denotes the restriction homomorphism

$$\rho_j : \Gamma(U_{i_0 \cdots \widehat{\imath_j} \cdots i_{p+1}}, \mathcal{F}) \longrightarrow \Gamma(U_{i_0 \cdots i_{p+1}}, \mathcal{F}).$$

Since $\partial \circ \partial = 0$, we have $d \circ d = 0$. Thus $C(\mathfrak{U}, \mathcal{F})$ is equipped with a coboundary operator which makes it a complex. The qth cohomology group of the complex $C(\mathfrak{U}, \mathcal{F})$ is denoted $H^q(\mathfrak{U}, \mathcal{F})$. We have:

Proposition 1. $H^0(\mathfrak{U}, \mathcal{F}) = \Gamma(X, \mathcal{F})$.

A 0-cochain is a system $(f_i)_{i\in I}$, each f_i being a section of $\mathcal{F}$ over U_i. Such a cochain is a cocycle if and only if $f_i - f_j = 0$ over $U_i \cap U_j$; equivalently, there is a section f of $\mathcal{F}$ over all of X which agrees with f_i on U_i for every $i \in I$. This proves the proposition.

(Thus $H^0(\mathfrak{U}, \mathcal{F})$ is independent of $\mathfrak{U}$; of course, in general the same is not true of $H^q(\mathfrak{U}, \mathcal{F})$.)

It follows immediately that df is alternating whenever f is alternating; in other words, d leaves $C'(\mathfrak{U}, \mathcal{F})$ stable, and this is a subcomplex of $C(\mathfrak{U}, \mathcal{F})$. The cohomology groups of $C'(\mathfrak{U}, \mathcal{F})$ are denoted $H'^q(\mathfrak{U}, \mathcal{F})$.

Proposition 2. *For every $q \geq 0$, the injection of $C'(\mathfrak{U}, \mathcal{F})$ into $C(\mathfrak{U}, \mathcal{F})$ induces an isomorphism from $H'^q(\mathfrak{U}, \mathcal{F})$ onto $H^q(\mathfrak{U}, \mathcal{F})$.*

Endow I with a total ordering, and let h be the simplicial endomorphism of $K(I)$ defined as follows:

$$h(i_0, \dots, i_q) = 0 \quad \text{if two of the indices } i_0, \dots, i_q \text{ are equal,}$$

and

$$h(i_0, \dots, i_q) = \varepsilon_\sigma(i_{\sigma 0}, \dots, i_{\sigma q})$$

if all the indices are distinct, where σ is the permutation of $\{0, \dots, q\}$ such that

$$i_{\sigma 0} < i_{\sigma 1} < \cdots < i_{\sigma q}.$$

One immediately verifies that h commutes with ∂ and that $h(s) = s$ if $\dim(s) = 0$. It follows (cf. [7], Chap. VI, §5) that there is a simplicial endomorphism k, raising dimension by one, such that

$$1 - h = \partial \circ k + k \circ \partial.$$

Passing to $C(\mathfrak{U}, \mathcal{F})$ gives

$$1 - {}^t h = {}^t k \circ d + d \circ {}^t k.$$

But ${}^t h$ is immediately seen to be a *projection* of $C(\mathfrak{U}, \mathcal{F})$ onto $C'(\mathfrak{U}, \mathcal{F})$; since the preceding formula shows that it is a homotopy operator, the proposition follows. (Compare [7], Chap. VI, Th. 6.10.)

Corollary. $H^q(\mathfrak{U}, \mathcal{F}) = 0$ *for* $q > \dim(\mathfrak{U})$.

By the definition of $\dim(\mathfrak{U})$, one has $U_{i_0 \cdots i_q} = \varnothing$ for $q > \dim(\mathfrak{U})$ whenever the indices $i_0, \dots, i_q$ are distinct. Hence $C'^q(\mathfrak{U}, \mathcal{F}) = 0$, and therefore

$$H^q(\mathfrak{U}, \mathcal{F}) = H'^q(\mathfrak{U}, \mathcal{F}) = 0.$$

21. Passage from a covering to a finer covering.

A covering $\mathfrak{U} = \{U_i\}_{i\in I}$ is said to be *finer* than a covering $\mathfrak{V} = \{V_j\}_{j\in J}$ if there is a map $\tau : I \to J$ such that $U_i \subset V_{\tau i}$ for every $i \in I$. If $f \in C^q(\mathfrak{V}, \mathcal{F})$, put

$$(\tau f)_{i_0 \cdots i_q} = \rho_U^V(f_{\tau i_0 \cdots \tau i_q}),$$

where ρ_U^V denotes the restriction homomorphism defined by the inclusion of $U_{i_0 \cdots i_q}$ in $V_{\tau i_0 \cdots \tau i_q}$. The map $f \mapsto \tau f$ is a homomorphism from $C^q(\mathfrak{V}, \mathcal{F})$ to $C^q(\mathfrak{U}, \mathcal{F})$, defined for every $q \geq 0$ and commuting with d; it therefore induces homomorphisms

$$\tau^* : H^q(\mathfrak{V}, \mathcal{F}) \longrightarrow H^q(\mathfrak{U}, \mathcal{F}).$$

Proposition 3. *The homomorphisms $\tau^* : H^q(\mathfrak{V}, \mathcal{F}) \to H^q(\mathfrak{U}, \mathcal{F})$ depend only on $\mathfrak{U}$ and $\mathfrak{V}$, and not on the chosen map τ.*

Let τ and τ' be two maps from I to J such that $U_i \subset V_{\tau i}$ and $U_i \subset V_{\tau' i}$. We must show that $\tau^* = \tau'^*$.

For $f \in C^q(\mathfrak{V}, \mathcal{F})$, put

$$(kf)_{i_0 \cdots i_{q-1}} = \sum_{h=0}^{h=q-1} (-1)^h \rho_h(f_{\tau i_0 \cdots \tau i_h\, \tau' i_h \cdots \tau' i_{q-1}}),$$

where ρ_h denotes the restriction homomorphism defined by the inclusion of $U_{i_0 \cdots i_{q-1}}$ in $V_{\tau i_0 \cdots \tau i_h\, \tau' i_h \cdots \tau' i_{q-1}}$.

A direct calculation (cf. [7], Chap. VI, §3) gives

$$dkf + k\,df = \tau' f - \tau f,$$

which proves the proposition.

Consequently, if $\mathfrak{U}$ is finer than $\mathfrak{V}$, then for every integer $q \geq 0$ there is a canonical homomorphism from $H^q(\mathfrak{V}, \mathcal{F})$ to $H^q(\mathfrak{U}, \mathcal{F})$. In what follows, this homomorphism is denoted $\sigma(\mathfrak{U}, \mathfrak{V})$.

22. Cohomology groups of X with coefficients in the sheaf $\mathcal{F}$.

The relation "$\mathfrak{U}$ is finer than $\mathfrak{V}$" (which we shall henceforth write $\mathfrak{U} < \mathfrak{V}$) is a *preorder* on the coverings of X. Moreover, this relation is *directed*: if $\mathfrak{U} = \{U_i\}_{i \in I}$ and $\mathfrak{V} = \{V_j\}_{j \in J}$ are two coverings, then

$$\mathfrak{W} = \{U_i \cap V_j\}_{(i,j) \in I \times J}$$

is a covering finer than both $\mathfrak{U}$ and $\mathfrak{V}$.

We shall say that two coverings $\mathfrak{U}$ and $\mathfrak{V}$ are equivalent if $\mathfrak{U} < \mathfrak{V}$ and $\mathfrak{V} < \mathfrak{U}$. Every covering $\mathfrak{U}$ is equivalent to a covering $\mathfrak{U}'$ whose index set is a subset of $\mathfrak{P}(X)$: indeed, one may take for $\mathfrak{U}'$ the *set* of open subsets of X belonging to the *family* $\mathfrak{U}$. Thus one may speak of the set of equivalence classes of coverings; it is a directed ordered set.[6]

If $\mathfrak{U} < \mathfrak{V}$, the end of the preceding number defined a well-determined homomorphism

$$\sigma(\mathfrak{U}, \mathfrak{V}) : H^q(\mathfrak{V}, \mathcal{F}) \longrightarrow H^q(\mathfrak{U}, \mathcal{F})$$

for every integer $q \geq 0$ and every sheaf $\mathcal{F}$ on X. Clearly $\sigma(\mathfrak{U}, \mathfrak{U})$ is the identity, and

$$\sigma(\mathfrak{U}, \mathfrak{V}) \circ \sigma(\mathfrak{V}, \mathfrak{W}) = \sigma(\mathfrak{U}, \mathfrak{W})$$

if $\mathfrak{U} < \mathfrak{V} < \mathfrak{W}$. It follows that if $\mathfrak{U}$ and $\mathfrak{V}$ are equivalent, then $\sigma(\mathfrak{U}, \mathfrak{V})$ and $\sigma(\mathfrak{V}, \mathfrak{U})$ are inverse isomorphisms; in other words, $H^q(\mathfrak{U}, \mathcal{F})$ depends only on the equivalence class of the covering $\mathfrak{U}$.

Definition. *The qth cohomology group of X with coefficients in the sheaf $\mathcal{F}$, denoted $H^q(X, \mathcal{F})$, is the direct limit of the groups $H^q(\mathfrak{U}, \mathcal{F})$, taken over the directed ordered set of equivalence classes of coverings of X by means of the homomorphisms $\sigma(\mathfrak{U}, \mathfrak{V})$.*

In other words, an element of $H^q(X, \mathcal{F})$ is a pair $(\mathfrak{U}, x)$ with $x \in H^q(\mathfrak{U}, \mathcal{F})$, where two pairs $(\mathfrak{U}, x)$ and $(\mathfrak{V}, y)$ are identified if there is a covering $\mathfrak{W}$ such that $\mathfrak{W} < \mathfrak{U}$, $\mathfrak{W} < \mathfrak{V}$, and

$$\sigma(\mathfrak{W}, \mathfrak{U})(x) = \sigma(\mathfrak{W}, \mathfrak{V})(y) \quad \text{in } H^q(\mathfrak{W}, \mathcal{F}).$$

Thus every covering $\mathfrak{U}$ of X has an associated canonical homomorphism

$$\sigma(\mathfrak{U}) : H^q(\mathfrak{U}, \mathcal{F}) \longrightarrow H^q(X, \mathcal{F}).$$

One may also define $H^q(X, \mathcal{F})$ as the direct limit of the $H^q(\mathfrak{U}, \mathcal{F})$ over any cofinal family of coverings $\mathfrak{U}$. Thus, if X is quasi-compact (respectively, quasi-paracompact), it is enough to consider finite coverings (respectively, locally finite coverings).

[6]By contrast, one could not speak of "the set" of all coverings, since a covering is a family whose index set is arbitrary.

For $q = 0$, Proposition 1 gives:
PROPOSITION 4. $H^0(X, \mathcal{F}) = \Gamma(X, \mathcal{F})$.

23. Homomorphisms of sheaves.

Let φ be a homomorphism from a sheaf $\mathcal{F}$ to a sheaf $\mathcal{G}$. If $\mathfrak{U}$ is a covering of X, associate to every $f \in C^q(\mathfrak{U}, \mathcal{F})$ the element $\varphi f \in C^q(\mathfrak{U}, \mathcal{G})$ defined by $(\varphi f)_s = \varphi(f_s)$. The map $f \mapsto \varphi f$ is a homomorphism from $C(\mathfrak{U}, \mathcal{F})$ to $C(\mathfrak{U}, \mathcal{G})$ which commutes with the coboundary, and hence induces homomorphisms

$$\varphi^* : H^q(\mathfrak{U}, \mathcal{F}) \longrightarrow H^q(\mathfrak{U}, \mathcal{G}).$$

We have

$$\varphi^* \circ \sigma(\mathfrak{U}, \mathfrak{V}) = \sigma(\mathfrak{U}, \mathfrak{V}) \circ \varphi^*,$$

and passage to the limit therefore gives homomorphisms

$$\varphi^* : H^q(X, \mathcal{F}) \longrightarrow H^q(X, \mathcal{G}).$$

When $q = 0$, φ^* coincides with the homomorphism from $\Gamma(X, \mathcal{F})$ to $\Gamma(X, \mathcal{G})$ naturally defined by φ.

In general, the homomorphisms φ^* have the usual formal properties:

$$(\varphi + \psi)^* = \varphi^* + \psi^*, \qquad (\varphi \circ \psi)^* = \varphi^* \circ \psi^*, \qquad 1^* = 1.$$

In other words, for every $q \geq 0$, $H^q(X, \mathcal{F})$ is a covariant additive functor of $\mathcal{F}$. In particular, if $\mathcal{F}$ is the direct sum of two sheaves $\mathcal{G}_1$ and $\mathcal{G}_2$, then $H^q(X, \mathcal{F})$ is the direct sum of $H^q(X, \mathcal{G}_1)$ and $H^q(X, \mathcal{G}_2)$.

Suppose that $\mathcal{F}$ is a sheaf of $\mathcal{A}$-modules. Every global section of the sheaf $\mathcal{A}$ defines an endomorphism of $\mathcal{F}$, and hence of each $H^q(X, \mathcal{F})$. It follows that the $H^q(X, \mathcal{F})$ are modules over the ring $\Gamma(X, \mathcal{A})$.

24. Exact sequence of sheaves: the general case.

Let

$$0 \longrightarrow \mathcal{A} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\beta} \mathcal{C} \longrightarrow 0$$

be an exact sequence of sheaves. If $\mathfrak{U}$ is a covering of X, the sequence

$$0 \longrightarrow C(\mathfrak{U}, \mathcal{A}) \xrightarrow{\alpha} C(\mathfrak{U}, \mathcal{B}) \xrightarrow{\beta} C(\mathfrak{U}, \mathcal{C})$$

is plainly exact, but the homomorphism β is not in general surjective. Denote by $C_0(\mathfrak{U}, \mathcal{C})$ the image of this homomorphism; it is a subcomplex of $C(\mathfrak{U}, \mathcal{C})$, whose cohomology groups will be denoted $H_0^q(\mathfrak{U}, \mathcal{C})$. The exact sequence of complexes

$$0 \longrightarrow C(\mathfrak{U}, \mathcal{A}) \longrightarrow C(\mathfrak{U}, \mathcal{B}) \longrightarrow C_0(\mathfrak{U}, \mathcal{C}) \longrightarrow 0$$

gives rise to an exact cohomology sequence

$$\cdots \longrightarrow H^q(\mathfrak{U}, \mathcal{B}) \longrightarrow H_0^q(\mathfrak{U}, \mathcal{C}) \xrightarrow{d} H^{q+1}(\mathfrak{U}, \mathcal{A}) \longrightarrow H^{q+1}(\mathfrak{U}, \mathcal{B}) \longrightarrow \cdots,$$

where the coboundary operator d is defined in the usual way.

Now let $\mathfrak{U} = \{U_i\}_{i \in I}$ and $\mathfrak{V} = \{V_j\}_{j \in J}$ be two coverings, and let $\tau : I \to J$ be a map such that $U_i \subset V_{\tau i}$; thus $\mathfrak{U} < \mathfrak{V}$. The commutative diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & C(\mathfrak{V}, \mathcal{A}) & \longrightarrow & C(\mathfrak{V}, \mathcal{B}) & \longrightarrow & C(\mathfrak{V}, \mathcal{C}) \\
 & & \downarrow{\scriptstyle\tau} & & \downarrow{\scriptstyle\tau} & & \downarrow{\scriptstyle\tau} \\
0 & \longrightarrow & C(\mathfrak{U}, \mathcal{A}) & \longrightarrow & C(\mathfrak{U}, \mathcal{B}) & \longrightarrow & C(\mathfrak{U}, \mathcal{C})
\end{array}$$

shows that τ maps $C_0(\mathfrak{V},\mathcal{C})$ into $C_0(\mathfrak{U},\mathcal{C})$, and hence defines homomorphisms

$$\tau^* : H_0^q(\mathfrak{V},\mathcal{C}) \longrightarrow H_0^q(\mathfrak{U},\mathcal{C}).$$

Moreover, the homomorphisms τ^* are independent of the choice of the map τ.[7] Indeed, if $f \in C_0^q(\mathfrak{V},\mathcal{C})$, then $kf \in C_0^{q-1}(\mathfrak{U},\mathcal{C})$, with the notation of the proof of Proposition 3. We have therefore obtained canonical homomorphisms

$$\sigma(\mathfrak{U},\mathfrak{V}) : H_0^q(\mathfrak{V},\mathcal{C}) \longrightarrow H_0^q(\mathfrak{U},\mathcal{C});$$

we may then define $H_0^q(X,\mathcal{C})$ as the direct limit, over $\mathfrak{U}$, of the groups $H_0^q(\mathfrak{U},\mathcal{C})$.

Since a direct limit of exact sequences is exact (cf. [7], Chap. VIII, Th. 5.4), we obtain:

Proposition 5. *The sequence*

$$\cdots \longrightarrow H^q(X,\mathcal{B}) \xrightarrow{\beta^*} H_0^q(X,\mathcal{C}) \xrightarrow{d} H^{q+1}(X,\mathcal{A}) \xrightarrow{\alpha^*} H^{q+1}(X,\mathcal{B}) \longrightarrow \cdots$$

is exact.

Here d denotes the homomorphism obtained by passage to the limit from

$$d : H_0^q(\mathfrak{U},\mathcal{C}) \longrightarrow H^{q+1}(\mathfrak{U},\mathcal{A}).$$

In order to apply the preceding proposition, it is useful to compare the groups $H_0^q(X,\mathcal{C})$ and $H^q(X,\mathcal{C})$. The injection of $C_0(\mathfrak{U},\mathcal{C})$ into $C(\mathfrak{U},\mathcal{C})$ defines homomorphisms

$$H_0^q(\mathfrak{U},\mathcal{C}) \longrightarrow H^q(\mathfrak{U},\mathcal{C}),$$

and hence, by passage to the limit over $\mathfrak{U}$, homomorphisms

$$H_0^q(X,\mathcal{C}) \longrightarrow H^q(X,\mathcal{C}).$$

Proposition 6. *The canonical homomorphism $H_0^q(X,\mathcal{C}) \to H^q(X,\mathcal{C})$ is bijective for $q = 0$ and injective for $q = 1$.*

We first prove a lemma.

Lemma 1. *Let $\mathfrak{V} = \{V_j\}_{j\in J}$ be a covering, and let $f = (f_j)$ be an element of $C^0(\mathfrak{V},\mathcal{C})$. There exist a covering $\mathfrak{U} = \{U_i\}_{i\in I}$ and a map $\tau : I \to J$ such that $U_i \subset V_{\tau i}$ and $\tau f \in C_0^0(\mathfrak{U},\mathcal{C})$.*

For every $x \in X$, choose $\tau x \in J$ such that $x \in V_{\tau x}$. Since $f_{\tau x}$ is a section of $\mathcal{C}$ over $V_{\tau x}$, there exist an open neighborhood U_x of x, contained in $V_{\tau x}$, and a section b_x of $\mathcal{B}$ over U_x such that $\beta(b_x) = f_{\tau x}$ on U_x. The $\{U_x\}_{x\in X}$ form a covering $\mathfrak{U}$ of X, and the b_x form a 0-cochain b of $\mathfrak{U}$ with values in $\mathcal{B}$; since $\tau f = \beta(b)$, we indeed have $\tau f \in C_0^0(\mathfrak{U},\mathcal{C})$.

We now show that $H_0^1(X,\mathcal{C}) \to H^1(X,\mathcal{C})$ is injective. An element of the kernel of this map can be represented by a 1-cocycle $z = (z_{j_0 j_1}) \in C_0'(\mathfrak{V},\mathcal{C})$ for which there exists $f = (f_j) \in C^0(\mathfrak{V},\mathcal{C})$ with $df = z$. Applying Lemma 1 to f, we find a covering $\mathfrak{U}$ such that $\tau f \in C_0^0(\mathfrak{U},\mathcal{C})$, which implies that τz is cohomologous to 0[8] in $C_0(\mathfrak{U},\mathcal{C})$, and therefore has image 0 in $H_0^1(X,\mathcal{C})$. The same argument shows that $H_0^0(X,\mathcal{C}) \to H^0(X,\mathcal{C})$ is bijective.

Corollary 1. *There is an exact sequence*

$$0 \longrightarrow H^0(X,\mathcal{A}) \longrightarrow H^0(X,\mathcal{B}) \longrightarrow H^0(X,\mathcal{C}) \longrightarrow H^1(X,\mathcal{A}) \longrightarrow H^1(X,\mathcal{B}) \longrightarrow H^1(X,\mathcal{C}).$$

This is an immediate consequence of Propositions 5 and 6.

Corollary 2. *If $H^1(X,\mathcal{A}) = 0$, then $\Gamma(X,\mathcal{B}) \to \Gamma(X,\mathcal{C})$ is surjective.*

[7] The printed source says "the choice of the map τ^*"; the auxiliary map being chosen is τ, whereas τ^* is the induced homomorphism.

[8] The printed source leaves a blank after "cohomologous to." The missing term must be 0: $\tau z = d(\tau f)$ inside $C_0(\mathfrak{U},\mathcal{C})$.

25. Exact sequence of sheaves: the case where X is paracompact. Recall that a space X is called paracompact if it is Hausdorff and every covering of X admits a finer locally finite covering. For such a space, Proposition 6 may be extended to every value of q (I do not know whether such an extension is possible for non-Hausdorff spaces):

PROPOSITION 7. *If X is paracompact, the canonical homomorphism*

$$H_0^q(X,\mathcal{C}) \longrightarrow H^q(X,\mathcal{C})$$

is bijective for every $q \geq 0$.

The proposition is an immediate consequence of the following lemma, analogous to Lemma 1:

LEMMA 2. *Let $\mathfrak{V} = \{V_j\}_{j\in J}$ be a covering, and let $f = (f_{j_0\dots j_q})$ be an element of $C^q(\mathfrak{V},\mathcal{C})$. There exist a covering $\mathfrak{U} = \{U_i\}_{i\in I}$ and a map $\tau : I \to J$ such that $U_i \subset V_{\tau i}$ and $\tau f \in C_0^q(\mathfrak{U},\mathcal{C})$.*

Since X is paracompact, we may suppose that $\mathfrak{V}$ is locally finite. There is then a covering $\{W_j\}_{j\in J}$ such that $\overline{W}_j \subset V_j$. For every $x \in X$, choose an open neighborhood U_x of x such that:

(a) if $x \in V_j$ (respectively, $x \in \overline{W}_j$), then $U_x \subset V_j$ (respectively, $U_x \subset W_j$);

(b) if $U_x \cap W_j \neq \varnothing$, then $U_x \subset V_j$;

(c) if $x \in V_{j_0\dots j_q}$, there is a section b of $\mathcal{B}$ over U_x such that

$$\beta(b) = f_{j_0\dots j_q} \quad \text{over } U_x.$$

Condition (c) can be achieved by the definition of a quotient sheaf and the fact that x belongs to only finitely many of the sets $V_{j_0\dots j_q}$. Once (c) has been obtained, it is enough to shrink U_x suitably to satisfy (a) and (b).

The family $\{U_x\}_{x\in X}$ forms a covering $\mathfrak{U}$; for every $x \in X$, choose $\tau x \in J$ such that $x \in W_{\tau x}$. We now verify that $\tau f \in C_0^q(\mathfrak{U},\mathcal{C})$,[9] that is, that $f_{\tau x_0\dots\tau x_q}$ is the image under β of a section of $\mathcal{B}$ over

$$U_{x_0} \cap \dots \cap U_{x_q}.$$

If this intersection is empty, the assertion is clear. Otherwise, $U_{x_0} \cap U_{x_k} \neq \varnothing$ for $0 \leq k \leq q$; since $U_{x_k} \subset W_{\tau x_k}$, we have $U_{x_0} \cap W_{\tau x_k} \neq \varnothing$. By (b), it follows that $U_{x_0} \subset V_{\tau x_k}$, and hence $x_0 \in V_{\tau x_0\dots\tau x_q}$. Applying (c), we see that there is a section b of $\mathcal{B}$ over U_{x_0} such that[10]

$$\beta(b) = f_{\tau x_0\dots\tau x_q} \quad \text{over } U_{x_0},$$

and therefore, a fortiori, over $U_{x_0} \cap \dots \cap U_{x_q}$. This completes the proof.

COROLLARY. *If X is paracompact, there is an exact sequence*

$$\cdots \longrightarrow H^q(X,\mathcal{B}) \xrightarrow{\beta^*} H^q(X,\mathcal{C}) \xrightarrow{d} H^{q+1}(X,\mathcal{A}) \xrightarrow{\alpha^*} H^{q+1}(X,\mathcal{B}) \longrightarrow \cdots.$$

Here d is the composite of the inverse of the isomorphism $H_0^q(X,\mathcal{C}) \to H^q(X,\mathcal{C})$ with $d : H_0^q(X,\mathcal{C}) \to H^{q+1}(X,\mathcal{A})$; thus

$$H^q(X,\mathcal{C}) \xrightarrow{\sim} H_0^q(X,\mathcal{C}) \xrightarrow{d} H^{q+1}(X,\mathcal{A}).$$

The preceding exact sequence is called the *cohomology exact sequence* defined by the given exact sequence of sheaves

$$0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{B} \longrightarrow \mathcal{C} \longrightarrow 0.$$

[9] The printed source has the unadorned U here. The covering introduced in the lemma is $\mathfrak{U}$; plain U denotes the individual neighborhoods U_x.

[10] The printed source writes $\beta(b)_x$ although no point x is defined in this assertion and the equality is asserted over the whole open set U_{x_0}.

More generally, it holds whenever one can prove that $H^q_0(X,\mathcal{C}) \to H^q(X,\mathcal{C})$ is bijective. We shall see in no. 47 that this is the case when X is an algebraic variety and $\mathcal{A}$ is a coherent algebraic sheaf.

26. Cohomology of a closed subspace. Let $\mathcal{F}$ be a sheaf on the space X, and let Y be a subspace of X. Let $\mathcal{F}(Y)$ be the sheaf induced by $\mathcal{F}$ on Y, in the sense of no. 4. If $\mathfrak{U} = \{U_i\}_{i\in I}$ is a covering of X, the sets $U'_i = Y \cap U_i$ form a covering $\mathfrak{U}'$ of Y. If $f_{i_0\ldots i_q}$ is a section of $\mathcal{F}$ over $U_{i_0\ldots i_q}$, its restriction to

$$U'_{i_0\ldots i_q} = Y \cap U_{i_0\ldots i_q}$$

is a section of $\mathcal{F}(Y)$.[11] Restriction is a homomorphism

$$\rho : C(\mathfrak{U},\mathcal{F}) \longrightarrow C(\mathfrak{U}',\mathcal{F}(Y))$$

which commutes with d, and hence defines

$$\rho^* : H^q(\mathfrak{U},\mathcal{F}) \longrightarrow H^q(\mathfrak{U}',\mathcal{F}(Y)).$$

If $\mathfrak{U} < \mathfrak{V}$, then $\mathfrak{U}' < \mathfrak{V}'$, and

$$\rho^* \circ \sigma(\mathfrak{U},\mathfrak{V}) = \sigma(\mathfrak{U}',\mathfrak{V}') \circ \rho^*.$$

Thus, by passage to the limit over $\mathfrak{U}$, the homomorphisms ρ^* define homomorphisms

$$\rho^* : H^q(X,\mathcal{F}) \longrightarrow H^q(Y,\mathcal{F}(Y)).$$

Proposition 8. *Suppose that Y is closed in X and that $\mathcal{F}$ is zero outside Y. Then the homomorphism*

$$\rho^* : H^q(X,\mathcal{F}) \longrightarrow H^q(Y,\mathcal{F}(Y))$$

is bijective for every $q \geq 0$.

The proposition follows from the following two facts:

(a) Every covering $\mathfrak{W} = \{W_i\}_{i\in I}$ of Y is of the form $\mathfrak{U}'$, where $\mathfrak{U}$ is a covering of X.

Indeed, it is enough to set $U_i = W_i \cup (X - Y)$, since Y is closed in X.

(b) For every covering $\mathfrak{U}$ of X, the homomorphism

$$\rho : C(\mathfrak{U},\mathcal{F}) \longrightarrow C(\mathfrak{U}',\mathcal{F}(Y))$$

is bijective.

Indeed, this follows from Proposition 5 of no. 5, applied to $U_{i_0\ldots i_q}$ and the sheaf $\mathcal{F}$.

Proposition 8 may also be expressed as follows. If $\mathcal{G}$ is a sheaf on Y, and if $\mathcal{G}^X$ is the sheaf obtained by extending $\mathcal{G}$ by zero outside Y, then

$$H^q(Y,\mathcal{G}) = H^q(X,\mathcal{G}^X) \qquad (q \geq 0).$$

In other words, the identification of $\mathcal{G}$ with $\mathcal{G}^X$ is compatible with passage to cohomology groups.

§4. Comparison of the cohomology groups of different coverings.

In this section, X denotes a topological space and $\mathcal{F}$ a sheaf on X. We propose to give conditions on a covering $\mathfrak{U}$ of X under which

$$H^n(\mathfrak{U},\mathcal{F}) = H^n(X,\mathcal{F}) \qquad \text{for every } n \geq 0.$$

[11] The printed source places a stray comma between $U'_{i_0\ldots i_q}$ and the defining equality. It is removed here and in the corrected French layer.

27. Double complexes. A double complex (cf. [6], Chap. IV, §4) is a bigraded abelian group

$$K = \sum_{p,q} K^{p,q}, \qquad p \geq 0, \quad q \geq 0,$$

equipped with two endomorphisms d' and d'' satisfying

$$\begin{cases} d' \text{ maps } K^{p,q} \text{ into } K^{p+1,q}, \text{ and } d'' \text{ maps } K^{p,q} \text{ into } K^{p,q+1}, \\ d' \circ d' = 0, \qquad d' \circ d'' + d'' \circ d' = 0, \qquad d'' \circ d'' = 0. \end{cases}$$

An element of $K^{p,q}$ is called bihomogeneous, of bidegree (p,q) and total degree $p+q$. The endomorphism $d = d' + d''$ satisfies $d \circ d = 0$, and the cohomology groups of K, equipped with this coboundary operator, will be denoted by $H^n(K)$, where n denotes the total degree.

We may also equip K with the coboundary operator d'. Since d' is compatible with the bigrading of K, this gives cohomology groups denoted by $H_I^{p,q}(K)$; with d'' we obtain the groups $H_{II}^{p,q}(K)$.

We shall denote by K_{II}^q the subgroup of $K^{0,q}$ consisting of the elements x such that $d'(x) = 0$, and by K_{II} the direct sum of the K_{II}^q $(q = 0, 1, \ldots)$. There is an analogous definition for

$$K_I = \sum_{p=0}^{\infty} K_I^p.$$

Notice that

$$K_{II}^q = H_I^{0,q}(K) \qquad \text{and} \qquad K_I^p = H_{II}^{p,0}(K).$$

The group K_{II} is a subcomplex of K, and on K_{II} the operator d coincides with d''.

Proposition 1. *If $H_I^{p,q}(K) = 0$ for $p > 0$ and $q \geq 0$, the injection $K_{II} \to K$ induces a bijection*

$$H^n(K_{II}) \longrightarrow H^n(K)$$

for every $n \geq 0$.

(Cf. [4], Exposé XVII-6, whose proof we reproduce below.)

Replacing K by K/K_{II}, we are reduced to proving that, if $H_I^{p,q}(K) = 0$ for $p \geq 0$ and $q \geq 0$, then $H^n(K) = 0$ for every $n \geq 0$. Set[12]

$$K_h = \sum_{p \geq 0,\, q \geq h} K^{p,q}.$$

The K_h $(h = 0, 1, \ldots)$ are nested subcomplexes of K, and K_h/K_{h+1} is isomorphic to

$$\sum_{p=0}^{\infty} K^{p,h},$$

equipped with the coboundary operator d'. Consequently,[13]

$$H^n(K_h/K_{h+1}) = H_I^{n-h,h}(K) = 0$$

for all n and h, and hence $H^n(K_h) = H^n(K_{h+1})$. Since $H^n(K_h) = 0$ when $h > n$, descending induction on h gives $H^n(K_h) = 0$ for all n and h. Finally $K_0 = K$, and the proposition is proved.

28. Double complex defined by two coverings. Let $\mathfrak{U} = \{U_i\}_{i \in I}$ and $\mathfrak{V} = \{V_j\}_{j \in J}$ be two coverings of X. If s is a p-simplex of $S(I)$ and s' is a q-simplex of $S(J)$, we shall denote by

[12] The printed definition sums only over $q \geq h$ while leaving p free. Since K_h is a subgroup of the whole double complex and K_h/K_{h+1} is the full row $\sum_{p=0}^{\infty} K^{p,h}$, the corrected definition sums over $p \geq 0$ as well.

[13] The printed source has $H_I^{h,n-h}(K)$. In the row $q = h$, an element of total degree n has horizontal degree $p = n - h$, so the correct term is $H_I^{n-h,h}(K)$.

U_s the intersection of the U_i for $i \in s$ (cf. no. 18), by $V_{s'}$ the intersection of the V_j for $j \in s'$, by $\mathfrak{V}_s$ the covering of U_s formed by the sets $\{U_s \cap V_j\}_{j \in J}$, and by $\mathfrak{U}_{s'}$ the covering of $V_{s'}$ formed by the sets $\{V_{s'} \cap U_i\}_{i \in I}$.

Define a double complex

$$C(\mathfrak{U}, \mathfrak{V}; \mathcal{F}) = \sum_{p,q} C^{p,q}(\mathfrak{U}, \mathfrak{V}; \mathcal{F})$$

as follows:

$$C^{p,q}(\mathfrak{U}, \mathfrak{V}; \mathcal{F}) = \prod \Gamma(U_s \cap V_{s'}, \mathcal{F}),$$

where the product is taken over all pairs (s, s') such that s is a simplex of dimension p in $S(I)$ and s' is a simplex of dimension q in $S(J)$.

Thus an element $f \in C^{p,q}(\mathfrak{U}, \mathfrak{V}; \mathcal{F})$ is a system $(f_{s,s'})$ of sections of $\mathcal{F}$ over the sets $U_s \cap V_{s'}$; equivalently, with the notation of no. 18, it is a system

$$f_{i_0 \dots i_p, j_0 \dots j_q} \in \Gamma(U_{i_0 \dots i_p} \cap V_{j_0 \dots j_q}, \mathcal{F}).$$

We may also identify $C^{p,q}(\mathfrak{U}, \mathfrak{V}; \mathcal{F})$ with

$$\prod_{s'} C^p(\mathfrak{U}_{s'}, \mathcal{F}).$$

For each s' there is a coboundary operation

$$d : C^p(\mathfrak{U}_{s'}, \mathcal{F}) \longrightarrow C^{p+1}(\mathfrak{U}_{s'}, \mathcal{F}),$$

and hence a homomorphism

$$d_{\mathfrak{U}} : C^{p,q}(\mathfrak{U}, \mathfrak{V}; \mathcal{F}) \longrightarrow C^{p+1,q}(\mathfrak{U}, \mathfrak{V}; \mathcal{F}).$$

Writing out the definition of $d_{\mathfrak{U}}$ gives

$$(d_{\mathfrak{U}} f)_{i_0 \dots i_{p+1}, j_0 \dots j_q} = \sum_{k=0}^{k=p+1} (-1)^k \rho_k\big(f_{i_0 \dots \hat{i}_k \dots i_{p+1}, j_0 \dots j_q}\big),$$

where ρ_k denotes the restriction homomorphism defined by the inclusion of[14]

$$U_{i_0 \dots i_{p+1}} \cap V_{j_0 \dots j_q} \quad \text{in} \quad U_{i_0 \dots \hat{i}_k \dots i_{p+1}} \cap V_{j_0 \dots j_q}.$$

Similarly, define

$$d_{\mathfrak{V}} : C^{p,q}(\mathfrak{U}, \mathfrak{V}; \mathcal{F}) \longrightarrow C^{p,q+1}(\mathfrak{U}, \mathfrak{V}; \mathcal{F});$$

then

$$(d_{\mathfrak{V}} f)_{i_0 \dots i_p, j_0 \dots j_{q+1}} = \sum_{h=0}^{h=q+1} (-1)^h \rho_h\big(f_{i_0 \dots i_p, j_0 \dots \hat{j}_h \dots j_{q+1}}\big).$$

It is clear that

$$d_{\mathfrak{U}} \circ d_{\mathfrak{U}} = 0, \qquad d_{\mathfrak{U}} \circ d_{\mathfrak{V}} = d_{\mathfrak{V}} \circ d_{\mathfrak{U}}, \qquad d_{\mathfrak{V}} \circ d_{\mathfrak{V}} = 0.$$

Setting

$$d' = d_{\mathfrak{U}}, \qquad d'' = (-1)^p d_{\mathfrak{V}},$$

endows $C(\mathfrak{U}, \mathfrak{V}; \mathcal{F})$ with the structure of a double complex. We may therefore apply to $K = C(\mathfrak{U}, \mathfrak{V}; \mathcal{F})$ the definitions of the preceding number. The groups or complexes denoted in the general case by

$$H^n(K), \quad H_I^{p,q}(K), \quad H_{II}^{p,q}(K), \quad K_I, \quad K_{II}$$

[14] The printed target intersection retains i_k. The section being restricted is indexed by the simplex with i_k omitted, so the larger target of the inclusion must omit i_k as well. The corrected French and English restore the hat; diplomatic French preserves the print.

will here be denoted respectively by

$$H^n(\mathfrak{U},\mathfrak{V};\mathcal{F}),\quad H_I^{p,q}(\mathfrak{U},\mathfrak{V};\mathcal{F}),\quad H_{II}^{p,q}(\mathfrak{U},\mathfrak{V};\mathcal{F}),\quad C_I(\mathfrak{U},\mathfrak{V};\mathcal{F}),\quad C_{II}(\mathfrak{U},\mathfrak{V};\mathcal{F}).$$

The definitions of d' and d'' immediately give:

PROPOSITION 2. *$H_I^{p,q}(\mathfrak{U},\mathfrak{V};\mathcal{F})$ is isomorphic to*

$$\prod_{s'} H^p(\mathfrak{U}_{s'},\mathcal{F}),$$

where the product is taken over all simplices of dimension q in $S(J)$. In particular,

$$C_{II}^q(\mathfrak{U},\mathfrak{V};\mathcal{F}) = H_I^{0,q}(\mathfrak{U},\mathfrak{V};\mathcal{F})$$

is isomorphic to

$$\prod_{s'} H^0(\mathfrak{U}_{s'},\mathcal{F}) = C^q(\mathfrak{V},\mathcal{F}).$$

We shall denote by ι'' the canonical isomorphism

$$\iota'' : C(\mathfrak{V},\mathcal{F}) \longrightarrow C_{II}(\mathfrak{U},\mathfrak{V};\mathcal{F}).$$

Thus, if $(f_{j_0\ldots j_q})$ is an element of $C^q(\mathfrak{V},\mathcal{F})$, then

$$(\iota'' f)_{i_0,j_0\ldots j_q} = \rho_{i_0}(f_{j_0\ldots j_q}),$$

where ρ_{i_0} denotes the restriction homomorphism defined by the inclusion of

$$U_{i_0}\cap V_{j_0\ldots j_q} \quad \text{in} \quad V_{j_0\ldots j_q}.$$

Of course, the analogue of Proposition 2 holds for $H_{II}^{p,q}(\mathfrak{U},\mathfrak{V};\mathcal{F})$, and there is an isomorphism

$$\iota' : C(\mathfrak{U},\mathcal{F}) \longrightarrow C_I(\mathfrak{U},\mathfrak{V};\mathcal{F}).$$

29. Applications. With the notation of the preceding number, we have:

PROPOSITION 3. *Suppose that*

$$H^p(\mathfrak{U}_{s'},\mathcal{F}) = 0$$

for every s' and every $p>0$. Then the homomorphism

$$H^n(\mathfrak{V},\mathcal{F}) \longrightarrow H^n(\mathfrak{U},\mathfrak{V};\mathcal{F})$$

defined by ι'' is bijective for every $n \geq 0$.

This is an immediate consequence of Propositions 1 and 2.

Before stating Proposition 4, let us prove a lemma.

LEMMA 1. *Let $\mathfrak{W} = \{W_i\}_{i\in I}$ be a covering of a space Y, and let $\mathcal{F}$ be a sheaf on Y. If there is an $i\in I$ such that $W_i = Y$, then*

$$H^p(\mathfrak{W},\mathcal{F}) = 0$$

for every $p>0$.

Let $\mathfrak{W}'$ be the covering of Y consisting of the single open set Y. Clearly $\mathfrak{W} < \mathfrak{W}'$, while the hypothesis on $\mathfrak{W}$ means that $\mathfrak{W}' < \mathfrak{W}$. It follows from no. 22 that

$$H^p(\mathfrak{W},\mathcal{F}) = H^p(\mathfrak{W}',\mathcal{F}) = 0 \qquad (p>0).$$

PROPOSITION 4. *Suppose that the covering $\mathfrak{V}$ is finer than the covering $\mathfrak{U}$. Then*

$$\iota'' : H^n(\mathfrak{V},\mathcal{F}) \longrightarrow H^n(\mathfrak{U},\mathfrak{V};\mathcal{F})$$

is bijective for every $n \geq 0$. Moreover, the homomorphism[15]

$$\iota''^{-1} \circ \iota' : H^n(\mathfrak{U}, \mathcal{F}) \longrightarrow H^n(\mathfrak{V}, \mathcal{F})$$

coincides with the homomorphism $\sigma(\mathfrak{V}, \mathfrak{U})$ of no. 21.

Applying Lemma 1 to $\mathfrak{W} = \mathfrak{U}_{s'}$ and $Y = V_{s'}$, we see that

$$H^p(\mathfrak{U}_{s'}, \mathcal{F}) = 0 \qquad (p > 0),$$

and Proposition 3 then shows that

$$\iota'' : H^n(\mathfrak{V}, \mathcal{F}) \longrightarrow H^n(\mathfrak{U}, \mathfrak{V}; \mathcal{F})$$

is bijective for every $n \geq 0$.

Let $\tau : J \to I$ be a map such that $V_j \subset U_{\tau j}$. To prove the second part of the proposition, we must show that, if f is an n-cocycle of $C(\mathfrak{U}, \mathcal{F})$, the cocycles $\iota'(f)$ and $\iota''(\tau f)$ are cohomologous in $C(\mathfrak{U}, \mathfrak{V}; \mathcal{F})$.

For every integer p with $0 \leq p \leq n-1$, define

$$g^p \in C^{p, n-p-1}(\mathfrak{U}, \mathfrak{V}; \mathcal{F})$$

by

$$g^p_{i_0 \ldots i_p, j_0 \ldots j_{n-p-1}} = \rho_p\big(f_{i_0 \ldots i_p\, \tau j_0 \ldots \tau j_{n-p-1}}\big).$$

[16] Here ρ_p denotes the restriction homomorphism defined by the inclusion of

$$U_{i_0 \ldots i_p} \cap V_{j_0 \ldots j_{n-p-1}} \quad \text{in} \quad U_{i_0 \ldots i_p\, \tau j_0 \ldots \tau j_{n-p-1}}.$$

A direct calculation, taking into account that f is a cocycle, gives

$$d''(g^0) = \iota''(\tau f), \quad \ldots, \quad d''(g^p) = d'(g^{p-1}), \quad \ldots, \quad d'(g^{n-1}) = (-1)^n \iota'(f).$$

It follows that

$$d\big(g^0 - g^1 + \cdots + (-1)^{n-1} g^{n-1}\big) = \iota''(\tau f) - \iota'(f),$$

which proves that $\iota''(\tau f)$ and $\iota'(f)$ are cohomologous.

Proposition 5. *Suppose that $\mathfrak{V}$ is finer than $\mathfrak{U}$ and that*

$$H^q(\mathfrak{V}_s, \mathcal{F}) = 0$$

for every s and every $q > 0$. Then the homomorphism

$$\sigma(\mathfrak{V}, \mathfrak{U}) : H^n(\mathfrak{U}, \mathcal{F}) \longrightarrow H^n(\mathfrak{V}, \mathcal{F})$$

is bijective for every $n \geq 0$.

Applying Proposition 3 with the roles of $\mathfrak{U}$ and $\mathfrak{V}$ interchanged, we see that[17]

$$\iota' : H^n(\mathfrak{U}, \mathcal{F}) \longrightarrow H^n(\mathfrak{U}, \mathfrak{V}; \mathcal{F})$$

is bijective. The proposition now follows directly from Proposition 4.

Theorem 1. *Let X be a topological space, let $\mathfrak{U} = \{U_i\}_{i \in I}$ be a covering of X, and let $\mathcal{F}$ be a sheaf on X. Suppose that there is a family of coverings $\mathfrak{V}^\alpha$, $\alpha \in A$, of X satisfying the following two conditions:*

[15] The printed source writes $\iota' \circ \iota''^{-1}$, which is not type-correct for the displayed map under the article's usual right-to-left composition convention. The map from $H^n(\mathfrak{U}, \mathcal{F})$ to $H^n(\mathfrak{V}, \mathcal{F})$ is $\iota''^{-1} \circ \iota'$.

[16] The printed formula ends with the undefined extra index τj_{n-p}. The left side and the following inclusion both stop at j_{n-p-1}; that terminal index also gives the required $n+1$ arguments of the n-cochain f.

[17] The printed source gives the domain of ι' as $H^n(\mathfrak{V}, \mathcal{F})$. But ι' was defined on $C(\mathfrak{U}, \mathcal{F})$, and the swapped application of Proposition 3 therefore has domain $H^n(\mathfrak{U}, \mathcal{F})$.

(a) *For every covering* $\mathfrak{W}$ *of* X, *there is an* $\alpha \in A$ *such that* $\mathfrak{V}^\alpha < \mathfrak{W}$.

(b) $H^q(\mathfrak{V}^\alpha_s, \mathcal{F}) = 0$ *for every* $\alpha \in A$, *every simplex* s *of* $S(I)$, *and every* $q > 0$.

Then

$$\sigma(\mathfrak{U}) : H^n(\mathfrak{U}, \mathcal{F}) \longrightarrow H^n(X, \mathcal{F})$$

is bijective for every $n \geq 0$.

Since the $\mathfrak{V}^\alpha$ are arbitrarily fine, we may suppose that they are finer than $\mathfrak{U}$. In that case, Proposition 5 shows that

$$\sigma(\mathfrak{V}^\alpha, \mathfrak{U}) : H^n(\mathfrak{U}, \mathcal{F}) \longrightarrow H^n(\mathfrak{V}^\alpha, \mathcal{F})$$

is bijective for every $n \geq 0$. Since the $\mathfrak{V}^\alpha$ are arbitrarily fine, $H^n(X, \mathcal{F})$ is the direct limit of the groups $H^n(\mathfrak{V}^\alpha, \mathcal{F})$, and the theorem follows immediately.

Remarks.

(1) It is probable that Theorem 1 remains valid if condition (b) is replaced by the following weaker condition:

$$\lim_{\alpha} H^q(\mathfrak{V}^\alpha_s, \mathcal{F}) = 0$$

for every simplex s of $S(I)$ and every $q > 0$.

(2) Theorem 1 is analogous to a theorem of Leray on acyclic coverings. Cf. [10], as well as [4], Exposé XVII-7.

Chapter II. Algebraic Varieties—Coherent Algebraic Sheaves on Affine Varieties

Throughout the remainder of this article, K denotes a commutative algebraically closed field of arbitrary characteristic.

§1. Algebraic varieties

30. Spaces satisfying condition (A). Let X be a topological space. Condition (A) is the following:

(A)—*Every decreasing sequence of closed subsets of X is stationary.*

In other words, if

$$F_1 \supset F_2 \supset F_3 \supset \cdots,$$

where the F_i are closed in X, there is an integer n such that $F_m = F_n$ for $m \geq n$. Equivalently:

(A′)—*The set of closed subsets of X, ordered by inclusion, satisfies the minimum condition.*

Examples. Equip a set X with the topology whose closed subsets are the finite subsets of X and X itself; condition (A) is then satisfied. More generally, every algebraic variety equipped with the Zariski topology satisfies (A) (cf. no. 34).

Proposition 1.

(a) *If X satisfies condition (A), then X is quasi-compact.*

(b) *If X satisfies condition (A), then every subspace of X also satisfies it.*

(c) *If X is the union of a finite family of subspaces Y_i satisfying condition (A), then X also satisfies condition (A).*

Let (F_i) be a downward directed family of closed subsets of X. If X satisfies (A′), one of the F_i is contained in all the others; hence, if $\bigcap_i F_i = \varnothing$, there is an i such that $F_i = \varnothing$. This proves (a).

Let

$$G_1 \supset G_2 \supset G_3 \supset \cdots$$

be a decreasing sequence of closed subsets of a subspace Y of X. If X satisfies (A), there is an n such that $\overline{G}_m = \overline{G}_n$ for $m \geq n$, whence

$$G_m = Y \cap \overline{G}_m = Y \cap \overline{G}_n = G_n.$$

This proves (b).

Let

$$F_1 \supset F_2 \supset F_3 \supset \cdots$$

be a decreasing sequence of closed subsets of a space X satisfying the hypotheses of (c). Since the Y_i satisfy (A), for each i there is an n_i such that

$$F_m \cap Y_i = F_{n_i} \cap Y_i \qquad (m \geq n_i).$$

If $n = \sup_i n_i$, then $F_m = F_n$ for $m \geq n$. This proves (c).

A space X is said to be *irreducible* if it is not the union of two closed subsets distinct from X itself; equivalently, any two nonempty open subsets of X have nonempty intersection. It follows that every finite family of nonempty open subsets of X has nonempty intersection, and every open subset of X is likewise irreducible.

Proposition 2. *Every space X satisfying condition (A) is the union of finitely many irreducible closed subspaces Y_i. If no Y_i is contained in Y_j for any pair (i, j) with $i \neq j$, the set of the Y_i is uniquely determined by X; the Y_i are then called the irreducible components of X.*

The existence of a decomposition $X = \bigcup_i Y_i$ follows immediately from (A). If $X = \bigcup_k Z_k$ is another such decomposition, then

$$Y_i = \bigcup_k (Y_i \cap Z_k).$$

Since Y_i is irreducible, there is an index k such that $Z_k \supset Y_i$. Interchanging the roles of Y_i and Z_k, we similarly obtain an index i' such that $Y_{i'} \supset Z_k$. Thus

$$Y_i \subset Z_k \subset Y_{i'}.$$

The hypothesis on the Y_i gives $i = i'$ and $Y_i = Z_k$, and the uniqueness of the decomposition follows at once.

Proposition 3. *Let X be a topological space which is the union of a finite family of nonempty open subsets V_i. Then X is irreducible if and only if the V_i are irreducible and $V_i \cap V_j \neq \varnothing$ for every pair (i, j).*

The necessity of these conditions was noted above; let us prove their sufficiency. Suppose that $X = Y \cup Z$, where Y and Z are closed. Then

$$V_i = (V_i \cap Y) \cup (V_i \cap Z),$$

so every V_i is contained either in Y or in Z. Suppose that both Y and Z are distinct from X. We can then find indices i, j such that V_i is not contained in Y and V_j is not contained in Z. By what precedes, $V_i \subset Z$ and $V_j \subset Y$. Put

$$T = V_j - (V_i \cap V_j).$$

The set T is closed in V_j, and

$$V_j = T \cup (Z \cap V_j).$$

Since V_j is irreducible, either $T = V_j$, which means $V_i \cap V_j = \varnothing$, or $Z \cap V_j = V_j$, which means $V_j \subset Z$. Both alternatives are contradictions, and the proof is complete.

31. Locally closed subsets of affine space. Let r be an integer ≥ 0, and let $X = K^r$ be the *affine space* of dimension r over K. We equip X with the *Zariski topology*; recall that a subset of X is closed in this topology if it is the common zero set of a family of polynomials

$$P^\alpha \in K[X_1, \ldots, X_r].$$

Since the polynomial ring is Noetherian, X satisfies condition (A) of the preceding number; moreover, it is easy to show that X is irreducible.

If $x = (x_1, \ldots, x_r)$ is a point of X, we denote by $\mathcal{O}_x$ the *local ring* at x. Recall that it is the subring of the field $K(X_1, \ldots, X_r)$ consisting of the rational functions R which can be written in the form

$$R = P/Q,$$

where P and Q are polynomials and $Q(x) \neq 0$.

Such a rational function is said to be *regular at x*. At every point $x \in X$ at which $Q(x) \neq 0$, the function $x \mapsto P(x)/Q(x)$ is a continuous K-valued function (where K is given the Zariski topology), and it may be identified with R because K is infinite. The rings $\mathcal{O}_x$, $x \in X$, therefore form a subsheaf $\mathcal{O}$ of the sheaf $\mathcal{F}(X)$ of germs of K-valued functions on X (cf. no. 3); the sheaf $\mathcal{O}$ is a sheaf of rings.

We shall extend this construction to locally closed subspaces of X. (A subset of a space X is said to be *locally closed* in X if it is the intersection of an open subset and a closed subset of X.) Let Y be such a subspace, and let $\mathcal{F}(Y)$ be the sheaf of germs of K-valued functions on Y. If x is a point of Y, restriction of a function defines a canonical homomorphism

$$\varepsilon_x : \mathcal{F}(X)_x \longrightarrow \mathcal{F}(Y)_x.$$

The image of $\mathcal{O}_x$ under ε_x is a subring of $\mathcal{F}(Y)_x$, which we denote by $\mathcal{O}_{x,Y}$. The $\mathcal{O}_{x,Y}$ form a subsheaf $\mathcal{O}_Y$ of $\mathcal{F}(Y)$, called the *sheaf of local rings* of Y.

Thus, by definition, a section of $\mathcal{O}_Y$ over an open subset V of Y is a map $f : V \to K$ which, in a neighborhood of each point $x \in V$, is equal to the restriction to V of a rational function regular at x. Such a function f is said to be *regular* on V. It is continuous when V has the topology induced from X and K has the Zariski topology. The set of functions regular at every point of V is a ring, namely $\Gamma(V, \mathcal{O}_Y)$. Notice also that, if $f \in \Gamma(V, \mathcal{O}_Y)$ and $f(x) \neq 0$ for every $x \in V$, then $1/f$ also belongs to $\Gamma(V, \mathcal{O}_Y)$.

The sheaf $\mathcal{O}_Y$ may also be characterized as follows.

PROPOSITION 4. *Let U (respectively F) be an open (respectively closed) subspace of X, and let $Y = U \cap F$. Let $I(F)$ be the ideal of $K[X_1, \ldots, X_r]$ consisting of the polynomials vanishing on F. If x is a point of Y, the kernel of the surjection*

$$\varepsilon_x : \mathcal{O}_x \longrightarrow \mathcal{O}_{x,Y}$$

is the ideal $I(F)\mathcal{O}_x$ of $\mathcal{O}_x$.

It is clear that every element of $I(F)\mathcal{O}_x$ belongs to the kernel of ε_x. Conversely, let $R = P/Q$ be an element of this kernel, where P and Q are polynomials and $Q(x) \neq 0$. By hypothesis, there is an open neighborhood W of x such that $P(y) = 0$ for every $y \in W \cap F$. Let F' be the complement of W, which is closed in X. Since $x \notin F'$, the definition of the Zariski topology gives a polynomial P_1 which vanishes on F' and does not vanish at x. Then $P\,P_1$ belongs to $I(F)$, and we may write[18]

$$R = \frac{P\,P_1}{Q\,P_1},$$

which proves that $R \in I(F)\mathcal{O}_x$.

COROLLARY. *The ring $\mathcal{O}_{x,Y}$ is isomorphic to the localization of $K[X_1, \ldots, X_r]/I(F)$ at the maximal ideal defined by the point x.*

This follows immediately from the construction of the localization of a quotient ring (cf., for example, [8], Chap. XV, §5, Th. XI).

32. Regular maps. Let U (respectively V) be a locally closed subspace of K^r (respectively K^s). A map $\varphi : U \to V$ is said to be *regular* on U (or simply regular) if:

(a) *φ is continuous;*

(b) *if $x \in U$ and $f \in \mathcal{O}_{\varphi(x),V}$, then $f \circ \varphi \in \mathcal{O}_{x,U}$.*

Denote the coordinates of the point $\varphi(x)$ by $\varphi_i(x)$, $1 \leq i \leq s$. We then have:

PROPOSITION 5. *The map $\varphi : U \to V$ is regular on U if and only if the functions $\varphi_i : U \to K$ are regular on U for every i, $1 \leq i \leq s$.*

Since the coordinate functions are regular on V, the condition is necessary. Conversely, suppose that

$$\varphi_i \in \Gamma(U, \mathcal{O}_U) \qquad (1 \leq i \leq s).$$

If $P(X_1, \ldots, X_s)$ is a polynomial, then $P(\varphi_1, \ldots, \varphi_s)$ belongs to $\Gamma(U, \mathcal{O}_U)$, since $\Gamma(U, \mathcal{O}_U)$ is a ring. It follows that this is a continuous function on U, so its zero locus is closed; this proves the continuity of φ.

If $x \in U$ and $f \in \mathcal{O}_{\varphi(x),V}$, then locally we may write $f = P/Q$, where P and Q are polynomials and $Q(\varphi(x)) \neq 0$. In a neighborhood of x we then have

$$f \circ \varphi = \frac{P \circ \varphi}{Q \circ \varphi}.$$

By what we have just seen, $P \circ \varphi$ and $Q \circ \varphi$ are regular near x. Since $(Q \circ \varphi)(x) \neq 0$, it follows that $f \circ \varphi$ is regular near x, as required.

[18] The printed source writes $R = P \cdot P_1/Q \cdot P_1$ without grouping. The argument multiplies both numerator and denominator of P/Q by P_1, so corrected French and English make the equal fraction explicit.

The *composite* of two regular maps is regular. A bijection $\varphi : U \to V$ is called a *biregular isomorphism* (or simply an isomorphism) if both φ and φ^{-1} are regular maps. Equivalently, φ is a homeomorphism from U onto V which carries the sheaf $\mathcal{O}_U$ to the sheaf $\mathcal{O}_V$.

33. Products. If r and r' are two integers ≥ 0, we identify the affine space $K^{r+r'}$ with the product $K^r \times K^{r'}$. The Zariski topology of $K^{r+r'}$ is *finer* than the product of the Zariski topologies of K^r and $K^{r'}$; it is in fact strictly finer if r and r' are both positive. It follows that, if U and U' are locally closed subspaces of K^r and $K^{r'}$, then $U \times U'$ is a locally closed subspace of $K^{r+r'}$, and the sheaf $\mathcal{O}_{U\times U'}$ is well defined.

Let W be a locally closed subspace of K^t, where $t \geq 0$, and let $\varphi : W \to U$ and $\varphi' : W \to U'$ be two maps. Proposition 5 immediately gives:

Proposition 6. *The map $x \mapsto (\varphi(x), \varphi'(x))$ from W to $U \times U'$ is regular if and only if φ and φ' are regular.*

Since every *constant* map is regular, the preceding proposition shows that every *section* $x \mapsto (x, x'_0)$, with $x'_0 \in U'$, is a regular map from U to $U \times U'$. Moreover, the *projections* $U \times U' \to U$ and $U \times U' \to U'$ are plainly regular.

Let V and V' be locally closed subspaces of K^s and $K^{s'}$, and let $\psi : U \to V$ and $\psi' : U' \to V'$ be two maps. The preceding remarks, together with Proposition 6, give the following (cf. [1], Chapter IV):

Proposition 7. *The map $\psi \times \psi' : U \times U' \to V \times V'$ is regular if and only if ψ and ψ' are regular.*

Hence:

Corollary. *The map $\psi \times \psi'$ is a biregular isomorphism if and only if ψ and ψ' are biregular isomorphisms.*

34. Definition of the structure of an algebraic variety.

Definition. *An algebraic variety over K (or simply an algebraic variety) is a set X equipped with:*

1° *a topology,*

2° *a subsheaf $\mathcal{O}_X$ of the sheaf $\mathcal{F}(X)$ of germs of K-valued functions on X,*

these data being required to satisfy the axioms (VA_I) *and* (VA_{II}) *stated below.*

First observe that, if X and Y are equipped with two structures of the preceding type, there is a notion of an *isomorphism* from X onto Y: it is a homeomorphism from X onto Y which carries $\mathcal{O}_X$ to $\mathcal{O}_Y$. Moreover, if X' is an open subset of X, we may equip X' with the induced topology and the induced sheaf; this gives the notion of an *induced structure* on an open subset. With this understood, we can state axiom (VA_I):

(VA_I)—*There exists a finite open covering $\mathfrak{V} = \{V_i\}_{i\in I}$ of X such that every V_i, equipped with the structure induced by that of X, is isomorphic to a locally closed subspace U_i of an affine space, equipped with the sheaf $\mathcal{O}_{U_i}$ defined in no. 31.*

For brevity, we call a topological space X equipped with a sheaf $\mathcal{O}_X$ satisfying axiom (VA_I) a *prealgebraic variety.* An isomorphism $\varphi_i : V_i \to U_i$ is called a *chart* of the open set V_i; thus condition (VA_I) says that X can be covered by finitely many open sets admitting charts. Proposition 1 of no. 30 shows that X then satisfies condition (A), and hence is quasi-compact, as are all its subspaces.

The topology of X is called the "Zariski topology" of X, and the sheaf $\mathcal{O}_X$ is called the *sheaf of local rings* of X.

Proposition 8. *Let X be a set which is the union of a finite family of subsets X_j, $j \in J$. Suppose that each X_j is equipped with the structure of a prealgebraic variety and that the following conditions hold:*

(a) *$X_i \cap X_j$ is open in X_i for all $i, j \in J$;*

(b) *the structures induced by X_i and X_j on $X_i \cap X_j$ agree for all $i, j \in J$.*

There then exists a unique structure of a prealgebraic variety on X such that the X_j are open in X and the structure induced on each X_j is the given structure.

The existence and uniqueness of the topology on X and of the sheaf $\mathcal{O}_X$ are immediate. It remains to verify that this topology and this sheaf satisfy (VA$_I$), which follows because there are only finitely many X_j and each satisfies (VA$_I$).

COROLLARY. *Let X and X' be two prealgebraic varieties. There exists a unique structure of a prealgebraic variety on $X \times X'$ satisfying the following condition: if $\varphi : V \to U$ and $\varphi' : V' \to U'$ are charts (with V open in X and V' open in X'), then $V \times V'$ is open in $X \times X'$ and $\varphi \times \varphi' : V \times V' \to U \times U'$ is a chart.*

Cover X by finitely many open sets V_i with charts $\varphi_i : V_i \to U_i$, and let (V'_j, U'_j, φ'_j) be an analogous system for X'. The set $X \times X'$ is the union of the $V_i \times V'_j$. Equip each $V_i \times V'_j$ with the structure of a prealgebraic variety transported from that of $U_i \times U'_j$ by $\varphi_i^{-1} \times \varphi_j'^{-1}$. By the corollary to Proposition 7, conditions (a) and (b) of Proposition 8 apply to this covering of $X \times X'$. We thereby obtain a structure of a prealgebraic variety on $X \times X'$ satisfying the required conditions.

We may apply the preceding corollary in the special case $X' = X$. Thus $X \times X$ is equipped with the structure of a prealgebraic variety, and in particular with a *topology*. We can now state axiom (VA$_{II}$):

(VA$_{II}$)—*The diagonal Δ of $X \times X$ is closed in $X \times X$.*

Suppose that X is a prealgebraic variety obtained by the "gluing" procedure of Proposition 8. Condition (VA$_{II}$) holds if and only if

$$X_{ij} = \Delta \cap (X_i \times X_j)$$

is closed in $X_i \times X_j$. Now X_{ij} is the set of pairs (x, x) with $x \in X_i \cap X_j$. Suppose there are charts $\varphi_i : X_i \to U_i$, and put

$$T_{ij} = (\varphi_i \times \varphi_j)(X_{ij}).$$

Then T_{ij} is the set of pairs $(\varphi_i(x), \varphi_j(x))$ as x ranges over $X_i \cap X_j$. Axiom (VA$_{II}$) therefore takes the following form:

(VA$'_{II}$)—*For every pair (i, j), T_{ij} is closed in $U_i \times U_j$.*

In this form we recognize Weil's axiom (A) (cf. [16], p. 167), except that Weil considers only irreducible varieties.

EXAMPLES of algebraic varieties: every locally closed subspace U of an affine space, equipped with the induced topology and the sheaf $\mathcal{O}_U$ defined in no. 31, is an algebraic variety. Every projective variety is an algebraic variety (cf. no. 51). Every algebraic fiber space (cf. [17]) whose base and fiber are algebraic varieties is an algebraic variety.

REMARKS. (1) Note the analogy between condition (VA$_{II}$) and the separation condition imposed on topological, differentiable, and analytic varieties.

(2) Simple examples show that condition (VA$_{II}$) *does not* follow from condition (VA$_I$).

35. Regular maps, induced structures, products.

Let X and Y be two algebraic varieties and let φ be a map from X to Y. The map φ is said to be *regular* if:

(a) *φ is continuous;*

(b) *if $x \in X$ and $f \in \mathcal{O}_{\varphi(x),Y}$, then $f \circ \varphi \in \mathcal{O}_{x,X}$.*

As in no. 32, the composite of two regular maps is regular; and a bijection $\varphi : X \to Y$ is an isomorphism if and only if φ and φ^{-1} are regular maps. Thus regular maps form a family of *morphisms* for the structure of algebraic varieties, in the sense of [1], Chapter IV.

Let X be an algebraic variety and let X' be a locally closed subset of X. Equip X' with the topology induced by that of X and with the sheaf $\mathcal{O}_{X'}$ induced by $\mathcal{O}_X$. More precisely, for every $x \in X'$, define $\mathcal{O}_{x,X'}$ to be the image of $\mathcal{O}_{x,X}$ under the canonical homomorphism

$$\mathcal{F}(X)_x \longrightarrow \mathcal{F}(X')_x.$$

Axiom (VA_I) holds: if $\varphi_i : V_i \to U_i$ is a system of charts such that $X = \bigcup V_i$, put

$$V_i' = X' \cap V_i, \qquad U_i' = \varphi_i(V_i');$$

then the restrictions $\varphi_i : V_i' \to U_i'$ form a system of charts such that $X' = \bigcup V_i'$. Axiom (VA_{II}) holds because the topology of $X' \times X'$ is induced by that of $X \times X$ (one could equally use (VA'_{II})). We thereby define an algebraic-variety structure on X' which is said to be *induced* by that of X; we also say that X' is a *subvariety* of X (Weil [16] reserves the term "subvariety" for what we call here a closed irreducible subvariety). If ι denotes the inclusion of X' into X, then ι is regular. Moreover, if φ is a map from an algebraic variety Y to X', then $\varphi : Y \to X'$ is regular if and only if $\iota \circ \varphi : Y \to X$ is regular. This justifies the term "induced structure" (cf. [1], loc. cit.).

If X and X' are two algebraic varieties, then $X \times X'$ is an algebraic variety, called the *product variety*. Indeed, it is enough to prove axiom (VA'_{II}). In other words, if $\varphi_i : V_i \to U_i$ and $\varphi'_{i'} : V'_{i'} \to U'_{i'}$ are systems of charts such that $X = \bigcup V_i$ and $X' = \bigcup V'_{i'}$, then

$$T_{ij} \times T'_{i',j'}$$

is closed in $U_i \times U_j \times U'_{i'} \times U'_{j'}$ (with the notation of no. 34). This follows immediately from the fact that T_{ij} and $T'_{i',j'}$ are closed in $U_i \times U_j$ and $U'_{i'} \times U'_{j'}$, respectively.

Propositions 6 and 7 hold without change for arbitrary algebraic varieties.

If $\varphi : X \to Y$ is a regular map, its graph Φ is *closed* in $X \times Y$, since it is the inverse image of the diagonal of $Y \times Y$ under the map

$$\varphi \times 1 : X \times Y \longrightarrow Y \times Y.$$

Moreover, the map $\psi : X \to \Phi$ defined by $\psi(x) = (x, \varphi(x))$ is an isomorphism: indeed, ψ is regular, as is ψ^{-1}, since the latter is the restriction of the projection $X \times Y \to X$.

36. Field of rational functions on an irreducible variety.

We shall first prove two lemmas of a purely topological nature.

Lemma 1. *Let X be a connected space, let G be an abelian group, and let $\mathcal{G}$ be the constant sheaf on X isomorphic to G. The canonical map*

$$G \longrightarrow \Gamma(X, \mathcal{G})$$

is bijective.

An element of $\Gamma(X, \mathcal{G})$ is nothing other than a continuous map from X to G endowed with the discrete topology. Since X is connected, such a map is constant, which proves the lemma.

We shall say that a sheaf $\mathcal{F}$ on a space X is *locally constant* if every point x has a neighborhood U such that the restriction of $\mathcal{F}$ to U is constant.

Lemma 2. *Every locally constant sheaf on an irreducible space is constant.*

Let $\mathcal{F}$ be the sheaf and X the space, and put

$$F = \Gamma(X, \mathcal{F}).$$

It is enough to show that the canonical homomorphism

$$\rho_x : F \longrightarrow \mathcal{F}_x$$

is bijective for every $x \in X$, for we shall then obtain an isomorphism from the constant sheaf isomorphic to F onto the given sheaf $\mathcal{F}$.

If $f \in F$, the set of points $x \in X$ such that $f(x) = 0$ is open, by the general properties of sheaves, and closed, because $\mathcal{F}$ is locally constant. Since an irreducible space is connected, this set is therefore either $\emptyset$ or X, which already proves that ρ_x is injective.

Now let $m \in \mathcal{F}_x$, and let s be a section of $\mathcal{F}$ over a neighborhood U of x such that $s(x) = m$. Cover X by nonempty open sets U_i on each of which $\mathcal{F}$ is constant. Since X is irreducible, $U \cap U_i \neq \emptyset$; choose a point $x_i \in U \cap U_i$. There is a section s_i of $\mathcal{F}$ over U_i such that $s_i(x_i) = s(x_i)$. Since s and s_i agree at x_i, they agree throughout $U \cap U_i$, because $U \cap U_i$ is irreducible and hence connected. Likewise, s_i and s_j agree on $U_i \cap U_j$, since they agree on the nonempty set $U \cap U_i \cap U_j$. Thus the sections s_i define a unique section s of $\mathcal{F}$ over X, and $\rho_x(s) = m$. This completes the proof.

Now let X be an irreducible algebraic variety. If U is a nonempty open subset of X, put

$$\mathcal{A}_U = \Gamma(U, \mathcal{O}_X).$$

The ring $\mathcal{A}_U$ is an *integral domain.* Indeed, suppose that $f\,g = 0$, where f and g are regular maps from U to K. If F (respectively G) is the set of points $x \in U$ such that $f(x) = 0$ (respectively $g(x) = 0$), then $U = F \cup G$, and F and G are closed in U because f and g are continuous. Since U is irreducible, it follows that $F = U$ or $G = U$; hence f or g is zero on U.

We may therefore form the field of fractions of $\mathcal{A}_U$, which we denote by $\mathcal{K}_U$. If $U \subset V$, the homomorphism

$$\rho_U^V : \mathcal{A}_V \longrightarrow \mathcal{A}_U$$

is injective because U is dense in V, and it induces a well-defined embedding

$$\varphi_U^V : \mathcal{K}_V \longrightarrow \mathcal{K}_U.$$

The system $\{\mathcal{K}_U, \varphi_U^V\}$ defines a *sheaf of fields* $\mathcal{K}$. Moreover, $\mathcal{K}_x$ is canonically isomorphic to the field of fractions of $\mathcal{O}_{x,X}$.

Proposition 9. *For every irreducible algebraic variety X, the sheaf $\mathcal{K}$ defined above is a constant sheaf.*

By Lemma 2, it is enough to prove the proposition when X is a locally closed subvariety of affine space K^r. Let F be the closure of X in K^r, and let $I(F)$ be the ideal of $K[X_1, \ldots, X_r]$ consisting of the polynomials which vanish on F (or on X, which is equivalent). Put

$$A = K[X_1, \ldots, X_r]/I(F).$$

The ring A is an integral domain because X is irreducible; let $K(A)$ be its field of fractions. By the corollary to Proposition 4, $\mathcal{O}_{x,X}$ may be identified with the ring of fractions of A with respect to the maximal ideal defined by x. This gives an isomorphism from the field $K(A)$ onto the field of fractions of $\mathcal{O}_{x,X}$, and it is easy to verify that these isomorphisms define an isomorphism from the constant sheaf equal to $K(A)$ onto the sheaf $\mathcal{K}$. This proves the proposition.

By Lemma 1, the sections of the sheaf $\mathcal{K}$ form a field, isomorphic to $\mathcal{K}_x$ for every $x \in X$, which we denote by $K(X)$. It is called the *field of rational functions* on X. It is a finitely generated extension of the field K, and its transcendence degree over K is the *dimension* of X. We extend this definition to reducible algebraic varieties by putting

$$\dim X = \sup_i \dim Y_i$$

when X is the union of the closed irreducible subvarieties Y_i.

In general we identify the field $K(X)$ with the fields $\mathcal{K}_x$. Since $\mathcal{O}_{x,X} \subset \mathcal{K}_x$, this identifies $\mathcal{O}_{x,X}$ with a *subring* of $K(X)$; it is the specialization ring of the point x in $K(X)$, in the sense of Weil [16], p. 77. If U is open in X, then $\Gamma(U, \mathcal{O}_X)$ is the intersection, inside $K(X)$, of the rings $\mathcal{O}_{x,X}$ as x ranges over U.

If Y is a subvariety of X, then $\dim Y \leq \dim X$. If, moreover, Y is closed and contains no irreducible component of X, then $\dim Y < \dim X$, as one sees by reducing to the case of subvarieties of K^r (cf., for example, [8], Chap. X, §5, Th. II).

§2. Coherent algebraic sheaves

37. The sheaf of local rings of an algebraic variety. Return to the notation of no. 31: let $X = K^r$, and let $\mathcal{O}$ be the sheaf of local rings of X.

LEMMA 1. *The sheaf $\mathcal{O}$ is a coherent sheaf of rings, in the sense of no. 15.*

Let $x \in X$, let U be a neighborhood of x, and let $f_1, \ldots, f_p$ be sections of $\mathcal{O}$ over U, that is, rational functions regular at every point of U. We must show that the sheaf of relations among $f_1, \ldots, f_p$ is a sheaf of finite type over $\mathcal{O}$.

After replacing U by a smaller neighborhood, we may suppose that

$$f_i = P_i/Q,$$

where the P_i and Q are polynomials and Q does not vanish on U. Now let $y \in U$, and let $g_i \in \mathcal{O}_y$ be such that

$$\sum_{i=1}^{i=p} g_i f_i = 0$$

in a neighborhood of y. We may write $g_i = R_i/S$, where the R_i and S are polynomials and S does not vanish at y. The relation "$\sum_{i=1}^{i=p} g_i f_i = 0$ in a neighborhood of y" is equivalent to the relation "$\sum_{i=1}^{i=p} R_i P_i = 0$ in a neighborhood of y," which in turn is equivalent to

$$\sum_{i=1}^{i=p} R_i P_i = 0.$$

Since the module of relations among the polynomials P_i is of finite type, because the polynomial ring is Noetherian, it follows that the sheaf of relations among the f_i is of finite type.

Now let V be a closed subvariety of $X = K^r$. For every $x \in X$, let $\mathcal{J}_x(V)$ be the ideal of $\mathcal{O}_x$ consisting of the elements $f \in \mathcal{O}_x$ whose restriction to V is zero in a neighborhood of x; thus $\mathcal{J}_x(V) = \mathcal{O}_x$ if $x \notin V$. The $\mathcal{J}_x(V)$ form a subsheaf $\mathcal{J}(V)$ of $\mathcal{O}$.

LEMMA 2. *The sheaf $\mathcal{J}(V)$ is a coherent sheaf of $\mathcal{O}$-modules.*

Let $I(V)$ be the ideal of $K[X_1, \ldots, X_r]$ consisting of the polynomials P which vanish on V. By Proposition 4 of no. 31,

$$\mathcal{J}_x(V) = I(V)\mathcal{O}_x$$

for every $x \in V$, and this formula also holds for $x \notin V$, as is immediately seen. Since the ideal $I(V)$ is generated by finitely many elements, the sheaf $\mathcal{J}(V)$ is of finite type and hence coherent by Lemma 1 and Proposition 8 of no. 15.

We shall now extend Lemma 1 to an arbitrary algebraic variety.

PROPOSITION 1. *If V is an algebraic variety, then the sheaf $\mathcal{O}_V$ is a coherent sheaf of rings on V.*

The question is local, so we may suppose that V is a closed subvariety of affine space K^r. By Lemma 2, $\mathcal{J}(V)$ is a coherent sheaf of ideals. Hence $\mathcal{O}/\mathcal{J}(V)$ is a coherent sheaf of rings on X, by Theorem 3 of no. 16. This sheaf of rings is zero outside V, and its restriction to V is precisely $\mathcal{O}_V$ (no. 31). It follows that $\mathcal{O}_V$ is a coherent sheaf of rings on V (no. 17, corollary to Proposition 11).

REMARK. It is clear that Proposition 1 holds, more generally, for every prealgebraic variety.

38. Coherent algebraic sheaves. Let V be an algebraic variety whose sheaf of local rings is $\mathcal{O}_V$. By an *algebraic sheaf* on V we shall mean any sheaf of $\mathcal{O}_V$-modules, in the sense of no. 6. If $\mathcal{F}$ and $\mathcal{G}$ are two algebraic sheaves, we shall say that $\varphi : \mathcal{F} \to \mathcal{G}$ is an *algebraic homomorphism*, or simply a homomorphism, if it is an $\mathcal{O}_V$-homomorphism. Recall that this is equivalent to saying that each

$$\varphi_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x$$

is $\mathcal{O}_{x,V}$-linear and that φ carries every local section of $\mathcal{F}$ to a local section of $\mathcal{G}$.

If $\mathcal{F}$ is an algebraic sheaf on V, the cohomology groups $H^q(V, \mathcal{F})$ are modules over $\Gamma(V, \mathcal{O}_V)$ (cf. no. 23); in particular, they are *vector spaces over* K.

An algebraic sheaf $\mathcal{F}$ on V is said to be *coherent* if it is a coherent sheaf of $\mathcal{O}_V$-modules, in the sense of no. 12. By Proposition 7 of no. 15 and Proposition 1 above, such a sheaf is characterized by being locally isomorphic to the cokernel of an algebraic homomorphism

$$\varphi : \mathcal{O}_V^q \longrightarrow \mathcal{O}_V^p.$$

We shall give some examples of coherent algebraic sheaves; others will appear later, notably in nos. 48 and 57.

39. Ideal sheaf defined by a closed subvariety. Let W be a closed subvariety of an algebraic variety V. For every $x \in V$, let $\mathcal{J}_x(W)$ be the ideal of $\mathcal{O}_{x,V}$ consisting of the elements f whose restriction to W is zero in a neighborhood of x, and let $\mathcal{J}(W)$ be the subsheaf of $\mathcal{O}_V$ formed by the $\mathcal{J}_x(W)$. We have the following proposition, which generalizes Lemma 2.

Proposition 2. *The sheaf $\mathcal{J}(W)$ is a coherent algebraic sheaf.*

The question is local, so we may suppose that V, and hence also W, is a closed subvariety of affine space K^r. It then follows from Lemma 2, applied to W, that the ideal sheaf defined by W in K^r is of finite type. Consequently $\mathcal{J}(W)$, which is its image under the canonical homomorphism $\mathcal{O} \to \mathcal{O}_V$, is also of finite type, and is therefore coherent by Proposition 8 of no. 15 and Proposition 1 of no. 37.

Let $\mathcal{O}_W$ be the sheaf of local rings of W, and let $\mathcal{O}_W^V$ be the sheaf on V obtained by extending $\mathcal{O}_W$ by zero outside W (cf. no. 5). This sheaf is canonically isomorphic to $\mathcal{O}_V/\mathcal{J}(W)$; in other words, there is an exact sequence

$$0 \longrightarrow \mathcal{J}(W) \longrightarrow \mathcal{O}_V \longrightarrow \mathcal{O}_W^V \longrightarrow 0.$$

Now let $\mathcal{F}$ be an algebraic sheaf on W, and let $\mathcal{F}^V$ be the sheaf obtained by extending $\mathcal{F}$ by zero outside W. We may regard $\mathcal{F}^V$ as a sheaf of $\mathcal{O}_W^V$-modules, and hence also as a sheaf of $\mathcal{O}_V$-modules whose annihilator contains $\mathcal{J}(W)$. We have:

Proposition 3. *If $\mathcal{F}$ is a coherent algebraic sheaf on W, then $\mathcal{F}^V$ is a coherent algebraic sheaf on V. Conversely, if $\mathcal{G}$ is a coherent algebraic sheaf on V whose annihilator contains $\mathcal{J}(W)$, then the restriction of $\mathcal{G}$ to W is a coherent algebraic sheaf on W.*

If $\mathcal{F}$ is a coherent algebraic sheaf on W, then $\mathcal{F}^V$ is a coherent sheaf of $\mathcal{O}_W^V$-modules (no. 17, Proposition 11), and hence a coherent sheaf of $\mathcal{O}_V$-modules (no. 16, Theorem 3). Conversely, if $\mathcal{G}$ is a coherent algebraic sheaf on V whose annihilator contains $\mathcal{J}(W)$, then $\mathcal{G}$ may be regarded as a sheaf of $\mathcal{O}_V/\mathcal{J}(W)$-modules, and it is coherent (no. 16, Theorem 3); the restriction of $\mathcal{G}$ to W is therefore a coherent sheaf of $\mathcal{O}_W$-modules (no. 17, Proposition 11).

Thus every coherent algebraic sheaf on W may be identified with a coherent algebraic sheaf on V; by Proposition 8 of no. 26, this identification does not change the cohomology groups. In particular, every coherent algebraic sheaf on an affine variety, respectively a projective variety, may be regarded as a coherent algebraic sheaf on affine space, respectively projective space. We shall frequently use this possibility below.

Remark. Let $\mathcal{G}$ be a coherent algebraic sheaf on V that is zero outside W. The annihilator of $\mathcal{G}$ *need not contain* $\mathcal{J}(W)$; in other words, $\mathcal{G}$ cannot always be regarded as a coherent algebraic sheaf on W. All that can be asserted is that its annihilator contains a *power* of $\mathcal{J}(W)$.

40. Sheaves of fractional ideals. Let V be an irreducible algebraic variety, and let $K(V)$ be the constant sheaf of rational functions on V (cf. no. 36). The sheaf $K(V)$ is algebraic, but it is not coherent if $\dim V > 0$. An algebraic subsheaf $\mathcal{F}$ of $K(V)$ may be called a "sheaf of fractional ideals," since each $\mathcal{F}_x$ is a fractional ideal of $\mathcal{O}_{x,V}$.

Proposition 4. *An algebraic subsheaf $\mathcal{F}$ of $K(V)$ is coherent if and only if it is of finite type.*

The necessity is trivial. To prove the sufficiency, it is enough to show that $K(V)$ satisfies condition (b) of Definition 2 in no. 12; in other words, if $f_1, \dots, f_p$ are rational functions, then the sheaf $\mathcal{R}(f_1, \dots, f_p)$ is of finite type. If x is a point of V, we can find functions g_i and h such that $f_i = g_i/h$, with the g_i and h regular on a neighborhood U of x and h nowhere zero on U. The sheaf $\mathcal{R}(f_1, \dots, f_p)$ is then equal to $\mathcal{R}(g_1, \dots, g_p)$, which is of finite type because $\mathcal{O}_V$ is a coherent sheaf of rings.

41. Sheaf associated with a fiber space with vector-space fiber. Let E be an algebraic fiber space with vector-space fiber of dimension r and base an algebraic variety V. By definition, the typical fiber of E is the vector space K^r, and its structure group is the linear group $\mathbf{GL}(r, K)$ acting on K^r in the usual way. (For the definition of an algebraic fiber space, cf. [17]; see also [15], no. 4, for *analytic* fiber spaces with vector-space fibers.)

If U is an open subset of V, let $\mathcal{S}(E)_U$ be the set of sections of E that are regular on U. If $V \supset U$, there is a restriction homomorphism $\varphi_U^V : \mathcal{S}(E)_V \to \mathcal{S}(E)_U$. This gives a sheaf $\mathcal{S}(E)$, called the *sheaf of germs of sections* of E. Since E is a fiber space with vector-space fiber, every $\mathcal{S}(E)_U$ is a $\Gamma(U, \mathcal{O}_V)$-module, and it follows that $\mathcal{S}(E)$ is an algebraic sheaf on V. Locally identifying E with $V \times K^r$, we obtain:

Proposition 5. *The sheaf $\mathcal{S}(E)$ is locally isomorphic to $\mathcal{O}_V^r$; in particular, it is a coherent algebraic sheaf.*

Conversely, it is easy to see that every algebraic sheaf $\mathcal{F}$ on V which is locally isomorphic to $\mathcal{O}_V^r$ is isomorphic to a sheaf $\mathcal{S}(E)$, where E is determined up to isomorphism (cf. [15] for the analytic case).

If V is nonsingular, we may take for E the fiber space of tangent p-covectors on V, where p is an integer ≥ 0. Let Ω^p be the corresponding sheaf $\mathcal{S}(E)$. An element of Ω_x^p, for $x \in V$, is nothing other than a differential form of degree p on V which is regular at x. Put

$$h^{p,q} = \dim_K H^q(V, \Omega^p).$$

In the classical case, when V is projective, $h^{p,q}$ is known to equal the dimension of the space of harmonic forms of type (p, q) (Dolbeault's theorem[3]). If B_n denotes the n-th Betti number of V, then

$$B_n = \sum_{p+q=n} h^{p,q}.$$

In the general case, one could take the preceding formula as the *definition* of the Betti numbers of a nonsingular projective variety; indeed, we shall see in no. 66 that the $h^{p,q}$ are finite. Their properties would then have to be studied, notably whether they coincide with those occurring in the Weil conjectures concerning varieties over finite fields.[4] We merely point out here that they satisfy "Poincaré duality,"

$$B_n = B_{2m-n},$$

when V is irreducible and of dimension m.

The cohomology groups $H^q(V, \mathcal{S}(E))$ also occur in other questions, notably in the Riemann–Roch theorem and in the classification of algebraic fiber spaces over V whose structure group is the affine group $x \mapsto ax + b$ (cf. [17], §4, where the case $\dim V = 1$ is treated).

§3. Coherent algebraic sheaves on affine varieties

42. Affine varieties. An algebraic variety V is said to be *affine* if it is isomorphic to a closed subvariety of an affine space. The product of two affine varieties is an affine variety, and every closed subvariety of an affine variety is an affine variety.

An open subset U of an algebraic variety X is said to be *affine* if, when endowed with the algebraic-variety structure induced by that of X, it is an affine variety.

Proposition 1. *Let U and V be two open subsets of an algebraic variety X. If U and V are affine, then $U \cap V$ is affine.*

[3] P. Dolbeault, *Sur la cohomologie des variétés analytiques complexes*, C. R. Paris, 236 (1953), 175–177.
[4] *Bull. Amer. Math. Soc.*, 55 (1949), 507.

Let Δ be the diagonal of $X \times X$. By no. 35, the map $x \mapsto (x,x)$ is a biregular isomorphism from X onto Δ; its restriction to $U \cap V$ is therefore a biregular isomorphism from $U \cap V$ onto $\Delta \cap (U \times V)$. Since U and V are affine varieties, $U \times V$ is also an affine variety. Moreover, Δ is closed in $X \times X$ by axiom (VA_{II}), so $\Delta \cap (U \times V)$ is closed in $U \times V$ and is indeed an affine variety, as required.

(It is easy to see that this proposition fails for prealgebraic varieties: axiom (VA_{II}) plays an essential role.)

We now introduce notation that will be used throughout the rest of this section. If V is an algebraic variety and f is a regular function on V, we denote by V_f the open subset of V consisting of the points $x \in V$ such that $f(x) \neq 0$.

Proposition 2. *If V is an affine algebraic variety and f is a regular function on V, then the open subset V_f is affine.*

Let W be the subset of $V \times K$ consisting of the pairs (x, λ) such that $\lambda f(x) = 1$. Plainly W is closed in $V \times K$, and hence is an affine variety. For every $(x, \lambda) \in W$, set $\pi(x, \lambda) = x$; the map π is a regular map from W to V_f. Conversely, for every $x \in V_f$, set $\omega(x) = (x, 1/f(x))$. The map $\omega : V_f \to W$ is regular, and $\pi \circ \omega = 1$ and $\omega \circ \pi = 1$. Thus V_f and W are isomorphic.

Proposition 3. *Let V be a closed subvariety of K^r, let F be a closed subset of V, and let $U = V - F$. As P ranges over the polynomials vanishing on F, the open subsets V_P form a basis for the topology of U.*

Let $U' = V - F'$ be an open subset of U, and let $x \in U'$. We must show that there is a polynomial P such that $V_P \subset U'$ and $x \in V_P$. In other words, P must vanish on F' and be nonzero at x. The existence of such a polynomial follows immediately from the definition of the topology of K^r.

Theorem 1. *The affine open subsets of an algebraic variety X form a basis of open subsets for the topology of X.*

Since the question is local, we may suppose that X is a locally closed subvariety of an affine space K^r; the theorem then follows immediately from Propositions 2 and 3.

Corollary. *Coverings of X by affine open subsets are arbitrarily fine.*

Note that if $\mathfrak{U} = \{U_i\}_{i \in I}$ is such a covering, then all the $U_{i_0 \ldots i_p}$ are affine open subsets, by Proposition 1.

43. Some preliminary properties of irreducible varieties. Let V be a closed subvariety of K^r, and let $I(V)$ be the ideal of $K[X_1, \ldots, X_r]$ consisting of the polynomials vanishing on V. Let

$$A = K[X_1, \ldots, X_r]/I(V).$$

There is a canonical homomorphism

$$\iota : A \longrightarrow \Gamma(V, \mathcal{O}_V),$$

which is injective by the very definition of $I(V)$.

Proposition 4. *If V is irreducible, then $\iota : A \to \Gamma(V, \mathcal{O}_V)$ is bijective.*

(In fact, this holds for *every* closed subvariety of K^r, as we shall show in the next number.)

Let $K(V)$ be the field of fractions of A. By no. 36, $\mathcal{O}_{x,V}$ may be identified with the localization of A at the maximal ideal $\mathfrak{m}_x$ consisting of the polynomials vanishing at x, and

$$\Gamma(V, \mathcal{O}_V) = A = \bigcap_{x \in V} \mathcal{O}_{x,V},$$

where all the $\mathcal{O}_{x,V}$ are regarded as subrings of $K(V)$. But every maximal ideal of A is one of the $\mathfrak{m}_x$, because K is algebraically closed (Hilbert's Nullstellensatz). It follows immediately (cf. [8], Chap. XV, §5, Theorem X) that

$$A = \bigcap_{x \in V} \mathcal{O}_{x,V} = \Gamma(V, \mathcal{O}_V),$$

as required.

PROPOSITION 5. *Let X be an irreducible algebraic variety, let Q be a regular function on X, and let P be a regular function on X_Q. Then, for every sufficiently large n, the rational function $Q^n P$ is regular on all of X.*

By the quasi-compactness of X, the question is local; by Theorem 1, we may therefore suppose that X is a closed subvariety of K^r. The preceding proposition then shows that Q is an element of

$$A = K[X_1, \dots, X_r]/I(X).$$

The hypothesis on P means that, for every point $x \in X_Q$, we can write $P = P_x/Q_x$, with $P_x, Q_x \in A$ and $Q_x(x) \neq 0$. Let $\mathfrak{a}$ be the ideal of A generated by the Q_x. The zero locus of $\mathfrak{a}$ is contained in the zero locus of Q; by Hilbert's Nullstellensatz, it follows that $Q^n \in \mathfrak{a}$ for all sufficiently large n. Hence

$$Q^n = \sum R_x Q_x, \qquad Q^n P = \sum R_x P_x, \qquad R_x \in A,$$

which shows that $Q^n P$ is regular on X.

(One could instead use the fact that X_Q is affine when X is affine and apply Proposition 4 to X_Q.)

PROPOSITION 6. *Let X be an irreducible algebraic variety, let Q be a regular function on X, let $\mathcal{F}$ be a coherent algebraic sheaf on X, and let s be a section of $\mathcal{F}$ over X whose restriction to X_Q is zero. Then, for every sufficiently large n, the section $Q^n s$ is zero on all of X.*

Again the question is local, so we may suppose:

(a) X is a closed subvariety of K^r;

(b) $\mathcal{F}$ is isomorphic to the cokernel of a homomorphism $\varphi : \mathcal{O}_X^p \to \mathcal{O}_X^q$;

(c) s is the image of a section σ of $\mathcal{O}_X^q$.

(Indeed, all these conditions hold locally.)

Set

$$A = \Gamma(X, \mathcal{O}_X) = K[X_1, \dots, X_r]/I(X).$$

The section σ may be identified with a system of q elements of A. On the other hand, let

$$t_1 = \varphi(1, 0, \dots, 0), \dots, t_p = \varphi(0, \dots, 0, 1).$$

The t_i, for $1 \leq i \leq p$, are sections of $\mathcal{O}_X^q$ over X, and hence may also be identified with systems of q elements of A. The hypothesis on s means that, for every $x \in X_Q$, we have $\sigma(x) \in \varphi(\mathcal{O}_{x,X}^p)$; that is, σ can be written in the form

$$\sigma = \sum_{i=1}^{i=p} f_i t_i, \qquad f_i \in \mathcal{O}_{x,X}.$$

Equivalently, after clearing denominators, there is a $Q_x \in A$, with $Q_x(x) \neq 0$, such that

$$Q_x \sigma = \sum_{i=1}^{i=p} R_i t_i, \qquad R_i \in A.$$

The preceding argument then shows that, for every sufficiently large n, Q^n belongs to the ideal generated by the Q_x. Consequently $Q^n \sigma(x) \in \varphi(\mathcal{O}_{x,X}^p)$ for every $x \in X$, which says precisely that $Q^n s$ is zero on all of X.

44. Vanishing of certain cohomology groups.

Proposition 7. *Let X be an irreducible affine variety, let Q_i be a finite family of regular functions on X having no common zero, and let $\mathfrak{U}$ be the open covering of X formed by the sets $X_{Q_i} = U_i$. If $\mathcal{F}$ is a coherent algebraic subsheaf of $\mathcal{O}_X^p$, then*

$$H^q(\mathfrak{U}, \mathcal{F}) = 0 \qquad \textit{for every } q > 0.$$

After replacing $\mathfrak{U}$ by an equivalent covering if necessary, we may suppose that none of the functions Q_i is identically zero; in other words, $U_i \neq \varnothing$ for every i.

Let $f = (f_{i_0 \ldots i_q})$ be a q-cocycle of $\mathfrak{U}$ with values in $\mathcal{F}$. Each $f_{i_0 \ldots i_q}$ is a section of $\mathcal{F}$ over $U_{i_0 \ldots i_q}$ and may therefore be identified with a system of p regular functions on $U_{i_0 \ldots i_q}$. Applying Proposition 5 to

$$Q = Q_{i_0} \cdots Q_{i_q},$$

we see that, for every sufficiently large n,

$$g_{i_0 \ldots i_q} = (Q_{i_0} \cdots Q_{i_q})^n f_{i_0 \ldots i_q}$$

is a system of p regular functions on all of X, that is, a section of $\mathcal{O}_X^p$ over X. Choose an integer n for which this holds for every tuple $i_0, \ldots, i_q$; this is possible because there are only finitely many such tuples. Consider the image of $g_{i_0 \ldots i_q}$ in the coherent sheaf $\mathcal{O}_X^p / \mathcal{F}$. It is a section which vanishes on $U_{i_0 \ldots i_q}$. Proposition 6 therefore shows that, for every sufficiently large m, the product of this section by $(Q_{i_0} \cdots Q_{i_q})^m$ is zero on all of X. Thus

$$(Q_{i_0} \cdots Q_{i_q})^m g_{i_0 \ldots i_q}$$

is a section of $\mathcal{F}$ over all of X. Setting $N = m + n$, we have consequently constructed sections $h_{i_0 \ldots i_q}$ of $\mathcal{F}$ over X which coincide with

$$(Q_{i_0} \cdots Q_{i_q})^N f_{i_0 \ldots i_q}$$

on $U_{i_0 \ldots i_q}$.

Since the Q_i^N have no common zero, there are functions

$$R_i \in \Gamma(X, \mathcal{O}_X)$$

such that $\sum R_i Q_i^N = 1$. For every tuple $i_0, \ldots, i_{q-1}$, set

$$k_{i_0 \ldots i_{q-1}} = \sum_i \frac{R_i h_{i i_0 \ldots i_{q-1}}}{(Q_{i_0} \cdots Q_{i_{q-1}})^N}.$$

This is defined because $Q_{i_0} \cdots Q_{i_{q-1}}$ is nonzero on $U_{i_0 \ldots i_{q-1}}$.

We have thereby defined a cochain $k \in C^{q-1}(\mathfrak{U}, \mathcal{F})$. I claim that $f = dk$, which will prove the proposition.

We must verify that $(dk)_{i_0 \ldots i_q} = f_{i_0 \ldots i_q}$. It is enough to show that these two sections agree on

$$U = \bigcap U_i,$$

for they will then agree everywhere: they are systems of p rational functions on X, and $U \neq \varnothing$. Over U we may write

$$k_{i_0 \ldots i_{q-1}} = \sum_i R_i Q_i^N f_{i i_0 \ldots i_{q-1}},$$

and hence[19]

$$(dk)_{i_0 \ldots i_q} = \sum_{j=0}^{j=q} (-1)^j \sum_i R_i Q_i^N f_{i i_0 \ldots \widehat{i_j} \ldots i_q}.$$

[19]The printed source has $(-1)^q$ in this summand. The coboundary formula alternates with the omitted index j, and the following use of the cocycle identity requires $(-1)^j$; corrected French and English therefore restore that exponent.

Since f is a cocycle, this gives

$$(dk)_{i_0\ldots i_q} = \sum_i R_i Q_i^N f_{i_0\ldots i_q} = f_{i_0\ldots i_q},$$

as required.

COROLLARY 1. *$H^q(X, \mathcal{F}) = 0$ for $q > 0$.*

Indeed, Proposition 3 shows that coverings of the type used in Proposition 7 are arbitrarily fine.

COROLLARY 2. *The homomorphism*

$$\Gamma(X, \mathcal{O}_X^p) \longrightarrow \Gamma(X, \mathcal{O}_X^p/\mathcal{F})$$

is surjective.

This follows from Corollary 1 above and Corollary 2 to Proposition 6 of no. 24.

COROLLARY 3. *Let V be a closed subvariety of K^r, and let*

$$A = K[X_1, \ldots, X_r]/I(V).$$

The homomorphism

$$\iota : A \longrightarrow \Gamma(V, \mathcal{O}_V)$$

is bijective.

Apply Corollary 2 above with $X = K^r$, $p = 1$, and $\mathcal{F} = \mathcal{J}(V)$, the ideal sheaf defined by V. It follows that every element of $\Gamma(V, \mathcal{O}_V)$ is the restriction of a section of $\mathcal{O}$ over X, that is, of a polynomial, by Proposition 4 applied to X.

45. Sections of a coherent algebraic sheaf on an affine variety.

THEOREM 2. *Let $\mathcal{F}$ be a coherent algebraic sheaf on an affine variety X. For every $x \in X$, the $\mathcal{O}_{x,X}$-module $\mathcal{F}_x$ is generated by the elements of $\Gamma(X, \mathcal{F})$.*

Since X is affine, we may embed it as a closed subvariety of an affine space K^r. Extending the sheaf $\mathcal{F}$ by zero outside X, we obtain a coherent algebraic sheaf on K^r (cf. no. 39), and are reduced to proving the theorem for this new sheaf. In other words, we may suppose that $X = K^r$.

By the definition of coherent sheaves, there is a covering of X by open sets over which $\mathcal{F}$ is isomorphic to a quotient of a sheaf $\mathcal{O}^p$. Using Proposition 3, we see that there are finitely many polynomials Q_i having no common zero and, over each $U_i = X_{Q_i}$, a surjective homomorphism

$$\varphi_i : \mathcal{O}^{p_i} \longrightarrow \mathcal{F}.$$

We may moreover suppose that none of these polynomials is identically zero.

The point x belongs to one of the U_i, say U_0. It is clear that $\mathcal{F}_x$ is generated by the sections of $\mathcal{F}$ over U_0. Since Q_0 is invertible in $\mathcal{O}_x$, it is therefore enough to prove the following lemma.

LEMMA 1. *If s_0 is a section of $\mathcal{F}$ over U_0, there are an integer N and a section s of $\mathcal{F}$ over X such that*

$$s = Q_0^N s_0 \qquad \textit{over } U_0.$$

By Proposition 2, $U_i \cap U_0$ is an affine variety, and is clearly irreducible. Applying Corollary 2 of Proposition 7 to this variety and to

$$\varphi_i : \mathcal{O}^{p_i} \longrightarrow \mathcal{F},$$

we find a section σ_{0i} of $\mathcal{O}^{p_i}$ over $U_i \cap U_0$ such that

$$\varphi_i(\sigma_{0i}) = s_0 \qquad \text{over } U_i \cap U_0.$$

Now $U_i \cap U_0$ is the locus of the points of U_i at which Q_0 does not vanish. We may therefore apply Proposition 5 with $X = U_i$ and $Q = Q_0$. Thus, for every sufficiently large n, there is a section σ_i of $\mathcal{O}^{p_i}$ over U_i which agrees with $Q_0^n \sigma_{0i}$ over $U_i \cap U_0$. Put

$$s'_i = \varphi_i(\sigma_i).$$

Then s'_i is a section of $\mathcal{F}$ over U_i which agrees with $Q_0^n s_0$ over $U_i \cap U_0$.

The sections s'_i and s'_j agree on $U_i \cap U_j \cap U_0$. Applying Proposition 6 to $s'_i - s'_j$, we see that, for every sufficiently large m,

$$Q_0^m(s'_i - s'_j) = 0 \qquad \text{on all of } U_i \cap U_j.$$

The sections $Q_0^m s'_i$ consequently define a unique section s of $\mathcal{F}$ over X, and

$$s = Q_0^{n+m} s_0 \qquad \text{over } U_0.$$

This proves the lemma and completes the proof of Theorem 2.

Corollary 1. *The sheaf $\mathcal{F}$ is isomorphic to a quotient sheaf of some $\mathcal{O}_X^p$.*

Since $\mathcal{F}_x$ is a finitely generated $\mathcal{O}_{x,X}$-module, the theorem shows that there are finitely many sections of $\mathcal{F}$ which generate $\mathcal{F}_x$. By Proposition 1 of no. 12, these sections also generate $\mathcal{F}_y$ when y is sufficiently near x. Since X is quasi-compact, we conclude that there are finitely many sections

$$s_1, \ldots, s_p$$

of $\mathcal{F}$ which generate $\mathcal{F}_x$ for every $x \in X$. This says precisely that $\mathcal{F}$ is isomorphic to a quotient sheaf of $\mathcal{O}_X^p$.

Corollary 2. *Let*

$$\mathcal{A} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\beta} \mathcal{C}$$

be an exact sequence of coherent algebraic sheaves on an affine variety X. Then the sequence

$$\Gamma(X, \mathcal{A}) \xrightarrow{\alpha} \Gamma(X, \mathcal{B}) \xrightarrow{\beta} \Gamma(X, \mathcal{C})$$

is exact.

As in the proof of Theorem 2, we may suppose that X is the affine space K^r and hence is irreducible. Set

$$\mathcal{J} = \operatorname{Im}(\alpha) = \operatorname{Ker}(\beta).$$

It remains to show that

$$\alpha : \Gamma(X, \mathcal{A}) \longrightarrow \Gamma(X, \mathcal{J})$$

is surjective. By Corollary 1 there is a surjective homomorphism

$$\varphi : \mathcal{O}_X^p \longrightarrow \mathcal{A},$$

and Corollary 2 to Proposition 7 shows that

$$\alpha \circ \varphi : \Gamma(X, \mathcal{O}_X^p) \longrightarrow \Gamma(X, \mathcal{J})$$

is surjective. It follows a fortiori that

$$\alpha : \Gamma(X, \mathcal{A}) \longrightarrow \Gamma(X, \mathcal{J})$$

is surjective, as required.

46. Cohomology groups of an affine variety with values in a coherent algebraic sheaf. We shall generalize Proposition 7.

THEOREM 3. *Let X be an affine variety, let Q_i be a finite family of regular functions on X having no common zero, and let $\mathfrak{U}$ be the open covering of X formed by the sets $X_{Q_i} = U_i$. If $\mathcal{F}$ is a coherent algebraic sheaf on X, then*

$$H^q(\mathfrak{U}, \mathcal{F}) = 0 \qquad \textit{for every } q > 0.$$

First suppose that X is irreducible. By Corollary 1 to Theorem 2, there is an exact sequence

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{O}_X^p \longrightarrow \mathcal{F} \longrightarrow 0.$$

The sequence of complexes

$$0 \longrightarrow C(\mathfrak{U}, \mathcal{R}) \longrightarrow C(\mathfrak{U}, \mathcal{O}_X^p) \longrightarrow C(\mathfrak{U}, \mathcal{F}) \longrightarrow 0$$

is exact. Indeed, this amounts to saying that every section of $\mathcal{F}$ over a set $U_{i_0 \dots i_q}$ is the image of a section of $\mathcal{O}_X^p$ over $U_{i_0 \dots i_q}$, which follows from Corollary 2 to Proposition 7 applied to the irreducible variety $U_{i_0 \dots i_q}$. This exact sequence of complexes gives rise to a long exact cohomology sequence

$$\cdots \longrightarrow H^q(\mathfrak{U}, \mathcal{O}_X^p) \longrightarrow H^q(\mathfrak{U}, \mathcal{F}) \longrightarrow H^{q+1}(\mathfrak{U}, \mathcal{R}) \longrightarrow \cdots .$$

For $q > 0$, Proposition 7 gives

$$H^q(\mathfrak{U}, \mathcal{O}_X^p) = H^{q+1}(\mathfrak{U}, \mathcal{R}) = 0,$$

and hence $H^q(\mathfrak{U}, \mathcal{F}) = 0$.

We now pass to the general case. We may embed X as a closed subvariety of an affine space K^r. By Corollary 3 to Proposition 7, the functions Q_i are induced by polynomials P_i. Let, on the other hand, the R_j be a finite system of generators of the ideal $I(X)$. The functions P_i and R_j have no common zero on K^r and hence define an open covering $\mathfrak{U}'$ of K^r. Let $\mathcal{F}'$ be the sheaf obtained by extending $\mathcal{F}$ by zero outside X. Applying what we have just proved to K^r, to the functions P_i, R_j, and to the sheaf $\mathcal{F}'$, we find that

$$H^q(\mathfrak{U}', \mathcal{F}') = 0 \qquad (q > 0).$$

The complex $C(\mathfrak{U}', \mathcal{F}')$ is immediately seen to be isomorphic to $C(\mathfrak{U}, \mathcal{F})$. It follows that

$$H^q(\mathfrak{U}, \mathcal{F}) = 0,$$

as required.

COROLLARY 1. *If X is an affine variety and $\mathcal{F}$ is a coherent algebraic sheaf on X, then*

$$H^q(X, \mathcal{F}) = 0 \qquad \textit{for every } q > 0.$$

Indeed, coverings of the type used in the preceding theorem are arbitrarily fine.

COROLLARY 2. *Let*

$$0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{B} \longrightarrow \mathcal{C} \longrightarrow 0$$

be an exact sequence of sheaves on an affine variety X. If $\mathcal{A}$ is coherent algebraic, then the homomorphism

$$\Gamma(X, \mathcal{B}) \longrightarrow \Gamma(X, \mathcal{C})$$

is surjective.

This follows from Corollary 1 with $q = 1$.

47. Coverings of algebraic varieties by affine open subsets.

PROPOSITION 8. *Let X be an affine variety, and let $\mathfrak{U} = \{U_i\}_{i \in I}$ be a finite covering of X by affine open subsets. If $\mathcal{F}$ is a coherent algebraic sheaf on X, then $H^q(\mathfrak{U}, \mathcal{F}) = 0$ for every $q > 0$.*

By Proposition 3, there are regular functions P_j on X such that the covering $\mathfrak{B} = \{X_{P_j}\}$ is finer than $\mathfrak{U}$. For every $(i_0, \ldots, i_p)$, the covering $\mathfrak{B}_{i_0 \ldots i_p}$ induced by $\mathfrak{B}$ on $U_{i_0 \ldots i_p}$ is defined by the restrictions of the P_j to $U_{i_0 \ldots i_p}$. Since $U_{i_0 \ldots i_p}$ is an affine variety by Proposition 1, Theorem 3 applies to it, and therefore $H^q(\mathfrak{B}_{i_0 \ldots i_p}, \mathcal{F}) = 0$ for every $q > 0$. Applying Proposition 5 of no. 29, we then obtain

$$H^q(\mathfrak{U}, \mathcal{F}) = H^q(\mathfrak{B}, \mathcal{F});$$

and since $H^q(\mathfrak{B}, \mathcal{F}) = 0$ for $q > 0$ by Theorem 3, the proposition is proved.

Theorem 4. *Let X be an algebraic variety, let $\mathcal{F}$ be a coherent algebraic sheaf on X, and let $\mathfrak{U} = \{U_i\}_{i \in I}$ be a finite covering of X by affine open subsets. The homomorphism $\sigma(\mathfrak{U}) : H^n(\mathfrak{U}, \mathcal{F}) \to H^n(X, \mathcal{F})$ is bijective for every $n \geq 0$.*

Consider the family $\mathfrak{B}^\alpha$ of finite coverings of X by affine open subsets. By the corollary to Theorem 1, these coverings are arbitrarily fine. Moreover, for every system $(i_0, \ldots, i_p)$, the covering $\mathfrak{B}^\alpha_{i_0 \ldots i_p}$ induced by $\mathfrak{B}^\alpha$ on $U_{i_0 \ldots i_p}$ is a covering by affine open subsets, by Proposition 1. Hence Proposition 8 gives $H^q(\mathfrak{B}^\alpha_{i_0 \ldots i_p}, \mathcal{F}) = 0$ for $q > 0$. Conditions (a) and (b) of Theorem 1 of no. 29 are therefore satisfied, and the theorem follows.

Theorem 5. *Let X be an algebraic variety, and let $\mathfrak{U} = \{U_i\}_{i \in I}$ be a finite covering of X by affine open subsets. Let $0 \to \mathcal{A} \to \mathcal{B} \to \mathcal{C} \to 0$ be an exact sequence of sheaves on X, where $\mathcal{A}$ is coherent algebraic. The canonical homomorphism $H^q_0(\mathfrak{U}, \mathcal{C}) \to H^q(\mathfrak{U}, \mathcal{C})$ (cf. no. 24) is bijective for every $q \geq 0$.*

It is clearly enough to show that $C_0(\mathfrak{U}, \mathcal{C}) = C(\mathfrak{U}, \mathcal{C})$; that is, every section of $\mathcal{C}$ over $U_{i_0 \ldots i_q}$ is the image of a section of $\mathcal{B}$ over $U_{i_0 \ldots i_q}$. This follows from Corollary 2 to Theorem 3.

Corollary 1. *Let X be an algebraic variety, and let $0 \to \mathcal{A} \to \mathcal{B} \to \mathcal{C} \to 0$ be an exact sequence of sheaves on X, where $\mathcal{A}$ is coherent algebraic. The canonical homomorphism $H^q_0(X, \mathcal{C}) \to H^q(X, \mathcal{C})$ is bijective for every $q \geq 0$.*

This is an immediate consequence of Theorems 1 and 5.

Corollary 2. *There is an exact sequence*

$$\cdots \longrightarrow H^q(X, \mathcal{B}) \longrightarrow H^q(X, \mathcal{C}) \longrightarrow H^{q+1}(X, \mathcal{A}) \longrightarrow H^{q+1}(X, \mathcal{B}) \longrightarrow \cdots .$$

ğ4. Correspondence between finitely generated modules and coherent algebraic sheaves

48. Sheaf associated with a module. Let V be an affine variety and let $\mathcal{O}$ be the sheaf of local rings of V. The ring $A = \Gamma(V, \mathcal{O})$ will be called the *coordinate ring* of V; it is a K-algebra whose only nilpotent element is 0. If V is embedded as a closed subvariety of an affine space K^r, we know (cf. no. 44) that A is identified with the quotient of $K[X_1, \ldots, X_r]$ by the ideal of polynomials vanishing on V. It follows that the algebra A is generated by finitely many elements.

Conversely, it is readily verified that if A is a commutative K-algebra without nilpotent elements other than 0, and is generated by finitely many elements, then there is an affine variety V such that A is isomorphic to $\Gamma(V, \mathcal{O})$. Moreover, V is determined up to isomorphism by this property (one may identify V with the set of characters of A, endowed with the usual topology).

Let M be an A-module. The module M defines a constant sheaf on V, which we shall again denote by M; similarly, A defines a constant sheaf, and the sheaf M may be regarded as a sheaf of A-modules. Put

$$\mathcal{A}(M) = \mathcal{O} \otimes_A M,$$

where $\mathcal{O}$ is also regarded as a sheaf of A-modules. It is clear that $\mathcal{A}(M)$ is an algebraic sheaf on V. Moreover, if $\varphi : M \to M'$ is an A-homomorphism, there is a homomorphism

$$\mathcal{A}(\varphi) = 1 \otimes \varphi : \mathcal{A}(M) \longrightarrow \mathcal{A}(M').$$

In other words, $M \mapsto \mathcal{A}(M)$ is a covariant functor.

Proposition 1. *The functor $M \mapsto \mathcal{A}(M)$ is exact.*

Let $M \to M' \to M''$ be an exact sequence of A-modules. We must show that the sequence $\mathcal{A}(M) \to \mathcal{A}(M') \to \mathcal{A}(M'')$ is exact; in other words, that for every $x \in V$ the sequence

$$\mathcal{O}_x \otimes_A M \longrightarrow \mathcal{O}_x \otimes_A M' \longrightarrow \mathcal{O}_x \otimes_A M''$$

is exact.

Now $\mathcal{O}_x$ is precisely the ring of fractions A_S of A, where S is the set of all $f \in A$ such that $f(x) \neq 0$ (for the definition of a ring of fractions, cf. [8], [12], or [13]). Proposition 1 is therefore a special case of the following result.

Lemma 1. *Let A be a ring, let S be a multiplicatively closed subset of A not containing* 0*, and let A_S be the ring of fractions of A with respect to S. If $M \to M' \to M''$ is an exact sequence of A-modules, then the sequence*

$$A_S \otimes_A M \longrightarrow A_S \otimes_A M' \longrightarrow A_S \otimes_A M''$$

is exact.

Let M_S denote the set of fractions m/s, with $m \in M$ and $s \in S$, where two fractions m/s and m'/s' are identified if there is an $s'' \in S$ such that

$$s''(s' \cdot m - s \cdot m') = 0.$$

It is readily seen that M_S is an A_S-module and that the map

$$a/s \otimes m \longmapsto a \cdot m/s$$

is an isomorphism from $A_S \otimes_A M$ onto M_S. We are therefore reduced to showing that the sequence

$$M_S \longrightarrow M'_S \longrightarrow M''_S$$

is exact, which is immediate.

Proposition 2. $\mathcal{A}(M) = 0$ *implies* $M = 0$.

Let m be an element of M. If $\mathcal{A}(M) = 0$, then $1 \otimes m = 0$ in $\mathcal{O}_x \otimes_A M$ for every $x \in V$. By the preceding discussion, $1 \otimes m = 0$ is equivalent to the existence of an element $s \in A$, with $s(x) \neq 0$, such that $s \cdot m = 0$. The annihilator of m is therefore contained in no maximal ideal of A; hence it is equal to A, and consequently $m = 0$.

Proposition 3. *If M is a finitely generated A-module, $\mathcal{A}(M)$ is a coherent algebraic sheaf on V.*

Since M is finitely generated and A is Noetherian, M is isomorphic to the cokernel of a homomorphism $\varphi : A^q \to A^p$, and $\mathcal{A}(M)$ is isomorphic to the cokernel of

$$\mathcal{A}(\varphi) : \mathcal{A}(A^q) \longrightarrow \mathcal{A}(A^p).$$

Since $\mathcal{A}(A^p) = \mathcal{O}^p$ and $\mathcal{A}(A^q) = \mathcal{O}^q$, it follows that $\mathcal{A}(M)$ is coherent.

49. Module associated with an algebraic sheaf. Let $\mathcal{F}$ be an algebraic sheaf on V, and put $\Gamma(\mathcal{F}) = \Gamma(V, \mathcal{F})$. Since $\mathcal{F}$ is a sheaf of $\mathcal{O}$-modules, $\Gamma(\mathcal{F})$ has a natural structure of an A-module. Every algebraic homomorphism $\varphi : \mathcal{F} \to \mathcal{G}$ defines an A-homomorphism

$$\Gamma(\varphi) : \Gamma(\mathcal{F}) \longrightarrow \Gamma(\mathcal{G}).$$

If

$$\mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H}$$

is an exact sequence of coherent algebraic sheaves, the sequence

$$\Gamma(\mathcal{F}) \longrightarrow \Gamma(\mathcal{G}) \longrightarrow \Gamma(\mathcal{H})$$

is exact (no. 45). Applying this to an exact sequence $\mathcal{O}^p \to \mathcal{F} \to 0$, we see that $\Gamma(\mathcal{F})$ is a finitely generated A-module if $\mathcal{F}$ is coherent.

The functors $M \mapsto \mathcal{A}(M)$ and $\mathcal{F} \mapsto \Gamma(\mathcal{F})$ are "inverse" to one another:

THEOREM 1. (a) *If M is a finitely generated A-module, $\Gamma(\mathcal{A}(M))$ is canonically isomorphic to M.*

(b) *If $\mathcal{F}$ is a coherent algebraic sheaf on V, $\mathcal{A}(\Gamma(\mathcal{F}))$ is canonically isomorphic to $\mathcal{F}$.*

We first prove (a). Every element $m \in M$ defines a section $\alpha(m)$ of $\mathcal{A}(M)$ by the formula

$$\alpha(m)(x) = 1 \otimes m \in \mathcal{O}_x \otimes_A M;$$

hence there is a homomorphism

$$\alpha : M \longrightarrow \Gamma\big(\mathcal{A}(M)\big).$$

When M is a finitely generated free module, α is bijective (it is enough to verify this for $M = A$, in which case it is evident). If M is an arbitrary finitely generated module, there is an exact sequence

$$L^1 \longrightarrow L^0 \longrightarrow M \longrightarrow 0,$$

where L^0 and L^1 are finitely generated free modules. The sequence

$$\mathcal{A}(L^1) \longrightarrow \mathcal{A}(L^0) \longrightarrow \mathcal{A}(M) \longrightarrow 0$$

is exact, and therefore so is

$$\Gamma(\mathcal{A}(L^1)) \longrightarrow \Gamma(\mathcal{A}(L^0)) \longrightarrow \Gamma(\mathcal{A}(M)) \longrightarrow 0.$$

The commutative diagram

$$\begin{tikzcd}
L^1 \arrow[r] \arrow[d, "\alpha"'] & L^0 \arrow[r] \arrow[d, "\alpha"'] & M \arrow[r] \arrow[d, "\alpha"'] & 0 \arrow[d, "\alpha"'] \\
\Gamma(\mathcal{A}(L^1)) \arrow[r] & \Gamma(\mathcal{A}(L^0)) \arrow[r] & \Gamma(\mathcal{A}(M)) \arrow[r] & 0
\end{tikzcd}$$

then shows that $\alpha : M \to \Gamma(\mathcal{A}(M))$ is bijective, which proves (a).

Now let $\mathcal{F}$ be a coherent algebraic sheaf on V. Associating to each $s \in \Gamma(\mathcal{F})$ the element $s(x) \in \mathcal{F}_x$ gives an A-homomorphism

$$\Gamma(\mathcal{F}) \longrightarrow \mathcal{F}_x,$$

which extends to an $\mathcal{O}_x$-homomorphism

$$\beta_x : \mathcal{O}_x \otimes_A \Gamma(\mathcal{F}) \longrightarrow \mathcal{F}_x.$$

It is readily verified that the β_x form a homomorphism of sheaves

$$\beta : \mathcal{A}(\Gamma(\mathcal{F})) \longrightarrow \mathcal{F}.$$

When $\mathcal{F} = \mathcal{O}^p$, the homomorphism β is bijective. By the same argument as above, it follows that β is bijective for every coherent algebraic sheaf $\mathcal{F}$, which proves (b).

REMARKS. (1) One may also deduce (b) from (a); cf. no. 65, proof of Proposition 6.

(2) We shall see in Chapter III how the preceding correspondence must be modified when coherent sheaves on projective space are studied.

50. Projective modules and fiber spaces with vector-space fiber. Recall ([6], Chap. I, Th. 2.2) that an A-module M is called *projective* if it is a direct summand of a free A-module.

PROPOSITION 4. *Let M be a finitely generated A-module. For M to be projective, it is necessary and sufficient that the $\mathcal{O}_x$-module $\mathcal{O}_x \otimes_A M$ be free for every $x \in V$.*

If M is projective, $\mathcal{O}_x \otimes_A M$ is $\mathcal{O}_x$-projective, and hence $\mathcal{O}_x$-free since $\mathcal{O}_x$ is a local ring (cf. [6], Chap. VIII, Th. 6.1′).

Conversely, if all the $\mathcal{O}_x \otimes_A M$ are free, then

$$\dim(M) = \sup_{x \in V} \dim\bigl(\mathcal{O}_x \otimes_A M\bigr) = 0 \qquad \text{(cf. [6], Chap. VII, Exer. 11),}$$

which implies that M is projective ([6], Chap. VI, §2).

Observe that if $\mathcal{F}$ is a coherent algebraic sheaf on V and $\mathcal{F}_x$ is isomorphic to $\mathcal{O}_x^p$, then $\mathcal{F}$ is isomorphic to $\mathcal{O}^p$ over a neighbourhood of x. If this property holds at every point $x \in V$, the sheaf $\mathcal{F}$ is therefore locally isomorphic to a sheaf $\mathcal{O}^p$, with the integer p remaining constant on every connected component of V. Applying this to the sheaf $\mathcal{A}(M)$ gives:

Corollary. *Let $\mathcal{F}$ be a coherent algebraic sheaf on a connected affine variety V. The following three properties are equivalent:*
(i) $\Gamma(\mathcal{F})$ *is a projective A-module.*
(ii) $\mathcal{F}$ *is locally isomorphic to a sheaf $\mathcal{O}^p$.*
(iii) $\mathcal{F}$ *is isomorphic to the sheaf of germs of sections of an algebraic fiber space with vector-space fiber and base V.*

In addition, the map $E \longmapsto \Gamma\bigl(\mathcal{S}(E)\bigr)$ (where E denotes a fiber space with vector-space fiber) establishes a one-to-one correspondence between the classes of fiber spaces and the classes of finitely generated projective A-modules. Under this correspondence, a *trivial* fiber space corresponds to a *free* module, and conversely.

Let us note that when $V = K^r$ (in which case $A = K[X_1, \ldots, X_r]$), it is unknown whether there exist finitely generated projective A-modules that are not free, or, equivalently, whether there exist nontrivial algebraic fiber spaces with vector-space fiber and base K^r.

Chapter III. Coherent Algebraic Sheaves on Projective Varieties

§1. Projective varieties

51. Notation. (The notation introduced below will be used without further reference throughout the remainder of this chapter.)

Let r be an integer ≥ 0, and let $Y = K^{r+1} \setminus \{0\}$. The multiplicative group K^* of the nonzero elements of K acts on Y by the formula

$$\lambda(\mu_0, \ldots, \mu_r) = (\lambda\mu_0, \ldots, \lambda\mu_r).$$

Two points y and y' are said to be equivalent if there exists $\lambda \in K^*$ such that $y' = \lambda y$. The quotient space of Y by this equivalence relation is denoted by $\mathbf{P}_r(K)$, or simply by X; it is the *projective space of dimension r over K*. The canonical projection of Y onto X is denoted by π.

Let $I = \{0, 1, \ldots, r\}$. For every $i \in I$, let t_i denote the ith coordinate function on K^{r+1}, defined by

$$t_i(\mu_0, \ldots, \mu_r) = \mu_i.$$

Let V_i denote the open subset of K^{r+1} consisting of the points at which $t_i \neq 0$, and let U_i be the image of V_i under π. The $\{U_i\}_{i \in I}$ form a covering $\mathfrak{U}$ of X. If $i, j \in I$, the function t_j/t_i is regular on V_i and invariant under K^*; it therefore defines a function on U_i, which we again denote by t_j/t_i. For fixed i, the functions t_j/t_i, $j \neq i$, define a bijection

$$\psi_i : U_i \longrightarrow K^r.$$

We equip K^{r+1} with its algebraic-variety structure and Y with the induced structure. We also equip X with the quotient topology of that on Y: a closed subset of X is therefore the image under π of a closed cone in K^{r+1}. If U is open in X, put

$$A_U = \Gamma\big(\pi^{-1}(U), \mathcal{O}_Y\big);$$

this is the ring of regular functions on $\pi^{-1}(U)$. Let A_U^0 be the subring of A_U consisting of the elements invariant under K^* (that is, the homogeneous functions of degree 0). When $V \supset U$, there is a restriction homomorphism

$$\varphi_U^V : A_V^0 \longrightarrow A_U^0,$$

and the system (A_U^0, φ_U^V) defines a sheaf $\mathcal{O}_X$, which may be regarded as a subsheaf of the sheaf $\mathfrak{F}(X)$ of germs of functions on X. A function f defined in a neighbourhood of x belongs to $\mathcal{O}_{x,X}$ if and only if it coincides locally with a function of the form P/Q, where P and Q are homogeneous polynomials of the same degree in $t_0, \dots, t_r$, with $Q(y) \neq 0$ for $y \in \pi^{-1}(x)$ (which we shall write more briefly as $Q(x) \neq 0$).

Proposition 1. *The projective space $X = \mathbf{P}_r(K)$, equipped with the preceding topology and sheaf, is an algebraic variety.*

The U_i, $i \in I$, are open subsets of X, and it is immediate that the bijections $\psi_i : U_i \to K^r$ defined above are biregular isomorphisms; this shows that axiom (VA_I) is satisfied. To prove that (VA_{II}) is also satisfied, one must see that the subset of $K^r \times K^r$ formed by the pairs $(\psi_i(x), \psi_j(x))$, where $x \in U_i \cap U_j$, is closed; this presents no difficulty.

In what follows, X will always be equipped with the algebraic-variety structure just defined, and the sheaf $\mathcal{O}_X$ will simply be denoted by $\mathcal{O}$. An algebraic variety V is said to be *projective* if it is isomorphic to a closed subvariety of a projective space. The study of coherent algebraic sheaves on projective varieties may be reduced to the study of coherent algebraic sheaves on the $\mathbf{P}_r(K)$; cf. no. 39.

52. Cohomology of subvarieties of projective space. Apply Theorem 4 of no. 47 to the covering $\mathfrak{U} = \{U_i\}_{i \in I}$ defined in the preceding number; this is possible because each U_i is isomorphic to K^r. We obtain:

Proposition 2. *If $\mathcal{F}$ is a coherent algebraic sheaf on $X = \mathbf{P}_r(K)$, the homomorphism*

$$\sigma(\mathfrak{U}) : H^n(\mathfrak{U}, \mathcal{F}) \longrightarrow H^n(X, \mathcal{F})$$

is bijective for every $n \geq 0$.

Since $\mathfrak{U}$ consists of $r+1$ open subsets, we have (cf. no. 20, corollary to Proposition 2):

Corollary.

$$H^n(X, \mathcal{F}) = 0 \qquad \textit{for } n > r.$$

This last result may be generalized as follows:

Proposition 3. *Let V be an algebraic variety isomorphic to a locally closed subvariety of a projective space X. Let $\mathcal{F}$ be a coherent algebraic sheaf on V, and let W be a subvariety of V such that $\mathcal{F}$ vanishes outside W. Then*

$$H^n(V, \mathcal{F}) = 0 \qquad \textit{for } n > \dim W.$$

In particular, on taking $W = V$, we obtain:

Corollary.

$$H^n(V, \mathcal{F}) = 0 \qquad \textit{for } n > \dim V.$$

Identify V with a locally closed subvariety of $X = \mathbf{P}_r(K)$. There is an open subset U of X such that V is closed in U. We may suppose that W is closed in V, which is plainly permissible; then W is closed in U. Put $F = X \setminus U$. Before proving Proposition 3, we establish two lemmas.

Lemma 1. *Let $k = \dim W$. There exist $k+1$ homogeneous polynomials $P_i(t_0, \ldots, t_r)$, of degrees > 0, which vanish on F and have no common zero on W.*

(By an abuse of language, a homogeneous polynomial P is said to vanish at a point x of $\mathbf{P}_r(K)$ if it vanishes on $\pi^{-1}(x)$.)

We argue by induction on k, the case $k = -1$ being trivial. Choose one point on each irreducible component of W, and let P_1 be a homogeneous polynomial, of degree > 0, which vanishes on F and vanishes at none of these points. (The existence of P_1 follows from the fact that F is closed, in view of the definition of the topology of $\mathbf{P}_r(K)$.) Let W' be the subvariety of W formed by the points $x \in W$ such that $P_1(x) = 0$. By the construction of P_1, no irreducible component of W is contained in W', and it follows (cf. no. 36) that $\dim W' < k$. Applying the induction hypothesis to W', we see that there exist k homogeneous polynomials $P_2, \ldots, P_{k+1}$ which vanish on F and have no common zero on W'. It is clear that $P_1, \ldots, P_{k+1}$ have the required properties.

Lemma 2. *Let $P(t_0, \ldots, t_r)$ be a homogeneous polynomial of degree $n > 0$. The set X_P of points $x \in X$ such that $P(x) \neq 0$ is an affine open subset of X.*

Associate with each point $y = (\mu_0, \ldots, \mu_r) \in Y$ the point of a suitable space K^N whose coordinates are all the monomials

$$\mu_0^{m_0} \cdots \mu_r^{m_r}, \qquad m_0 + \cdots + m_r = n.$$

Passing to the quotient gives a map

$$\varphi_n : X \longrightarrow \mathbf{P}_{N-1}(K).$$

It is classical, and moreover easy to verify, that φ_n is a biregular isomorphism of X onto a closed subvariety of $\mathbf{P}_{N-1}(K)$ (the "Veronese variety"). The map φ_n transforms the open subset X_P into the locus of points of $\varphi_n(X)$ which do not lie on a certain hyperplane of $\mathbf{P}_{N-1}(K)$. Since the complement of a hyperplane is isomorphic to an affine space, it follows that X_P is indeed isomorphic to a closed subvariety of an affine space.

We now prove Proposition 3. Extend the sheaf $\mathcal{F}$ by zero on $U \setminus V$. We obtain a coherent algebraic sheaf on U, which we again denote by $\mathcal{F}$, and we know (cf. no. 26) that

$$H^n(U, \mathcal{F}) = H^n(V, \mathcal{F}).$$

Let $P_1, \ldots, P_{k+1}$ be homogeneous polynomials satisfying the conditions of Lemma 1, and let $P_{k+2}, \ldots, P_h$ be homogeneous polynomials of degrees > 0 which vanish on $W \cup F$ and have no common zero at any point of $U \setminus W$. (To obtain such polynomials, it is enough to take a system of homogeneous generators of the ideal defined by $W \cup F$ in $K[t_0, \ldots, t_r]$.) For every i, $1 \leq i \leq h$, let V_i be the set of points $x \in X$ such that $P_i(x) \neq 0$. Then $V_i \subset U$, and the preceding hypotheses show that

$$\mathfrak{B} = \{V_i\}$$

is an open covering of U. Moreover, Lemma 2 shows that the V_i are affine open subsets, whence

$$H^n(\mathfrak{B}, \mathcal{F}) = H^n(U, \mathcal{F}) = H^n(V, \mathcal{F}) \qquad \text{for every } n \geq 0.$$

On the other hand, if $n > k$ and the indices $i_0, \ldots, i_n$ are distinct, one of these indices is $> k+1$, and $V_{i_0 \cdots i_n}$ does not meet W. It follows that the group of alternating cochains

$$C'^n(\mathfrak{B}, \mathcal{F})$$

is zero if $n > k$. Therefore $H^n(\mathfrak{B}, \mathcal{F}) = 0$, by Proposition 2 of no. 20, which proves Proposition 3.

53. Cohomology of irreducible algebraic curves. If V is an irreducible algebraic variety of dimension 1, the closed subsets of V distinct from V are precisely the *finite* subsets. If F is a finite subset of V and x is a point of F, put

$$V_x^F = (V \setminus F) \cup \{x\}.$$

The sets V_x^F, $x \in F$, form a finite open covering $\mathfrak{V}^F$ of V.

LEMMA 3. *The coverings $\mathfrak{V}^F$ of the preceding type are arbitrarily fine.*

Let $\mathfrak{U} = \{U_i\}_{i \in I}$ be an open covering of V. We may suppose it finite, since V is quasi-compact, and we may also suppose that $U_i \neq \varnothing$ for every $i \in I$. Put $F_i = V \setminus U_i$. Each F_i is finite, and so is

$$F = \bigcup_{i \in I} F_i.$$

We show that $\mathfrak{V}^F < \mathfrak{U}$, which will prove the lemma. Let $x \in F$. Since the U_i cover V, there is an $i \in I$ such that $x \notin F_i$. As $F \supset F_i$, we then have $F \setminus \{x\} \supset F_i$; equivalently, $V_x^F \subset U_i$. Thus $\mathfrak{V}^F < \mathfrak{U}$, as required.

LEMMA 4. *Let $\mathcal{F}$ be a sheaf on V, and let F be a finite subset of V. Then*

$$H^n(\mathfrak{V}^F, \mathcal{F}) = 0$$

for $n \geq 2$.

Put $W = V \setminus F$. If $x_0, \ldots, x_n$ are distinct and $n \geq 1$, then

$$V_{x_0}^F \cap \cdots \cap V_{x_n}^F = W.$$

Consequently, if $G = \Gamma(W, \mathcal{F})$, the alternating complex $C'(\mathfrak{V}^F, \mathcal{F})$ is isomorphic, in dimensions ≥ 1, to $C'(S(F), G)$, where $S(F)$ denotes the simplex whose vertex set is F. It follows that

$$H^n(\mathfrak{V}^F, \mathcal{F}) = H^n(S(F), G) = 0 \qquad \text{for } n \geq 2,$$

since the cohomology of a simplex is trivial.

Lemmas 3 and 4 plainly imply:

PROPOSITION 4. *If V is an irreducible algebraic curve and $\mathcal{F}$ is any sheaf on V, then*

$$H^n(V, \mathcal{F}) = 0 \qquad \textit{for } n \geq 2.$$

REMARK. I do not know whether a result analogous to the preceding one is valid for varieties of arbitrary dimension.

§2. Graded modules and coherent algebraic sheaves on projective space

54. The operation $\mathcal{F}(n)$. Let $\mathcal{F}$ be an algebraic sheaf on $X = \mathbf{P}_r(K)$, and let $\mathcal{F}_i = \mathcal{F}(U_i)$ be the restriction of $\mathcal{F}$ to U_i (cf. no. 51). For any integer n, let

$$\theta_{ij}(n) : \mathcal{F}_j(U_i \cap U_j) \longrightarrow \mathcal{F}_i(U_i \cap U_j)$$

be the isomorphism defined by multiplication by the function t_j^n/t_i^n. This makes sense because t_j/t_i is a regular function on $U_i \cap U_j$ with values in K^*. At every point of $U_i \cap U_j \cap U_k$ we have

$$\theta_{ij}(n) \circ \theta_{jk}(n) = \theta_{ik}(n).$$

We may therefore apply Proposition 4 of no. 4 and obtain an algebraic sheaf, denoted $\mathcal{F}(n)$, by gluing the sheaves $\mathcal{F}_i = \mathcal{F}(U_i)$ by means of the isomorphisms $\theta_{ij}(n)$.

There are canonical isomorphisms

$$\mathcal{F}(0) \simeq \mathcal{F}, \qquad \mathcal{F}(n)(m) \simeq \mathcal{F}(n+m).$$

Moreover, $\mathcal{F}(n)$ is locally isomorphic to $\mathcal{F}$, and is therefore coherent if $\mathcal{F}$ is. It also follows that every exact sequence

$$\mathcal{F} \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F}''$$

of algebraic sheaves gives rise, for every $n \in \mathbf{Z}$, to an exact sequence

$$\mathcal{F}(n) \longrightarrow \mathcal{F}'(n) \longrightarrow \mathcal{F}''(n).$$

Applying the preceding construction to $\mathcal{F} = \mathcal{O}$ gives the sheaves $\mathcal{O}(n)$, $n \in \mathbf{Z}$. We now give another description of these sheaves. If U is open in X, let A_U^n be the subset of

$$A_U = \Gamma(\pi^{-1}(U), \mathcal{O}_Y)$$

formed by the homogeneous functions of degree n, that is, the functions satisfying

$$f(\lambda y) = \lambda^n f(y) \qquad (\lambda \in K^*,\ y \in \pi^{-1}(U)).$$

The A_U^n are A_U^0-modules and hence give rise to an algebraic sheaf, which we denote by $\mathcal{O}'(n)$. An element of $\mathcal{O}'(n)_x$, $x \in X$, may therefore be identified with a rational fraction P/Q, where P and Q are homogeneous polynomials such that $Q(x) \neq 0$ and

$$\deg P - \deg Q = n.$$

Proposition 1. *The sheaves $\mathcal{O}(n)$ and $\mathcal{O}'(n)$ are canonically isomorphic.*

By definition, a section of $\mathcal{O}(n)$ over an open subset $U \subset X$ is a system (f_i) of sections of $\mathcal{O}$ over the $U \cap U_i$ such that

$$f_i = (t_j^n/t_i^n) f_j \qquad \text{on } U \cap U_i \cap U_j.$$

The f_i may be identified with regular homogeneous functions of degree 0 on the $\pi^{-1}(U) \cap \pi^{-1}(U_i)$. Put $g_i = t_i^n f_i$. Then $g_i = g_j$ at every point of $\pi^{-1}(U) \cap \pi^{-1}(U_i) \cap \pi^{-1}(U_j)$, and hence the g_i are the restrictions of a unique regular function g on $\pi^{-1}(U)$, homogeneous of degree n. Conversely, such a function g defines a system (f_i) by $f_i = g/t_i^n$. Thus the map $(f_i) \mapsto g$ is an isomorphism from $\mathcal{O}(n)$ onto $\mathcal{O}'(n)$.

In what follows we shall most often identify $\mathcal{O}(n)$ and $\mathcal{O}'(n)$ by means of the preceding isomorphism. A section of $\mathcal{O}'(n)$ over X is simply a regular function on Y homogeneous of degree n. If $r \geq 1$, such a function is identically zero for $n < 0$, and is a homogeneous *polynomial* of degree n for $n \geq 0$.

Proposition 2. *For every algebraic sheaf $\mathcal{F}$, the sheaves $\mathcal{F}(n)$ and $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{O}(n)$ are canonically isomorphic.*

Since $\mathcal{O}(n)$ is obtained from the $\mathcal{O}_i$ by gluing with the $\theta_{ij}(n)$, the sheaf $\mathcal{F} \otimes \mathcal{O}(n)$ is obtained from the $\mathcal{F}_i \otimes \mathcal{O}_i$ by gluing with the isomorphisms $1 \otimes \theta_{ij}(n)$. On identifying $\mathcal{F}_i \otimes \mathcal{O}_i$ with $\mathcal{F}_i$, we recover exactly the definition of $\mathcal{F}(n)$.

In what follows we shall likewise identify $\mathcal{F}(n)$ with $\mathcal{F} \otimes \mathcal{O}(n)$.

55. Sections of $\mathcal{F}(n)$.

We first prove a lemma on affine varieties, entirely analogous to Lemma 1 of no. 45.

Lemma 1. *Let V be an affine variety, let Q be a regular function on V, and let V_Q be the set of points $x \in V$ such that $Q(x) \neq 0$. Let $\mathcal{F}$ be a coherent algebraic sheaf on V, and let s be a section of $\mathcal{F}$ over V_Q. Then, for every sufficiently large n, there exists a section s' of $\mathcal{F}$ over all of V such that $s' = Q^n s$ over V_Q.*

By embedding V in an affine space and extending $\mathcal{F}$ by zero outside V, we reduce to the case in which V is an affine space and hence is irreducible. By Corollary 1 to Theorem 2 of no. 45, there is a surjective homomorphism

$$\varphi : \mathcal{O}_V^p \longrightarrow \mathcal{F}.$$

By Proposition 2 of no. 42, V_Q is an affine open subset, and hence (no. 44, Corollary 2 to Proposition 7) there is a section σ of $\mathcal{O}_V^p$ over V_Q such that $\varphi(\sigma) = s$. We may identify σ with a system of p regular functions on V_Q. Applying Proposition 5 of no. 43 to each of these functions, we see that there is a section σ' of $\mathcal{O}_V^p$ over V such that $\sigma' = Q^n \sigma$ over V_Q, provided that n is sufficiently large. Putting $s' = \varphi(\sigma')$, we obtain a section of $\mathcal{F}$ over V such that $s' = Q^n s$ over V_Q.

THEOREM 1. *Let $\mathcal{F}$ be a coherent algebraic sheaf on $X = \mathbf{P}_r(K)$. There is an integer $n(\mathcal{F})$ such that, for every $n \geq n(\mathcal{F})$ and every $x \in X$, the $\mathcal{O}_x$-module $\mathcal{F}(n)_x$ is generated by the elements of $\Gamma(X, \mathcal{F}(n))$.*

By definition of $\mathcal{F}(n)$, a section s of $\mathcal{F}(n)$ over X is a system (s_i) of sections of $\mathcal{F}$ over the U_i satisfying the compatibility conditions

$$s_i = (t_j^n/t_i^n)s_j \qquad \text{on } U_i \cap U_j.$$

We shall call s_i the *i-th component of s*.

On the other hand, since U_i is isomorphic to K^r, there are finitely many sections s_i^α of $\mathcal{F}$ over U_i that generate $\mathcal{F}_x$ for every $x \in U_i$ (no. 45, Corollary 1 to Theorem 2). If, for some integer n, we can find sections s^α of $\mathcal{F}(n)$ whose i-th components are the s_i^α, it is clear that $\Gamma(X, \mathcal{F}(n))$ generates $\mathcal{F}(n)_x$ for every $x \in U_i$. Theorem 1 will therefore follow once we prove the following lemma.

LEMMA 2. *Let s_i be a section of $\mathcal{F}$ over U_i. For every sufficiently large n, there exists a section s of $\mathcal{F}(n)$ whose i-th component is equal to s_i.*

Apply Lemma 1 to the affine variety $V = U_j$, the function $Q = t_i/t_j$, and the restriction of s_i to $U_i \cap U_j$. This is legitimate, since t_i/t_j is a regular function on U_j whose zero locus is $U_j - (U_i \cap U_j)$. It follows that there exist an integer p and a section s'_j of $\mathcal{F}$ over U_j such that

$$s'_j = (t_i^p/t_j^p)s_i \qquad \text{on } U_i \cap U_j.$$

For $j = i$, this gives $s'_i = s_i$, so the preceding formula may be written

$$s'_j = (t_i^p/t_j^p)s'_i.$$

Having defined the s'_j for every index j (with the same exponent p), consider

$$s'_j - (t_k^p/t_j^p)s'_k.$$

This is a section of $\mathcal{F}$ over $U_j \cap U_k$ whose restriction to $U_i \cap U_j \cap U_k$ is zero. Applying Proposition 6 of no. 43 to it, we see that, for every sufficiently large integer q,

$$(t_i^q/t_j^q)\big(s'_j - (t_k^p/t_j^p)s'_k\big) = 0 \qquad \text{on } U_j \cap U_k.$$

If we now put $s_j = (t_i^q/t_j^q)s'_j$ and $n = p + q$, the preceding formula becomes

$$s_j = (t_k^n/t_j^n)s_k,$$

and the system $s = (s_j)$ is a section of $\mathcal{F}(n)$ whose i-th component is equal to s_i, as required.

COROLLARY. *Every coherent algebraic sheaf $\mathcal{F}$ on $X = \mathbf{P}_r(K)$ is isomorphic to a quotient sheaf of a sheaf $\mathcal{O}(n)^p$, for suitable integers n and p.*

By the preceding theorem, there is an integer n such that $\mathcal{F}(-n)_x$ is generated by $\Gamma(X, \mathcal{F}(-n))$ for every $x \in X$. In view of the quasi-compactness of X, this is equivalent to saying that $\mathcal{F}(-n)$ is isomorphic to a quotient sheaf of $\mathcal{O}^p$, for a suitable integer $p \geq 0$. It follows that

$$\mathcal{F} \simeq \mathcal{F}(-n)(n)$$

is isomorphic to a quotient sheaf of $\mathcal{O}(n)^p \simeq \mathcal{O}^p(n)$.

56. Graded modules.

Let $S = K[t_0, \dots, t_r]$ be the polynomial algebra in $t_0, \dots, t_r$. For every integer $n \geq 0$, let S_n be the vector subspace of S formed by the homogeneous polynomials of degree n; for $n < 0$, put $S_n = 0$. The algebra S is the direct sum of the S_n, $n \in \mathbf{Z}$, and

$$S_p S_q \subset S_{p+q};$$

in other words, S is a *graded algebra*.

Recall that an S-module M is called *graded* when a decomposition of M as a direct sum

$$M = \sum_{n \in \mathbf{Z}} M_n$$

has been specified, where the M_n are subgroups of M such that $S_p M_q \subset M_{p+q}$ for every pair of integers (p, q). An element of M_n is called *homogeneous* of degree n. A submodule N of M is called *homogeneous* if it is the direct sum of the $N \cap M_n$, in which case it is a graded S-module.

If M and M' are two graded S-modules, an S-homomorphism

$$\varphi : M \longrightarrow M'$$

is called *homogeneous of degree s* if $\varphi(M_n) \subset M'_{n+s}$ for every $n \in \mathbf{Z}$. A homogeneous S-homomorphism of degree zero will simply be called a *homomorphism*.

If M is a graded S-module and n is an integer, we denote by $M(n)$ the graded S-module

$$M(n) = \sum_{p \in \mathbf{Z}} M(n)_p, \qquad M(n)_p = M_{n+p}.$$

Thus $M(n) = M$ as an S-module, but an element homogeneous of degree p in $M(n)$ is homogeneous of degree $n + p$ in M; in other words, $M(n)$ is obtained from M by lowering the degrees by n units.

We denote by $\mathcal{C}$ the class of graded S-modules M such that $M_n = 0$ for every sufficiently large n. If

$$A \longrightarrow B \longrightarrow C$$

is an exact sequence of homomorphisms of graded S-modules, the relations $A \in \mathcal{C}$ and $C \in \mathcal{C}$ clearly imply $B \in \mathcal{C}$; in other words, $\mathcal{C}$ is indeed a class in the sense of [14], Chapter I.

More generally, we shall use the terminology introduced in the cited article. In particular, a homomorphism $\varphi : A \to B$ is called *$\mathcal{C}$-injective* (respectively *$\mathcal{C}$-surjective*) if $\mathrm{Ker}(\varphi) \in \mathcal{C}$ (respectively $\mathrm{Coker}(\varphi) \in \mathcal{C}$), and *$\mathcal{C}$-bijective* if it is both $\mathcal{C}$-injective and $\mathcal{C}$-surjective.

A graded S-module M is called *finitely generated* if it is generated by finitely many elements. We say that M *satisfies condition (TF)* if there is an integer p such that the submodule

$$\sum_{n \geq p} M_n$$

of M is finitely generated. Equivalently, M is $\mathcal{C}$-isomorphic to a finitely generated module. The modules satisfying (TF) form a class containing $\mathcal{C}$.

A graded S-module L is called *free* (respectively *finitely generated free*) if it has a basis (respectively a finite basis) formed by homogeneous elements; in other words, if it is isomorphic to a direct sum (respectively a finite direct sum) of modules $S(n_i)$.

57. Algebraic sheaf associated with a graded S-module.

If U is a nonempty subset of X, we denote by $S(U)$ the subset of $S = K[t_0, \ldots, t_r]$ consisting of the homogeneous polynomials Q such that $Q(x) \neq 0$ for every $x \in U$. The subset $S(U)$ is multiplicatively closed and does not contain 0. For $U = \{x\}$, we write $S(x)$ instead of $S(\{x\})$.

Let M be a graded S-module. We denote by M_U the set of fractions m/Q, where $m \in M$, $Q \in S(U)$, and m and Q are homogeneous of the *same* degree. Two fractions m/Q and m'/Q' are identified if there exists $Q'' \in S(U)$ such that

$$Q''(Q'm - Qm') = 0.$$

This indeed defines an equivalence relation on the pairs (m, Q). For $U = \{x\}$, we write M_x instead of $M_{\{x\}}$.

Applying this construction to $M = S$, we find that S_U is the ring of rational fractions P/Q, where P and Q are homogeneous polynomials of the same degree and $Q \in S(U)$. If M is any graded S-module, then M_U becomes an S_U-module by setting

$$\frac{m}{Q} + \frac{m'}{Q'} = \frac{Q'm + Qm'}{QQ'}, \qquad \frac{P}{Q} \cdot \frac{m}{Q'} = \frac{Pm}{QQ'}.$$

If $U \subset V$, then $S(V) \subset S(U)$, and hence there are canonical homomorphisms

$$\varphi_U^V : M_V \longrightarrow M_U.$$

The system (M_U, φ_U^V), with U and V ranging over the nonempty open subsets of X, therefore defines a sheaf, which we denote by $\mathcal{A}(M)$. One immediately verifies that

$$\varinjlim_{x \in U} M_U = M_x;$$

that is, $\mathcal{A}(M)_x = M_x$.

In particular, $\mathcal{A}(S) = \mathcal{O}$. Since the M_U are S_U-modules, it follows that $\mathcal{A}(M)$ is a sheaf of $\mathcal{A}(S)$-modules, that is, an *algebraic sheaf* on X. Every homomorphism $\varphi : M \to M'$ naturally defines S_U-linear homomorphisms $\varphi_U : M_U \to M'_U$, and hence a homomorphism of sheaves

$$\mathcal{A}(\varphi) : \mathcal{A}(M) \longrightarrow \mathcal{A}(M'),$$

which we shall often denote simply by φ. Evidently,

$$\mathcal{A}(\varphi + \psi) = \mathcal{A}(\varphi) + \mathcal{A}(\psi), \qquad \mathcal{A}(1) = 1, \qquad \mathcal{A}(\varphi \circ \psi) = \mathcal{A}(\varphi) \circ \mathcal{A}(\psi).$$

Thus the operation $M \mapsto \mathcal{A}(M)$ is an *additive covariant functor* from the category of graded S-modules to the category of algebraic sheaves on X.

(The preceding definitions are entirely analogous to those of ğ4 of Chapter II. Notice, however, that S_U is not the ring of fractions of S with respect to $S(U)$, but only its homogeneous component of degree zero.)

58. First properties of the functor $\mathcal{A}(M)$.

Proposition 3. *The functor $M \mapsto \mathcal{A}(M)$ is exact.*

Let

$$M \xrightarrow{\alpha} M' \xrightarrow{\beta} M''$$

be an exact sequence of graded S-modules, and let us show that the sequence

$$M_x \xrightarrow{\alpha} M'_x \xrightarrow{\beta} M''_x$$

is also exact. Let $m'/Q \in M'_x$ be an element of the kernel of β. By the definition of M''_x, there exists $R \in S(x)$ such that $R\beta(m') = 0$. But then there exists $m \in M$ such that $\alpha(m) = Rm'$, and

$$\alpha\left(\frac{m}{RQ}\right) = \frac{m'}{Q},$$

as required. (Compare no. 48, Lemma 1.)

Proposition 4. *If M is a graded S-module and n is an integer, then $\mathcal{A}(M(n))$ is canonically isomorphic to $\mathcal{A}(M)(n)$.*

Let $i \in I$, $x \in U_i$, and $m/Q \in M(n)_x$, with $m \in M(n)_p$, $Q \in S(x)$, and $\deg Q = p$. Set

$$\eta_{i,x}(m/Q) = \frac{m}{t_i^n Q} \in M_x,$$

which is legitimate since $m \in M_{n+p}$ and $t_i^n Q \in S(x)$. It is immediate that $\eta_{i,x} : M(n)_x \to M_x$ is bijective for every $x \in U_i$, and defines an isomorphism η_i from $\mathcal{A}(M(n))$ onto $\mathcal{A}(M)$ over U_i. Moreover,

$$\eta_i \circ \eta_j^{-1} = \theta_{ij}(n)$$

over $U_i \cap U_j$. In view of the definition of the operation $\mathcal{F}(n)$ in no. 54, and Proposition 4 of no. 4, this proves that $\mathcal{A}(M(n))$ is isomorphic to $\mathcal{A}(M)(n)$.

Corollary. *$\mathcal{A}(S(n))$ is canonically isomorphic to $\mathcal{O}(n)$.*

Indeed, we have already said that $\mathcal{A}(S)$ is isomorphic to $\mathcal{O}$.

(It is also directly evident that $\mathcal{A}(S(n))$ is isomorphic to $\mathcal{O}'(n)$, since $\mathcal{O}'(n)_x$ consists precisely of the rational fractions P/Q such that $\deg P - \deg Q = n$ and $Q \in S(x)$.)

Proposition 5. *Let M be a graded S-module satisfying condition (TF). Then the algebraic sheaf $\mathcal{A}(M)$ is coherent, and $\mathcal{A}(M) = 0$ if and only if $M \in \mathcal{C}$.*

If $M \in \mathcal{C}$, then for every $m \in M$ and every $x \in X$ there exists $Q \in S(x)$ such that $Qm = 0$: it is enough to take Q of sufficiently large degree. Thus $M_x = 0$, whence $\mathcal{A}(M) = 0$.

Now let M be a graded S-module satisfying condition (TF). There is a finitely generated homogeneous submodule N of M such that $M/N \in \mathcal{C}$. By what precedes and Proposition 3, the map $\mathcal{A}(N) \to \mathcal{A}(M)$ is bijective, so it is enough to prove that $\mathcal{A}(N)$ is coherent. Since N is finitely generated, there is an exact sequence

$$L^1 \longrightarrow L^0 \longrightarrow N \longrightarrow 0,$$

where L^0 and L^1 are finitely generated *free* modules. By Proposition 3, the sequence

$$\mathcal{A}(L^1) \longrightarrow \mathcal{A}(L^0) \longrightarrow \mathcal{A}(N) \longrightarrow 0$$

is exact. By the corollary to Proposition 4, however, $\mathcal{A}(L^0)$ and $\mathcal{A}(L^1)$ are isomorphic to finite direct sums of sheaves $\mathcal{O}(n_i)$, and are therefore coherent. It follows that $\mathcal{A}(N)$ is coherent.

Finally, let M be a graded S-module satisfying (TF) and such that $\mathcal{A}(M) = 0$. By what precedes, we may suppose that M is finitely generated. If m is a homogeneous element of M, let $\mathfrak{a}_m$ be the annihilator of m, that is, the set of polynomials $Q \in S$ such that $Qm = 0$; clearly $\mathfrak{a}_m$ is a homogeneous ideal. Moreover, the assumption $M_x = 0$ for every $x \in X$ implies that the zero locus of $\mathfrak{a}_m$ in K^{r+1} is empty or reduced to $\{0\}$. Hilbert's Nullstellensatz then shows that every homogeneous polynomial of sufficiently large degree belongs to $\mathfrak{a}_m$. Applying this to a finite system of generators of M, we conclude at once that $M_p = 0$ for all sufficiently large p, which completes the proof.

Combining Propositions 3 and 5, we obtain:

Proposition 6. *Let M and M' be two graded S-modules satisfying condition (TF), and let $\varphi : M \to M'$ be a homomorphism. The homomorphism*

$$\mathcal{A}(\varphi) : \mathcal{A}(M) \longrightarrow \mathcal{A}(M')$$

is injective (respectively surjective, bijective) if and only if φ is $\mathcal{C}$-injective (respectively $\mathcal{C}$-surjective, $\mathcal{C}$-bijective).

59. Graded S-module associated with an algebraic sheaf.

Let $\mathcal{F}$ be an algebraic sheaf on X, and set

$$\Gamma(\mathcal{F}) = \sum_{n \in \mathbf{Z}} \Gamma(\mathcal{F})_n, \qquad \Gamma(\mathcal{F})_n = \Gamma(X, \mathcal{F}(n)).$$

The group $\Gamma(\mathcal{F})$ is graded; we shall endow it with the structure of an S-module. Let $s \in \Gamma(X, \mathcal{F}(q))$ and $P \in S_p$. We may identify P with a section of $\mathcal{O}(p)$ (compare no. 54); hence $P \otimes s$ is a section of

$$\mathcal{O}(p) \otimes \mathcal{F}(q) = \mathcal{F}(q)(p) = \mathcal{F}(p+q),$$

where we have used the isomorphisms of no. 54. We have thus defined a section of $\mathcal{F}(p+q)$, which we denote by $P.s$ rather than $P \otimes s$. The map $(P, s) \mapsto P.s$ gives $\Gamma(\mathcal{F})$ the structure of an S-module compatible with its grading.

One can also define $P.s$ in terms of its components over the U_i. If the components of s are $s_i \in \Gamma(U_i, \mathcal{F})$, with $s_i = (t_j^q/t_i^q).s_j$ on $U_i \cap U_j$, then

$$(P.s)_i = (P/t_i^p).s_i.$$

This is well defined, since P/t_i^p is a regular function on U_i.

To compare the functors $\mathcal{A}(M)$ and $\Gamma(\mathcal{F})$, we shall define two canonical homomorphisms:

$$\alpha : M \longrightarrow \Gamma(\mathcal{A}(M)), \qquad \beta : \mathcal{A}(\Gamma(\mathcal{F})) \longrightarrow \mathcal{F}.$$

Definition of α. Let M be a graded S-module, and let $m \in M_0$ be a homogeneous element of degree zero in M. The element $m/1$ is a well-defined element of M_x, and varies continuously with $x \in X$; hence m defines a section $\alpha(m)$ of $\mathcal{A}(M)$. If now m is homogeneous of degree n, then m is homogeneous of degree zero in $M(n)$, and therefore defines a section $\alpha(m)$ of $\mathcal{A}(M(n)) = \mathcal{A}(M)(n)$ (compare Proposition 4). This defines $\alpha : M \to \Gamma(\mathcal{A}(M))$, and it is immediate that α is a homomorphism.

Definition of β. Let $\mathcal{F}$ be an algebraic sheaf on X, and let s/Q be an element of $\Gamma(\mathcal{F})_x$, with $s \in \Gamma(X, \mathcal{F}(n))$, $Q \in S_n$, and $Q(x) \neq 0$. The function $1/Q$ is homogeneous of degree $-n$ and regular at x, hence is a section of $\mathcal{O}(-n)$ in a neighborhood of x. It follows that $1/Q \otimes s$ is a section of $\mathcal{O}(-n) \otimes \mathcal{F}(n) = \mathcal{F}$ in a neighborhood of x, and thus defines an element of $\mathcal{F}_x$. We denote this element by $\beta_x(s/Q)$, since it depends only on s/Q. The map β_x can also be defined using the components s_i of s: if $x \in U_i$, then

$$\beta_x(s/Q) = (t_i^n/Q).s_i(x).$$

The collection of homomorphisms β_x defines the homomorphism $\beta : \mathcal{A}(\Gamma(\mathcal{F})) \to \mathcal{F}$.

The homomorphisms α and β are related by the following propositions, which follow by a direct calculation.

Proposition 7. *Let M be a graded S-module. The composite of the homomorphisms*

$$\mathcal{A}(M) \longrightarrow \mathcal{A}(\Gamma(\mathcal{A}(M))) \longrightarrow \mathcal{A}(M)$$

is the identity.

(The first homomorphism is defined by $\alpha : M \to \Gamma(\mathcal{A}(M))$, and the second is β, applied to $\mathcal{F} = \mathcal{A}(M)$.)

Proposition 8. *Let $\mathcal{F}$ be an algebraic sheaf on X. The composite of the homomorphisms*

$$\Gamma(\mathcal{F}) \longrightarrow \Gamma(\mathcal{A}(\Gamma(\mathcal{F}))) \longrightarrow \Gamma(\mathcal{F})$$

is the identity.

(The first homomorphism is α, applied to $M = \Gamma(\mathcal{F})$, while the second is defined by $\beta : \mathcal{A}(\Gamma(\mathcal{F})) \to \mathcal{F}$.)

We shall show in no. 65 that $\beta : \mathcal{A}(\Gamma(\mathcal{F})) \to \mathcal{F}$ is bijective if $\mathcal{F}$ is coherent, and that $\alpha : M \to \Gamma(\mathcal{A}(M))$ is $\mathcal{C}$-bijective if M satisfies condition (TF).

60. The case of coherent algebraic sheaves.

Let us first prove a preliminary result.

Proposition 9. *Let $\mathcal{L}$ be an algebraic sheaf on X that is a direct sum of finitely many sheaves $\mathcal{O}(n_i)$. Then $\Gamma(\mathcal{L})$ satisfies condition (TF), and $\beta : \mathcal{A}(\Gamma(\mathcal{L})) \to \mathcal{L}$ is bijective.*

We reduce at once to $\mathcal{L} = \mathcal{O}(n)$, and then to $\mathcal{L} = \mathcal{O}$. In this case we know that $\Gamma(\mathcal{O}(p)) = S_p$ for $p \geq 0$, and hence $S \subset \Gamma(\mathcal{O})$, with the quotient belonging to $\mathcal{C}$. It follows first that $\Gamma(\mathcal{O})$ satisfies (TF), and then that

$$\mathcal{A}(\Gamma(\mathcal{O})) = \mathcal{A}(S) = \mathcal{O},$$

which proves the proposition.

(Observe that $\Gamma(\mathcal{O}) = S$ if $r \geq 1$; if $r = 0$, however, $\Gamma(\mathcal{O})$ is not even a finitely generated S-module.)

THEOREM 2. *For every coherent algebraic sheaf $\mathcal{F}$ on X, there is a graded S-module M satisfying condition (TF) such that $\mathcal{A}(M)$ is isomorphic to $\mathcal{F}$.*

By the corollary to Theorem 1, there is an exact sequence of algebraic sheaves

$$\mathcal{L}^1 \xrightarrow{\varphi} \mathcal{L}^0 \longrightarrow \mathcal{F} \longrightarrow 0,$$

where $\mathcal{L}^1$ and $\mathcal{L}^0$ satisfy the hypotheses of the preceding proposition. Let M be the cokernel of the homomorphism

$$\Gamma(\varphi) : \Gamma(\mathcal{L}^1) \longrightarrow \Gamma(\mathcal{L}^0).$$

By Proposition 9, M satisfies condition (TF). Applying the functor $\mathcal{A}$ to the exact sequence

$$\Gamma(\mathcal{L}^1) \longrightarrow \Gamma(\mathcal{L}^0) \longrightarrow M \longrightarrow 0,$$

we obtain the exact sequence

$$\mathcal{A}(\Gamma(\mathcal{L}^1)) \longrightarrow \mathcal{A}(\Gamma(\mathcal{L}^0)) \longrightarrow \mathcal{A}(M) \longrightarrow 0.$$

Consider the following commutative diagram:

$$\begin{array}{ccccccc}
\mathcal{A}(\Gamma(\mathcal{L}^1)) & \longrightarrow & \mathcal{A}(\Gamma(\mathcal{L}^0)) & \longrightarrow & \mathcal{A}(M) & \longrightarrow & 0 \\
\downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\beta} & & & & \\
\mathcal{L}^1 & \longrightarrow & \mathcal{L}^0 & \longrightarrow & \mathcal{F} & \longrightarrow & 0.
\end{array}$$

By Proposition 9, the two vertical homomorphisms are bijective. It follows that $\mathcal{A}(M)$ is isomorphic to $\mathcal{F}$, as required.

§3. Cohomology of projective space with values in a coherent algebraic sheaf

61. The complexes $C_k(M)$ and $C(M)$. We retain the notation of nos. 51 and 56. In particular, I will denote the interval $\{0, 1, \ldots, r\}$, and S will denote the graded algebra $K[t_0, \ldots, t_r]$.

Let M be a graded S-module, and let k and q be two integers ≥ 0. We shall define a group $C^q_k(M)$. An element of $C^q_k(M)$ is a mapping

$$(i_0, \ldots, i_q) \longmapsto m\langle i_0 \cdots i_q \rangle$$

that assigns to every sequence $(i_0, \ldots, i_q)$ of $q+1$ elements of I a homogeneous element of M of degree $k(q+1)$ and is alternating in $i_0, \ldots, i_q$. In particular, $m\langle i_0 \cdots i_q \rangle = 0$ if two of the indices $i_0, \ldots, i_q$ are equal. Addition in $C^q_k(M)$ and multiplication by an element $\lambda \in K$ are defined in the evident way, and $C^q_k(M)$ is a *vector space over* K.

If m is an element of $C^q_k(M)$, define $dm \in C^{q+1}_k(M)$ by

$$(dm)\langle i_0 \cdots i_{q+1} \rangle = \sum_{j=0}^{j=q+1} (-1)^j t^k_{i_j} . m\langle i_0 \cdots \hat{i}_j \cdots i_{q+1} \rangle.$$

A direct calculation shows that $d \circ d = 0$. Hence the direct sum $C_k(M) = \sum_{q=0}^{q=r} C^q_k(M)$, equipped with the coboundary operator d, is a *complex*; its q-th cohomology group will be denoted by $H^q_k(M)$.

(Following [11], let us mention another interpretation of the elements of $C_k^q(M)$. Introduce $r+1$ differential symbols $dx_0, \dots, dx_r$, and associate with every $m \in C_k^q(M)$ the "differential form" of degree $q+1$

$$\omega_m = \sum_{i_0 < \cdots < i_q} m\langle i_0 \cdots i_q\rangle \, dx_{i_0} \wedge \cdots \wedge dx_{i_q}.$$

If $\alpha_k = \sum_{i=0}^{i=r} t_i^k \, dx_i$, then

$$\omega_{dm} = \alpha_k \wedge \omega_m;$$

in other words, the coboundary operation becomes exterior multiplication by the form α_k.)

If h is an integer $\geq k$, let $\rho_k^h : C_k^q(M) \to C_h^q(M)$ be the homomorphism defined by

$$\rho_k^h(m)\langle i_0 \cdots i_q\rangle = (t_{i_0} \cdots t_{i_q})^{h-k} m\langle i_0 \cdots i_q\rangle.$$

We have $\rho_k^h \circ d = d \circ \rho_k^h$, and $\rho_h^l \circ \rho_k^h = \rho_k^l$ if $k \leq h \leq l$. We may therefore define the complex $C(M)$, the direct limit of the system $(C_k(M), \rho_k^h)$ as $k \to +\infty$. The cohomology groups of this complex will be denoted by $H^q(M)$. Since cohomology commutes with direct limits (cf. [6], Chapter V, Proposition 9.3*),

$$H^q(M) = \varinjlim_{k \to \infty} H_k^q(M).$$

Every homomorphism $\varphi : M \to M'$ defines a homomorphism

$$\varphi : C_k(M) \longrightarrow C_k(M')$$

by the formula $\varphi(m)\langle i_0 \cdots i_q\rangle = \varphi(m\langle i_0 \cdots i_q\rangle)$, and hence, by passage to the limit, a homomorphism $\varphi : C(M) \to C(M')$. Moreover, these homomorphisms commute with the coboundary and therefore define homomorphisms

$$\varphi : H_k^q(M) \longrightarrow H_k^q(M') \qquad \text{and} \qquad \varphi : H^q(M) \longrightarrow H^q(M').$$

If $0 \to M \to M' \to M'' \to 0$ is exact, then so is the sequence of complexes $0 \to C_k(M) \to C_k(M') \to C_k(M'') \to 0$, and hence there is an exact cohomology sequence

$$\cdots \longrightarrow H_k^q(M') \longrightarrow H_k^q(M'') \longrightarrow H_k^{q+1}(M) \longrightarrow H_k^{q+1}(M') \longrightarrow \cdots.$$

The same results hold for $C(M)$ and the $H^q(M)$.

Remark. We shall see later (cf. no. 69) that the $H_k^q(M)$ can be expressed in terms of the Ext_S^q.

62. Computation of $H_k^q(M)$ for certain modules M. Let M be a graded S-module, and let $m \in M$ be a homogeneous element of degree 0. The system $(t_i^k.m)$ is a 0-cocycle of $C_k(M)$, which we shall denote by $\alpha^k(m)$ and identify with its cohomology class. This gives a K-linear homomorphism

$$\alpha^k : M_0 \longrightarrow H_k^0(M).$$

Since $\alpha^h = \rho_k^h \circ \alpha^k$ when $h \geq k$, passage to the limit gives a homomorphism

$$\alpha : M_0 \longrightarrow H^0(M).$$

We introduce two further notations.

If $P_0, \dots, P_h$ are elements of S, we denote by $(P_0, \dots, P_h)M$ the submodule of M consisting of the elements

$$\sum_{i=0}^{i=h} P_i.m_i, \qquad m_i \in M.$$

If the P_i are homogeneous, this submodule is homogeneous.

If P is an element of S and N a submodule of M, we denote by $N : P$ the submodule of M consisting of the elements $m \in M$ such that $P.m \in N$. Evidently $N : P \supset N$; if N and P are homogeneous, then $N : P$ is homogeneous.

With these notations fixed, we have:

Proposition 1. *Let M be a graded S-module and let k be an integer ≥ 0. Suppose that, for every $i \in I$,*

$$(t_0^k, \ldots, t_{i-1}^k)M : t_i^k = (t_0^k, \ldots, t_{i-1}^k)M.$$

Then:

(a) $\alpha^k : M_0 \to H_k^0(M)$ is bijective if $r \geq 1$.

(b) $H_k^q(M) = 0$ for $0 < q < r$.

(For $i = 0$, the hypothesis means that $t_0^k.m = 0$ implies $m = 0$.)

This proposition is a special case of a result due to de Rham [11] (de Rham's result remains valid even when the $m\langle i_0 \cdots i_q\rangle$ are not assumed homogeneous). See also [6], Chapter VIII, §4, where a special case sufficient for our applications is treated.

We apply Proposition 1 to the graded S-module $S(n)$.

Proposition 2. *Let k be an integer ≥ 0, and let n be any integer. Then:*

(a) $\alpha^k : S_n \to H_k^0(S(n))$ is bijective if $r \geq 1$.

(b) $H_k^q(S(n)) = 0$ for $0 < q < r$.

(c) $H_k^r(S(n))$ has as a basis over K the cohomology classes of the monomials $t_0^{\alpha_0} \cdots t_r^{\alpha_r}$ such that $0 \leq \alpha_i < k$ and $\sum_{i=0}^{i=r} \alpha_i = k(r+1) + n$.

It is clear that the graded S-module $S(n)$ satisfies the hypotheses of Proposition 1; this proves (a) and (b). On the other hand, for every graded S-module M,

$$H_k^r(M) = M_{k(r+1)}/(t_0^k, \ldots, t_r^k)M_{kr}.$$

Now the monomials

$$t_0^{\alpha_0} \cdots t_r^{\alpha_r}, \qquad \alpha_i \geq 0, \qquad \sum_{i=0}^{i=r} \alpha_i = k(r+1) + n,$$

form a basis of $S(n)_{k(r+1)}$, and those among them for which at least one α_i is $\geq k$ form a basis of $(t_0^k, \ldots, t_r^k)S(n)_{kr}$. This proves (c).

It is convenient to write the exponents α_i in the form $\alpha_i = k - \beta_i$. The conditions in (c) then become

$$0 < \beta_i \leq k \qquad \text{and} \qquad \sum_{i=0}^{i=r} \beta_i = -n.$$

The second condition, together with $\beta_i > 0$, implies $\beta_i \leq -n - r$. Hence, if $k \geq -n - r$, the condition $\beta_i \leq k$ follows from the preceding two. Thus:

Corollary 1. *For $k \geq -n - r$, $H_k^r(S(n))$ has as a basis the cohomology classes of the monomials*

$$\frac{(t_0 \cdots t_r)^k}{t_0^{\beta_0} \cdots t_r^{\beta_r}}, \qquad \beta_i > 0, \qquad \sum_{i=0}^{i=r} \beta_i = -n.$$

We also have:

Corollary 2. *If $h \geq k \geq -n - r$, the homomorphism*

$$\rho_k^h : H_k^q(S(n)) \longrightarrow H_h^q(S(n))$$

is bijective for every $q \geq 0$.

For $q \neq r$, this follows from assertions (a) and (b) of Proposition 2. For $q = r$, it follows from Corollary 1, since ρ_k^h sends

$$\frac{(t_0 \cdots t_r)^k}{t_0^{\beta_0} \cdots t_r^{\beta_r}} \quad \text{to} \quad \frac{(t_0 \cdots t_r)^h}{t_0^{\beta_0} \cdots t_r^{\beta_r}}.$$

Corollary 3. *The homomorphism $\alpha : S_n \to H^0(S(n))$ is bijective if $r \geq 1$ or if $n \geq 0$. We have $H^q(S(n)) = 0$ for $0 < q < r$, and $H^r(S(n))$ is a vector space of dimension $\binom{-n-1}{r}$ over K.*

The assertion concerning α follows from Proposition 2(a) when $r \geq 1$; it is immediate when $r = 0$ and $n \geq 0$. The rest of the corollary follows at once from Corollaries 1 and 2, with the convention that the binomial coefficient $\binom{a}{r}$ is zero when $a < r$.

63. General properties of $H^q(M)$.

Proposition 3. *Let M be a graded S-module satisfying condition* (TF). *Then:*

(a) There exists an integer $k(M)$ such that $\rho_k^h : H_k^q(M) \to H_h^q(M)$ is bijective for $h \geq k \geq k(M)$ and every q.

(b) $H^q(M)$ is a finite-dimensional vector space over K for every $q \geq 0$.

(c) There exists an integer $n(M)$ such that, for $n \geq n(M)$, $\alpha : M_n \to H^0(M(n))$ is bijective and $H^q(M(n)) = 0$ for every $q > 0$.

We reduce at once to the case in which M is of finite type. We shall then say that M has *dimension* $\leq s$ (where s is an integer ≥ 0) if there is an exact sequence

$$0 \longrightarrow L^s \longrightarrow L^{s-1} \longrightarrow \cdots \longrightarrow L^0 \longrightarrow M \longrightarrow 0,$$

in which the L^i are graded free S-modules of finite type. By Hilbert's syzygy theorem (see [6], Chapter VIII, Theorem 6.5), this dimension is always $\leq r + 1$.

We prove the proposition by induction on the dimension of M. If that dimension is zero, M is free of finite type, that is, a direct sum of modules $S(n_i)$, and the proposition follows from Corollaries 2 and 3 to Proposition 2. Suppose that M has dimension $\leq s$, and let N be the kernel of $L^0 \to M$. The graded S-module N has dimension $\leq s - 1$, and there is an exact sequence

$$0 \longrightarrow N \longrightarrow L^0 \longrightarrow M \longrightarrow 0.$$

By the induction hypothesis, the proposition holds for N and L^0. Applying the five lemma ([7], Chapter I, Lemma 4.3) to the commutative diagram

$$\begin{tikzcd}
H_k^q(N) \arrow[r] \arrow[d] & H_k^q(L^0) \arrow[r] \arrow[d] & H_k^q(M) \arrow[r] \arrow[d] & H_k^{q+1}(N) \arrow[r] \arrow[d] & H_k^{q+1}(L^0) \arrow[d] \\
H_h^q(N) \arrow[r] & H_h^q(L^0) \arrow[r] & H_h^q(M) \arrow[r] & H_h^{q+1}(N) \arrow[r] & H_h^{q+1}(L^0),
\end{tikzcd}$$

for $h \geq k \geq \mathrm{Sup}(k(N), k(L^0))$, proves (a), and hence also (b), since the $H_k^q(M)$ are finite-dimensional over K. On the other hand, the exact sequence

$$H^q(L^0(n)) \longrightarrow H^q(M(n)) \longrightarrow H^{q+1}(N(n))$$

shows that $H^q(M(n)) = 0$ for $n \geq \mathrm{Sup}(n(L^0), n(N))$. Finally, consider the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] \arrow[d] & N_n \arrow[r] \arrow[d, "\alpha"] & L_n^0 \arrow[r] \arrow[d, "\alpha"] & M_n \arrow[r] \arrow[d, "\alpha"] & 0 \arrow[d] \\
0 \arrow[r] & H^0(N(n)) \arrow[r] & H^0(L^0(n)) \arrow[r] & H^0(M(n)) \arrow[r] & H^1(N(n));
\end{tikzcd}$$

The printed upper middle term is L_n, with its superscript 0 omitted. For $n \geq n(N)$ we have $H^1(N(n)) = 0$; it follows that $\alpha : M_n \to H^0(M(n))$ is bijective for $n \geq \mathrm{Sup}(n(L^0), n(N))$, which completes the proof of the proposition.

64. Comparison of the groups $H^q(M)$ and $H^q(X, \mathcal{A}(M))$.

Let M be a graded S-module, and let $\mathcal{A}(M)$ be the algebraic sheaf on $X = \mathbf{P}_r(K)$ constructed from M by the procedure of no. 57. We shall compare $C(M)$ with $C'(\mathfrak{U}, \mathcal{A}(M))$, the complex of alternating cochains of the covering $\mathfrak{U} = \{U_i\}_{i \in I}$ with values in the sheaf $\mathcal{A}(M)$.

Let $m \in C^q_k(M)$, and let $(i_0, \dots, i_q)$ be a sequence of $q+1$ elements of I. The polynomial $(t_{i_0} \cdots t_{i_q})^k$ plainly belongs to $S(U_{i_0 \cdots i_q})$, with the notation of no. 57. It follows that

$$\frac{m\langle i_0 \cdots i_q\rangle}{(t_{i_0} \cdots t_{i_q})^k}$$

belongs to M_U, where $U = U_{i_0 \cdots i_q}$, and hence defines a section of $\mathcal{A}(M)$ over $U_{i_0 \cdots i_q}$. As $(i_0, \dots, i_q)$ varies, the system formed by these sections is an alternating q-cochain of $\mathfrak{U}$ with values in $\mathcal{A}(M)$, which we denote by $\iota_k(m)$. It is immediate that ι_k commutes with d, and that $\iota_k = \iota_h \circ \rho^h_k$ if $h \geq k$. Passing to the direct limit, the ι_k therefore define a homomorphism

$$\iota : C(M) \longrightarrow C'(\mathfrak{U}, \mathcal{A}(M))$$

which commutes with d.

Proposition 4. *If M satisfies condition* (TF)*, the homomorphism $\iota : C(M) \to C'(\mathfrak{U}, \mathcal{A}(M))$ is bijective.*

If $M \in \mathcal{C}$, then $M_n = 0$ for $n \geq n_0$, whence $C_k(M) = 0$ for $k \geq n_0$ and $C(M) = 0$. Since every S-module satisfying (TF) is $\mathcal{C}$-isomorphic to a module of finite type, this shows that we may restrict ourselves to the case in which M is of finite type. We can then find an exact sequence

$$L^1 \longrightarrow L^0 \longrightarrow M \longrightarrow 0,$$

where L^1 and L^0 are free of finite type. By Propositions 3 and 5 of no. 58, the sequence

$$\mathcal{A}(L^1) \longrightarrow \mathcal{A}(L^0) \longrightarrow \mathcal{A}(M) \longrightarrow 0$$

is an exact sequence of coherent algebraic sheaves. Since the $U_{i_0 \cdots i_q}$ are affine open subsets, the sequence

$$C'(\mathfrak{U}, \mathcal{A}(L^1)) \longrightarrow C'(\mathfrak{U}, \mathcal{A}(L^0)) \longrightarrow C'(\mathfrak{U}, \mathcal{A}(M)) \longrightarrow 0$$

is exact (cf. no. 45, Corollary 2 to Theorem 2). The commutative diagram

$$\begin{array}{ccccccc}
C(L^1) & \longrightarrow & C(L^0) & \longrightarrow & C(M) & \longrightarrow & 0 \\
{\scriptstyle\iota}\downarrow & & {\scriptstyle\iota}\downarrow & & {\scriptstyle\iota}\downarrow & & \downarrow \\
C'(\mathfrak{U}, \mathcal{A}(L^1)) & \longrightarrow & C'(\mathfrak{U}, \mathcal{A}(L^0)) & \longrightarrow & C'(\mathfrak{U}, \mathcal{A}(M)) & \longrightarrow & 0
\end{array}$$

then shows that, if the proposition holds for L^1 and L^0, it holds for M. We are therefore reduced to the special case of a free module of finite type and then, by decomposition into a direct sum, to the case $M = S(n)$.

In this case $\mathcal{A}(S(n)) = \mathcal{O}(n)$. By the very definition of this sheaf, a section $f_{i_0 \cdots i_q}$ of $\mathcal{O}(n)$ over $U_{i_0 \cdots i_q}$ is a regular function on $V_{i_0} \cap \cdots \cap V_{i_q}$, homogeneous of degree n. Since $V_{i_0} \cap \cdots \cap V_{i_q}$ is the set of points of K^{r+1} at which $t_{i_0} \cdots t_{i_q}$ is nonzero, there is an integer k such that

$$f_{i_0 \cdots i_q} = \frac{P\langle i_0 \cdots i_q\rangle}{(t_{i_0} \cdots t_{i_q})^k},$$

where $P\langle i_0 \cdots i_q\rangle$ is a homogeneous polynomial of degree $n + k(q+1)$, that is, of degree $k(q+1)$ in $S(n)$. Thus every alternating cochain $f \in C'(\mathfrak{U}, \mathcal{O}(n))$ defines a system $P\langle i_0 \cdots i_q\rangle$ which is an element of $C_k(S(n))$, and hence a homomorphism

$$\nu : C'(\mathfrak{U}, \mathcal{O}(n)) \longrightarrow C(S(n)).$$

Since one immediately verifies that $\iota \circ \nu = 1$ and $\nu \circ \iota = 1$, it follows that ι is bijective, which completes the proof.

Corollary. *The homomorphism ι defines an isomorphism from $H^q(M)$ onto $H^q(X, \mathcal{A}(M))$ for every $q \geq 0$.*

Indeed, we know that $H'^q(\mathfrak{U}, \mathcal{A}(M)) = H^q(\mathfrak{U}, \mathcal{A}(M))$ (no. 20, Proposition 2), and $H^q(\mathfrak{U}, \mathcal{A}(M)) = H^q(X, \mathcal{A}(M))$ (no. 52, Proposition 2, which applies since $\mathcal{A}(M)$ is coherent).

Remark. It is easy to see that $\iota : C(M) \to C'(\mathfrak{U}, \mathcal{A}(M))$ is *injective* even when M does not satisfy condition (TF).

65. Applications.

Proposition 5. *If M is a graded S-module satisfying condition* (TF)*, the homomorphism $\alpha : M \to \Gamma(\mathcal{A}(M))$, defined in no. 59, is $\mathcal{C}$-bijective.*

We must show that $\alpha : M_n \to \Gamma(X, \mathcal{A}(M(n)))$ is bijective for all sufficiently large n. By Proposition 4, however, $\Gamma(X, \mathcal{A}(M(n)))$ is identified with $H^0(M(n))$. The proposition therefore follows from Proposition 3(c), taking into account that, under the preceding identification, the homomorphism α is transformed into the homomorphism defined at the beginning of no. 62, which is also denoted by α.

Proposition 6. *Let $\mathcal{F}$ be a coherent algebraic sheaf on X. The graded S-module $\Gamma(\mathcal{F})$ satisfies condition* (TF)*, and the homomorphism $\beta : \mathcal{A}(\Gamma(\mathcal{F})) \to \mathcal{F}$, defined in no. 59, is bijective.*

By Theorem 2 of no. 60, we may suppose that $\mathcal{F} = \mathcal{A}(M)$, where M is a module satisfying (TF). By the preceding proposition, $\alpha : M \to \Gamma(\mathcal{A}(M))$ is $\mathcal{C}$-bijective. Since M satisfies (TF), it follows that $\Gamma(\mathcal{A}(M))$ does also. Applying Proposition 6 of no. 58, we see that

$$\alpha : \mathcal{A}(M) \longrightarrow \mathcal{A}(\Gamma(\mathcal{A}(M)))$$

is bijective. Since the composite

$$\mathcal{A}(M) \xrightarrow{\alpha} \mathcal{A}(\Gamma(\mathcal{A}(M))) \xrightarrow{\beta} \mathcal{A}(M)$$

is the identity (no. 59, Proposition 7), it follows that β is bijective, which proves the proposition.

Proposition 7. *Let $\mathcal{F}$ be a coherent algebraic sheaf on X. The groups $H^q(X, \mathcal{F})$ are finite-dimensional vector spaces over K for every $q \geq 0$, and $H^q(X, \mathcal{F}(n)) = 0$ for $q > 0$ and all sufficiently large n.*

As above, we may suppose that $\mathcal{F} = \mathcal{A}(M)$, where M is a module satisfying (TF). The proposition then follows from Proposition 3 and the corollary to Proposition 4.

Proposition 8. *We have $H^q(X, \mathcal{O}(n)) = 0$ for $0 < q < r$, and $H^r(X, \mathcal{O}(n))$ is a vector space of dimension $\binom{-n-1}{r}$ over K, with basis given by the cohomology classes of the alternating cocycles of $\mathfrak{U}$*

$$f_{01\cdots r} = \frac{1}{t_0^{\beta_0} \cdots t_r^{\beta_r}}, \qquad \textit{where} \qquad \beta_i > 0 \quad \textit{and} \quad \sum_{i=0}^{i=r} \beta_i = -n.$$

We have $\mathcal{O}(n) = \mathcal{A}(S(n))$, whence $H^q(X, \mathcal{O}(n)) = H^q(S(n))$ by the corollary to Proposition 4. The proposition now follows immediately from this and the corollaries to Proposition 2.

In particular, $H^r(X, \mathcal{O}(-r-1))$ is a one-dimensional vector space over K, with basis given by the cohomology class of the cocycle

$$f_{01\cdots r} = \frac{1}{t_0 \cdots t_r}.$$

66. Coherent algebraic sheaves on projective varieties.

Let V be a closed subvariety of the projective space $X = \mathbf{P}_r(K)$, and let $\mathcal{F}$ be a coherent algebraic sheaf on V. Extending $\mathcal{F}$ by zero outside V, we obtain a coherent algebraic sheaf on X (cf. no. 39), denoted by $\mathcal{F}^X$; we know that $H^q(X, \mathcal{F}^X) = H^q(V, \mathcal{F})$. The results of the preceding number therefore apply to the groups $H^q(V, \mathcal{F})$. We first obtain (taking no. 52 into account):

Theorem 1. *The groups $H^q(V, \mathcal{F})$ are finite-dimensional vector spaces over K, and are zero for $q > \dim V$.*

In particular, for $q = 0$, we have:

Corollary. *$\Gamma(V, \mathcal{F})$ is a finite-dimensional vector space over K.*

(It is natural to conjecture that the theorem above holds for every *complete* variety, in the sense of Weil [16].)

Let $U'_i = U_i \cap V$; the U'_i form an open covering $\mathfrak{U}'$ of V. If $\mathcal{F}$ is an algebraic sheaf on V, let $\mathcal{F}_i = \mathcal{F}(U'_i)$, and let $\theta_{ij}(n)$ be the isomorphism from $\mathcal{F}_j(U'_i \cap U'_j)$ onto $\mathcal{F}_i(U'_i \cap U'_j)$ defined by multiplication by $(t_j/t_i)^n$. We denote by $\mathcal{F}(n)$ the sheaf obtained from the $\mathcal{F}_i$ by gluing by means of the $\theta_{ij}(n)$. The operation $\mathcal{F}(n)$ has the same properties as the one defined in no. 54, which it generalizes; in particular, $\mathcal{F}(n)$ is canonically isomorphic to $\mathcal{F} \otimes \mathcal{O}_V(n)$.

We have $\mathcal{F}^X(n) = \mathcal{F}(n)^X$. Applying Theorem 1 of no. 55, together with Proposition 7 of no. 65, we obtain:

Theorem 2. *Let $\mathcal{F}$ be a coherent algebraic sheaf on V. There exists an integer $m(\mathcal{F})$ such that, for every $n \geq m(\mathcal{F})$:*

(a) For every $x \in V$, the $\mathcal{O}_{x,V}$-module $\mathcal{F}(n)_x$ is generated by the elements of $\Gamma(V, \mathcal{F}(n))$;

(b) $H^q(V, \mathcal{F}(n)) = 0$ for every $q > 0$.

Remark. It is essential to observe that the sheaf $\mathcal{F}(n)$ depends not only on $\mathcal{F}$ and n, but also on the *embedding* of V into the projective space X. More precisely, let P be the principal fiber space $\pi^{-1}(V)$, whose structure group is K^*; for an integer n, let K^* act on K by the formula

$$(\lambda, \mu) \longrightarrow \lambda^{-n}\mu \qquad \text{if} \qquad \lambda \in K^* \quad \text{and} \quad \mu \in K.$$

Let $E^n = P \times_{K^*} K$ be the fiber space associated with P, with typical fiber K and endowed with the preceding operators; let $\mathcal{S}(E^n)$ be the sheaf of germs of sections of E^n (cf. no. 41). Taking account of the fact that the t_i/t_j form a system of changes of charts for P, we immediately verify that $\mathcal{S}(E^n)$ is canonically isomorphic to $\mathcal{O}_V(n)$. The formula

$$\mathcal{F}(n) = \mathcal{F} \otimes \mathcal{O}_V(n) = \mathcal{F} \otimes \mathcal{S}(E^n)$$

then shows that the operation $\mathcal{F} \longmapsto \mathcal{F}(n)$ depends only on *the class of the fiber space P defined by the embedding $V \to X$*. In particular, if V is normal, $\mathcal{F}(n)$ depends only on the linear-equivalence class of the hyperplane sections of V in the embedding under consideration (cf. [17]).

67. A supplement.

If M is a graded S-module satisfying (TF), we shall denote by $M^\natural$ the graded S-module $\Gamma(\mathcal{A}(M))$. We saw in no. 65 that $\alpha : M \longrightarrow M^\natural$ is $\mathcal{C}$-bijective. We shall give conditions under which α is bijective.

Proposition 9. *The homomorphism $\alpha : M \longrightarrow M^\natural$ is bijective if and only if the following conditions hold:*

(i) If $m \in M$ is such that $t_i \cdot m = 0$ for every $i \in I$, then $m = 0$.

(ii) If homogeneous elements $m_i \in M$ of the same degree satisfy $t_j \cdot m_i - t_i \cdot m_j = 0$ for every pair (i, j), then there exists $m \in M$ such that $m_i = t_i \cdot m$.

Let us show that conditions (i) and (ii) hold for $M^\natural$, which will prove their necessity. For (i), we may suppose that m is homogeneous, that is, that it is a section of $\mathcal{A}(M(n))$; in this case, the condition $t_i \cdot m = 0$ implies that m is zero on U_i, and, since this holds for every $i \in I$, we indeed have $m = 0$. For (ii), let n be the degree of the m_i; thus $m_i \in \Gamma(\mathcal{A}(M(n)))$. Since $1/t_i$ is a section of $\mathcal{O}(-1)$ on U_i, m_i/t_i is a section of $\mathcal{A}(M(n-1))$ on U_i, and the condition $t_j \cdot m_i - t_i \cdot m_j = 0$ shows that these various sections are the restrictions of a unique section m of $\mathcal{A}(M(n-1))$ on X. It remains to compare the sections $t_i \cdot m$ and m_i; to show that they coincide on U_j, it suffices to see that $t_j(t_i \cdot m - m_i) = 0$ on U_j, which follows from the formula $t_j \cdot m_i = t_i \cdot m_j$ and the definition of m.

Let us now show that (i) implies that α is injective. For all sufficiently large n, we know that $\alpha : M_n \longrightarrow M^\natural_n$ is bijective, and we may therefore argue by descending induction on n. If $\alpha(m) = 0$, with $m \in M_n$, then $t_i\alpha(m) = \alpha(t_i \cdot m) = 0$. The induction hypothesis applies because $t_i \cdot m \in M_{n+1}$, and gives $t_i \cdot m = 0$ for every i; condition (i) then gives $m = 0$.

Finally, let us show that (i) and (ii) imply that α is surjective. As before, we may argue by descending induction on n. If $m' \in M^\natural_n$, the induction hypothesis shows that there exist $m_i \in M_{n+1}$ such that $\alpha(m_i) = t_i \cdot m'$. We have $\alpha(t_j \cdot m_i - t_i \cdot m_j) = 0$, whence $t_j \cdot m_i - t_i \cdot m_j = 0$, since α is injective. Condition (ii) then implies the existence of $m \in M_n$ such that $t_i \cdot m = m_i$; we have $t_i(m' - \alpha(m)) = 0$ for every i, and condition (i), already proved for $M^\natural$, shows that $m' = \alpha(m)$ and completes the proof.

Remarks. (1) The proof shows that condition (i) is necessary and sufficient for α to be injective.

(2) Conditions (i) and (ii) may be expressed as follows:
the homomorphism $\alpha^1 : M_n \longrightarrow H^0_1(M(n))$ is bijective for every $n \in \mathbf{Z}$. Moreover, Proposition 4 shows that $M^\natural$ may be identified with the graded S-module $\sum_{n\in\mathbf{Z}} H^0(M(n))$, and it would be easy to deduce from this a purely algebraic proof of Proposition 9 (without using the sheaf $\mathcal{A}(M)$).

§4. Relations with the functors Ext^q_S

68. The functors Ext^q_S. We retain the notation of no. 56. If M and N are two graded S-modules, we shall denote by $\mathrm{Hom}_S(M,N)_n$ the group of homogeneous S-homomorphisms of degree n from M to N, and by $\mathrm{Hom}_S(M,N)$ the graded group

$$\sum_{n\in\mathbf{Z}} \mathrm{Hom}_S(M,N)_n.$$

It is a graded S-module; when M is of finite type, it coincides with the S-module of all S-homomorphisms from M to N.

The derived functors (cf. [6], Chapter V) of the functor $\mathrm{Hom}_S(M,N)$ are the functors $\mathrm{Ext}^q_S(M,N)$, $q = 0, 1, \ldots$. Let us briefly recall their definition.[20]

One chooses a "resolution" of M, that is, an exact sequence

$$\cdots \longrightarrow L^{q+1} \longrightarrow L^q \longrightarrow \cdots \longrightarrow L^0 \longrightarrow M \longrightarrow 0,$$

where the L^q are free graded S-modules and the maps are homomorphisms (that is, as usual, homogeneous S-homomorphisms of degree 0). Put $C^q = \mathrm{Hom}_S(L^q, N)$. The homomorphism $L^{q+1} \longrightarrow L^q$ defines by transposition a homomorphism $d : C^q \longrightarrow C^{q+1}$ satisfying $d \circ d = 0$. Thus $C = \sum_{q\geq 0} C^q$ has the structure of a complex, and its q-th cohomology group is, by definition, $\mathrm{Ext}^q_S(M,N)$; one shows that it does not depend on the chosen resolution. Since the C^q are graded S-modules and $d : C^q \longrightarrow C^{q+1}$ is homogeneous of degree 0, the $\mathrm{Ext}^q_S(M,N)$ are graded S-modules, with homogeneous subspaces $\mathrm{Ext}^q_S(M,N)_n$. These latter spaces are the cohomology groups of the complex formed by the $\mathrm{Hom}_S(L^q,N)_n$; in other words, they are the derived functors of $\mathrm{Hom}_S(M,N)_n$.

Let us recall the principal properties of Ext^q_S: $\mathrm{Ext}^0_S(M,N) = \mathrm{Hom}_S(M,N)$; $\mathrm{Ext}^q_S(M,N) = 0$ for $q > r+1$ if M is of finite type (by Hilbert's syzygy theorem, cf. [6], Chapter VIII, Theorem 6.5); $\mathrm{Ext}^q_S(M,N)$ is an S-module of finite type if M and N are of finite type (for one may choose a resolution in which the L^q are of finite type); and for every $n \in \mathbf{Z}$ there are canonical isomorphisms

$$\mathrm{Ext}^q_S(M(n),N) \simeq \mathrm{Ext}^q_S(M,N(-n)) \simeq \mathrm{Ext}^q_S(M,N)(-n).$$

The exact sequences

$$0 \longrightarrow N \longrightarrow N' \longrightarrow N'' \longrightarrow 0 \qquad \text{and} \qquad 0 \longrightarrow M \longrightarrow M' \longrightarrow M'' \longrightarrow 0$$

[20] When M is not a module of finite type, the $\mathrm{Ext}^q_S(M,N)$ defined here may differ from the $\mathrm{Ext}^q_S(M,N)$ defined in [6], because $\mathrm{Hom}_S(M,N)$ does not have the same meaning in the two cases. Nevertheless, all the proofs in [6] remain valid without change in the case considered here; this may be seen either directly or by applying the Appendix of [6].

give rise to the exact sequences

$$\cdots \longrightarrow \mathrm{Ext}_S^q(M,N) \longrightarrow \mathrm{Ext}_S^q(M,N') \longrightarrow \mathrm{Ext}_S^q(M,N'') \longrightarrow \mathrm{Ext}_S^{q+1}(M,N) \longrightarrow \cdots$$

and

$$\cdots \longrightarrow \mathrm{Ext}_S^q(M'',N) \longrightarrow \mathrm{Ext}_S^q(M',N) \longrightarrow \mathrm{Ext}_S^q(M,N) \longrightarrow \mathrm{Ext}_S^{q+1}(M'',N) \longrightarrow \cdots .$$

69. Interpretation of $H_k^q(M)$ by means of Ext_S^q. Let M be a graded S-module, and let k be an integer ≥ 0. Put

$$B_k^q(M) = \sum_{n\in\mathbf{Z}} H_k^q(M(n)),$$

with the notation of no. 61.

This gives a graded group, isomorphic to the q-th cohomology group of the complex $\sum_{n\in\mathbf{Z}} C_k(M(n))$. This complex may be given an S-module structure compatible with its grading by putting

$$(P\cdot m)\langle i_0\cdots i_q\rangle = P\cdot m\langle i_0\cdots i_q\rangle, \quad \text{if } P\in S_p \text{ and } m\langle i_0\cdots i_q\rangle \in C_k^q(M(n)).$$

Since the coboundary operator is a homogeneous S-homomorphism of degree 0, it follows that the $B_k^q(M)$ are themselves graded S-modules.

We shall put

$$B^q(M) = \varinjlim_{k\to\infty} B_k^q(M) = \sum_{n\in\mathbf{Z}} H^q(M(n)).$$

The $B^q(M)$ are graded S-modules. For $q=0$ we have

$$B^0(M) = \sum_{n\in\mathbf{Z}} H^0(M(n)),$$

and recover the module denoted by $M^\natural$ in no. 67 (when M satisfies condition (TF)). For each $n\in\mathbf{Z}$, we defined in no. 62 a linear map $\alpha : M_n \longrightarrow H^0(M(n))$; it follows at once that the sum of these maps defines a homomorphism, again denoted by α, from M to $B^0(M)$.

Proposition 1. *Let k be an integer ≥ 0, and let J_k be the ideal $(t_0^k,\ldots,t_r^k)$ of S. For every graded S-module M, the graded S-modules $B_k^q(M)$ and $\mathrm{Ext}_S^q(J_k,M)$ are isomorphic.*

Let L_k^q, $q=0,\ldots,r$, be the free graded S-module with basis elements $e\langle i_0\cdots i_q\rangle$, $0\leq i_0 < i_1 < \cdots < i_q \leq r$, of degree $k(q+1)$. Define an operator $d : L_k^{q+1} \longrightarrow L_k^q$ and an operator $\varepsilon : L_k^0 \longrightarrow J_k$ by the formulas

$$d(e\langle i_0\cdots i_{q+1}\rangle) = \sum_{j=0}^{j=q+1} (-1)^j t_{i_j}^k \cdot e\langle i_0\cdots \widehat{\imath}_j\cdots i_{q+1}\rangle$$

and

$$\varepsilon(e\langle i\rangle) = t_i^k.$$

Lemma 1. *The sequence of homomorphisms*

$$0 \longrightarrow L_k^r \xrightarrow{d} L_k^{r-1} \longrightarrow \cdots \longrightarrow L_k^0 \xrightarrow{\varepsilon} J_k \longrightarrow 0$$

is exact.

For $k=1$ this result is well known (cf. [6], Chapter VIII, §4); the general case is proved in the same way (or reduced to it). One may also use the theorem proved in [11].

Proposition 1 follows immediately from the lemma, on observing that the complex formed by the $\mathrm{Hom}_S(L_k^q,M)$ and the transpose of d is precisely the complex $\sum_{n\in\mathbf{Z}} C_k(M(n))$.

Corollary 1. *$H_k^q(M)$ is isomorphic to $\mathrm{Ext}_S^q(J_k,M)_0$.*

Indeed, these two groups are the degree-zero components of the graded groups $B^q_k(M)$ and $\mathrm{Ext}^q_S(J_k, M)$.

Corollary 2. *$H^q(M)$ is isomorphic to* $\varinjlim_{k\to\infty} \mathrm{Ext}^q_S(J_k, M)_0$.

It is readily seen that the homomorphism $\rho^h_k : H^q_k(M) \longrightarrow H^q_h(M)$ of no. 61 is carried by the isomorphism of Corollary 1 to the homomorphism

$$\mathrm{Ext}^q_S(J_k, M)_0 \longrightarrow \mathrm{Ext}^q_S(J_h, M)_0$$

defined by the inclusion $J_h \longrightarrow J_k$; this proves Corollary 2.

Remark. Let M be a graded S-module of finite type. By no. 48, M defines a coherent algebraic sheaf $\mathfrak{F}'$ on K^{r+1}, hence on $Y = K^{r+1} - \{0\}$, and one may verify that $H^q(Y, \mathfrak{F}')$ is isomorphic to $B^q(M)$.

70. Definition of the functors $T^q(M)$. We first define the notion of the *dual* module of a graded S-module. Let M be a graded S-module. For every $n \in \mathbf{Z}$, M_n is a vector space over K; we denote its dual vector space by $(M_n)'$. Put

$$M^* = \sum_{n\in\mathbf{Z}} M^*_n, \qquad M^*_n = (M_{-n})'.$$

We shall give M^* an S-module structure compatible with its grading. For every $P \in S_p$, the map $m \mapsto Pm$ is a K-linear map from M_{-n-p} to M_{-n} and therefore defines by transposition a K-linear map from $(M_{-n})' = M^*_n$ to $(M_{-n-p})' = M^*_{n+p}$. This defines the S-module structure of M^*. We could equally have defined

$$M^* = \mathrm{Hom}_S(M, K), \qquad K = S/(t_0, \dots, t_r),$$

where K denotes the indicated graded S-module.

The graded S-module M^* is called the *dual* of M. We have $M^{**} = M$ if each M_n is finite-dimensional over K; this is the case if $M = \Gamma(\mathcal{F})$, with $\mathcal{F}$ a coherent algebraic sheaf on X, or if M is of finite type. Every homomorphism $\varphi : M \longrightarrow N$ defines by transposition a homomorphism from N^* to M^*.

If $M \longrightarrow N \longrightarrow P$ is exact, then $P^* \longrightarrow N^* \longrightarrow M^*$ is exact as well. In other words, $M \mapsto M^*$ is a contravariant exact functor. When I is a homogeneous ideal of S, the dual of S/I is precisely the “inverse system” of I in the sense of Macaulay (cf. [9], no. 25).

Now let M be a graded S-module, and let q be an integer ≥ 0. In the preceding number we defined the graded S-module $B^q(M)$; its dual module will be denoted by $T^q(M)$. Thus, by definition,

$$T^q(M) = \sum_{n\in\mathbf{Z}} T^q(M)_n, \qquad T^q(M)_n = (H^q(M(-n)))'.$$

Every homomorphism $\varphi : M \longrightarrow N$ defines a homomorphism from $B^q(M)$ to $B^q(N)$, hence a homomorphism from $T^q(N)$ to $T^q(M)$. Thus the $T^q(M)$ are contravariant functors of M (we shall see in no. 72 that they can be expressed very simply in terms of Ext_S).

Every exact sequence

$$0 \longrightarrow M \longrightarrow N \longrightarrow P \longrightarrow 0$$

gives rise to an exact sequence

$$\cdots \longrightarrow B^q(M) \longrightarrow B^q(N) \longrightarrow B^q(P) \longrightarrow B^{q+1}(M) \longrightarrow \cdots,$$

and hence, by transposition, to an exact sequence

$$\cdots \longrightarrow T^{q+1}(M) \longrightarrow T^q(P) \longrightarrow T^q(N) \longrightarrow T^q(M) \longrightarrow \cdots.$$

The homomorphism $\alpha : M \longrightarrow B^0(M)$ defines by transposition a homomorphism

$$\alpha^* : T^0(M) \longrightarrow M^*.$$

Since $B^q(M) = 0$ for $q > r$, we have

$$T^q(M) = 0 \qquad (q > r).$$

71. Determination of $T^r(M)$. (In this number, as in the following one, we shall assume that $r \geq 1$; the case $r = 0$ leads to somewhat different, though trivial, statements.)

We shall denote by Ω the graded S-module $S(-r-1)$; it is a free module with a basis element of degree $r+1$. We saw in no. 62 that $H^r(\Omega) = H^r_k(\Omega)$ for all sufficiently large k, and that $H^r_k(\Omega)$ has over K the basis consisting of the single element

$$\frac{(t_0 \cdots t_r)^k}{t_0 \cdots t_r}.$$

The image of this element in $H^r(\Omega)$ will be denoted by ξ; thus ξ is a basis of $H^r(\Omega)$.

We shall now define a scalar product $\langle h, \varphi \rangle$ between elements

$$h \in B^r(M)_{-n} \qquad \text{and} \qquad \varphi \in \mathrm{Hom}_S(M, \Omega)_n,$$

where M is an arbitrary graded S-module. The element φ can be identified with an element of $\mathrm{Hom}_S(M(-n), \Omega)_0$, that is, with a homomorphism from $M(-n)$ to Ω. Passing to cohomology groups, it therefore defines a homomorphism

$$H^r(M(-n)) = B^r(M)_{-n} \longrightarrow H^r(\Omega),$$

which we shall again denote by φ. The image of h under this homomorphism is a scalar multiple of ξ, and we define $\langle h, \varphi \rangle$ by the formula

$$\varphi(h) = \langle h, \varphi \rangle \, \xi.$$

For every $\varphi \in \mathrm{Hom}_S(M, \Omega)_n$, the function $h \mapsto \langle h, \varphi \rangle$ is a linear form on $B^r(M)_{-n}$, and hence can be identified with an element $\nu(\varphi)$ of the dual of $B^r(M)_{-n}$, which is precisely $T^r(M)_n$. We have thus defined a homogeneous map of degree zero

$$\nu : \mathrm{Hom}_S(M, \Omega) \longrightarrow T^r(M),$$

and the formula

$$\langle Ph, \varphi \rangle = \langle h, P\varphi \rangle$$

shows that ν is an S-homomorphism.

Proposition 2. *The homomorphism $\nu : \mathrm{Hom}_S(M, \Omega) \longrightarrow T^r(M)$ is bijective.*

We first prove the proposition when M is a *free* module. If M is a direct sum of homogeneous submodules M^α, then

$$\mathrm{Hom}_S(M, \Omega)_n = \prod_\alpha \mathrm{Hom}_S(M^\alpha, \Omega)_n, \qquad T^r(M)_n = \prod_\alpha T^r(M^\alpha)_n.$$

Thus, if the proposition is true for the M^α, it is true for M. This reduces the case of free modules to that of a free module on one generator, that is, to the case $M = S(m)$.

We can then identify $\mathrm{Hom}_S(M, \Omega)_n$ with $\mathrm{Hom}_S(S, S(n-m-r-1))_0$, that is, with the vector space of homogeneous polynomials of degree $n-m-r-1$. Hence $\mathrm{Hom}_S(M, \Omega)_n$ has as a basis the monomials

$$t_0^{\gamma_0} \cdots t_r^{\gamma_r}, \qquad \gamma_i \geq 0, \qquad \sum_{i=0}^{i=r} \gamma_i = n-m-r-1.$$

On the other hand, we saw in no. 62 that $H^r_k(S(m-n))$ has as a basis, for all sufficiently large k, the monomials

$$\frac{(t_0 \cdots t_r)^k}{t_0^{\beta_0} \cdots t_r^{\beta_r}}, \qquad \beta_i > 0, \qquad \sum_{i=0}^{i=r} \beta_i = n-m.$$

Putting $\beta_i = \gamma'_i + 1$, these monomials can be written as

$$\frac{(t_0 \cdots t_r)^{k-1}}{t_0^{\gamma'_0} \cdots t_r^{\gamma'_r}}, \qquad \gamma'_i \geq 0, \qquad \sum_{i=0}^{i=r} \gamma'_i = n - m - r - 1.$$

Returning to the definition of $\langle h, \varphi \rangle$, we find that the scalar product

$$\left\langle \frac{(t_0 \cdots t_r)^{k-1}}{t_0^{\gamma'_0} \cdots t_r^{\gamma'_r}}, t_0^{\gamma_0} \cdots t_r^{\gamma_r} \right\rangle$$

is always zero unless $\gamma_i = \gamma'_i$ for every i, in which case it is equal to 1. This means that ν sends the basis of the $t_0^{\gamma_0} \cdots t_r^{\gamma_r}$ to the basis dual to the basis of the $(t_0 \cdots t_r)^{k-1}/(t_0^{\gamma_0} \cdots t_r^{\gamma_r})$. Consequently ν is bijective, which proves the proposition when M is free.

We now pass to the general case. Choose an exact sequence

$$L^1 \longrightarrow L^0 \longrightarrow M \longrightarrow 0$$

in which L^0 and L^1 are free. Consider the following commutative diagram:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathrm{Hom}_S(M, \Omega) & \longrightarrow & \mathrm{Hom}_S(L^0, \Omega) & \longrightarrow & \mathrm{Hom}_S(L^1, \Omega) \\
{\scriptstyle\nu}\big\downarrow & & {\scriptstyle\nu}\big\downarrow & & {\scriptstyle\nu}\big\downarrow & & {\scriptstyle\nu}\big\downarrow \\
0 & \longrightarrow & T^r(M) & \longrightarrow & T^r(L^0) & \longrightarrow & T^r(L^1).
\end{array}$$

The first row of this diagram is exact by the general properties of the functor Hom_S; the second row is also exact, because it is dual to the sequence

$$B^r(L^1) \longrightarrow B^r(L^0) \longrightarrow B^r(M) \longrightarrow 0,$$

which is itself exact by the exact cohomology sequence for the B^q and the fact that $B^{r+1}(M) = 0$ for every graded S-module M. Moreover, the two vertical homomorphisms

$$\nu : \mathrm{Hom}_S(L^0, \Omega) \longrightarrow T^r(L^0) \quad \text{and} \quad \nu : \mathrm{Hom}_S(L^1, \Omega) \longrightarrow T^r(L^1)$$

are bijective, as we have just seen. It follows that

$$\nu : \mathrm{Hom}_S(M, \Omega) \longrightarrow T^r(M)$$

is also bijective, which completes the proof.

72. Determination of the modules $T^q(M)$. We shall prove the following theorem, which generalizes Proposition 2.

THEOREM 1. *Let M be a graded S-module. For $q \neq r, r+1$,*[21] *the graded S-modules $T^{r-q}(M)$ and $\mathrm{Ext}^q_S(M, \Omega)$ are isomorphic. Moreover, there is an exact sequence*

$$0 \longrightarrow \mathrm{Ext}^r_S(M, \Omega) \longrightarrow T^0(M) \xrightarrow{\alpha^*} M^* \longrightarrow \mathrm{Ext}^{r+1}_S(M, \Omega) \longrightarrow 0.$$

We shall use the axiomatic characterization of derived functors given in [6], Chapter III, §5. To this end, first define new functors $E^q(M)$ as follows:

$$\begin{aligned}
&\text{For } q \neq r, r+1, && E^q(M) = T^{r-q}(M), \\
&\text{For } q = r, && E^r(M) = \mathrm{Ker}(\alpha^*), \\
&\text{For } q = r+1, && E^{r+1}(M) = \mathrm{Coker}(\alpha^*).
\end{aligned}$$

[21]The printed statement says only $q \neq r$. The immediately following definition treats both $q = r$ and $q = r+1$ separately, with $E^{r+1}(M) = \mathrm{Coker}(\alpha^*)$, and the displayed exact sequence identifies that cokernel with $\mathrm{Ext}^{r+1}_S(M, \Omega)$. Thus $r+1$ is necessarily also excluded from the direct T^{r-q} isomorphism. Corrected French and English restore the omitted exception; diplomatic French preserves the print.

The $E^q(M)$ are additive contravariant functors of M with the following properties.

(i) $E^0(M)$ is isomorphic to $\mathrm{Hom}_S(M, \Omega)$.

This is the assertion of Proposition 2.

(ii) If L is free, $E^q(L) = 0$ for every $q > 0$.

It suffices to verify this for $L = S(n)$, in which case it follows from no. 62.

(iii) With every exact sequence $0 \longrightarrow M \longrightarrow N \longrightarrow P \longrightarrow 0$ there is associated a sequence of coboundary operators $d^q : E^q(M) \longrightarrow E^{q+1}(P)$, and the sequence

$$\cdots \longrightarrow E^q(P) \longrightarrow E^q(N) \longrightarrow E^q(M) \xrightarrow{d^q} E^{q+1}(P) \longrightarrow \cdots$$

is exact.

The definition of d^q is evident if $q \neq r-1, r$: it is the homomorphism from $T^{r-q}(M)$ to $T^{r-q-1}(P)$ defined in no. 70. For $q = r-1$ or r, use the following commutative diagram:

$$\begin{tikzcd}
T^1(M) \arrow[r] & T^0(P) \arrow[r] \arrow[d, "\alpha^*"'] & T^0(N) \arrow[r] \arrow[d, "\alpha^*"'] & T^0(M) \arrow[r] \arrow[d, "\alpha^*"'] & 0 \arrow[d, "\alpha^*"'] \\
0 \arrow[r] & P^* \arrow[r] & N^* \arrow[r] & M^* \arrow[r] & 0.
\end{tikzcd}$$

This diagram first shows that the image of $T^1(M)$ is contained in the kernel of $\alpha^* : T^0(P) \longrightarrow P^*$, which is precisely $E^r(P)$. This defines

$$d^{r-1} : E^{r-1}(M) \longrightarrow E^r(P).$$

To define

$$d^r : \mathrm{Ker}(T^0(M) \longrightarrow M^*) \longrightarrow \mathrm{Coker}(T^0(P) \longrightarrow P^*),$$

we use the procedure of [6], Chapter III, Lemma 3.3. If $x \in \mathrm{Ker}(T^0(M) \longrightarrow M^*)$, there exist $y \in P^*$ and $z \in T^0(N)$ such that x is the image of z and y and z have the same image in N^*; one then puts $d^r(x) = y$.

The exactness of the sequence in (iii) follows from the exactness of

$$\cdots \longrightarrow T^{r-q}(P) \longrightarrow T^{r-q}(N) \longrightarrow T^{r-q}(M) \longrightarrow T^{r-q-1}(P) \longrightarrow \cdots$$

and from [6], loc. cit.

(iv) The isomorphism in (i) and the operators d^q in (iii) are "natural."

This follows immediately from their definitions.

Since properties (i)–(iv) characterize the derived functors of $\mathrm{Hom}_S(M, \Omega)$, we have

$$E^q(M) \simeq \mathrm{Ext}^q_S(M, \Omega),$$

which proves the theorem.

Corollary 1. *If M satisfies condition (TF), then $H^q(M)$ is isomorphic to the vector-space dual of* $\mathrm{Ext}^{r-q}_S(M, \Omega)_0$ *for every $q \geq 1$.*

Indeed, we know that $H^q(M)$ is a finite-dimensional vector space whose dual is isomorphic to $\mathrm{Ext}^{r-q}_S(M, \Omega)_0$.

Corollary 2. *If M satisfies condition (TF), the $T^q(M)$ are graded S-modules of finite type for $q \geq 1$, and $T^0(M)$ satisfies condition (TF).*

One may replace M by a module of finite type without changing the $B^q(M)$, and hence without changing the $T^q(M)$. The $\mathrm{Ext}^{r-q}_S(M, \Omega)$ are then S-modules of finite type, and $M^* \in \mathfrak{C}$, which proves the corollary.

73. Relations between the functors Ext^q_S and $\mathrm{Ext}^q_{\mathcal{O}_x}$. Let M and N be two graded S-modules. If x is a point of $X = \mathbf{P}_r(K)$, we defined the $\mathcal{O}_x$-modules M_x and N_x in no. 57; we shall relate the $\mathrm{Ext}^q_{\mathcal{O}_x}(M_x, N_x)$ to the graded S-module $\mathrm{Ext}^q_S(M, N)$.

Proposition 1. *Suppose that M is finitely generated. Then:*

(a) The sheaf $\mathfrak{A}(\mathrm{Hom}_S(M,N))$ *is isomorphic to the sheaf* $\mathrm{Hom}_{\mathcal{O}}(\mathfrak{A}(M),\mathfrak{A}(N))$.

(b) For every $x \in X$, *the* $\mathcal{O}_x$*-module* $\mathrm{Ext}^q_S(M,N)_x$ *is isomorphic to the* $\mathcal{O}_x$*-module* $\mathrm{Ext}^q_{\mathcal{O}_x}(M_x,N_x)$.

First define a homomorphism

$$\iota_x : \mathrm{Hom}_S(M,N)_x \longrightarrow \mathrm{Hom}_{\mathcal{O}_x}(M_x,N_x).$$

An element of the first module is a fraction φ/P, where $\varphi \in \mathrm{Hom}_S(M,N)_n$, $P \in S(x)$, and P is homogeneous of degree n. If m/P' is an element of M_x, then $\varphi(m)/(PP')$ is an element of N_x depending only on φ/P and m/P', and the assignment

$$\frac{m}{P'} \longmapsto \frac{\varphi(m)}{PP'}$$

is a homomorphism $\iota_x(\varphi/P) : M_x \longrightarrow N_x$; this defines ι_x. By Proposition 5 of no. 14, $\mathrm{Hom}_{\mathcal{O}_x}(M_x,N_x)$ may be identified with

$$\mathrm{Hom}_{\mathcal{O}}(\mathfrak{A}(M),\mathfrak{A}(N))_x.$$

Under this identification, ι_x becomes

$$\iota_x : \mathfrak{A}(\mathrm{Hom}_S(M,N))_x \longrightarrow \mathrm{Hom}_{\mathcal{O}}(\mathfrak{A}(M),\mathfrak{A}(N))_x,$$

and one readily checks that the collection of the ι_x is a homomorphism

$$\iota : \mathfrak{A}(\mathrm{Hom}_S(M,N)) \longrightarrow \mathrm{Hom}_{\mathcal{O}}(\mathfrak{A}(M),\mathfrak{A}(N)).$$

When M is a finitely generated free module, ι_x is bijective: indeed, it suffices to verify this when $M = S(n)$, in which case it is immediate.

Now let M be an arbitrary finitely generated graded S-module, and choose a resolution of M:

$$\cdots \longrightarrow L^{q+1} \longrightarrow L^q \longrightarrow \cdots \longrightarrow L^0 \longrightarrow M \longrightarrow 0,$$

in which the L^q are finitely generated free modules. Consider the complex C formed by the $\mathrm{Hom}_S(L^q,N)$. Its cohomology groups are the $\mathrm{Ext}^q_S(M,N)$. In other words, if B^q and Z^q denote the submodules of C^q consisting respectively of coboundaries and cocycles, there are exact sequences

$$0 \longrightarrow Z^q \longrightarrow C^q \longrightarrow B^{q+1} \longrightarrow 0$$

and

$$0 \longrightarrow B^q \longrightarrow Z^q \longrightarrow \mathrm{Ext}^q_S(M,N) \longrightarrow 0.$$

Since the functor $M \mapsto \mathfrak{A}(M)$ is exact, the sequences

$$0 \longrightarrow Z^q_x \longrightarrow C^q_x \longrightarrow B^{q+1}_x \longrightarrow 0$$

and

$$0 \longrightarrow B^q_x \longrightarrow Z^q_x \longrightarrow \mathrm{Ext}^q_S(M,N)_x \longrightarrow 0$$

are also exact.

But, by what precedes, C^q_x is isomorphic to $\mathrm{Hom}_{\mathcal{O}_x}(L^q_x,N_x)$. The modules $\mathrm{Ext}^q_S(M,N)_x$ are therefore isomorphic to the cohomology groups of the complex formed by the $\mathrm{Hom}_{\mathcal{O}_x}(L^q_x,N_x)$; and since the L^q_x are plainly free $\mathcal{O}_x$-modules, this is precisely the definition of $\mathrm{Ext}^q_{\mathcal{O}_x}(M_x,N_x)$. This proves (b). For $q=0$, the preceding argument shows that ι_x is bijective; hence ι is an isomorphism, which proves (a).

74. Vanishing of the cohomology groups $H^q(X,\mathfrak{F}(-n))$ as $n \longrightarrow +\infty$.

THEOREM 1. *Let $\mathfrak{F}$ be a coherent algebraic sheaf on X, and let q be an integer ≥ 0. The following two conditions are equivalent:*

(a) $H^q(X, \mathfrak{F}(-n)) = 0$ for all sufficiently large n.

(b) $\mathrm{Ext}^{r-q}_{\mathcal{O}_x}(\mathfrak{F}_x, \mathcal{O}_x) = 0$ for every $x \in X$.

By Theorem 2 of no. 60, we may suppose that $\mathfrak{F} = \mathfrak{A}(M)$, where M is a finitely generated graded S-module; and by no. 64, $H^q(X, \mathfrak{F}(-n))$ is isomorphic to $H^q(M(-n)) = B^q(M)_{-n}$. Thus condition (a) is equivalent to

$$T^q(M)_n = 0$$

for all sufficiently large n, that is, to $T^q(M) \in \mathfrak{C}$. By Theorem 1 of no. 72 and the fact that $M^* \in \mathfrak{C}$, since M is finitely generated, this last condition is equivalent to $\mathrm{Ext}^{r-q}_S(M, \Omega) \in \mathfrak{C}$. Since $\mathrm{Ext}^{r-q}_S(M, \Omega)$ is a finitely generated S-module,

$$\mathrm{Ext}^{r-q}_S(M, \Omega) \in \mathfrak{C}$$

is equivalent, by Proposition 5 of no. 58, to $\mathrm{Ext}^{r-q}_S(M, \Omega)_x = 0$ for every $x \in X$. Finally, Proposition 1 shows that

$$\mathrm{Ext}^{r-q}_S(M, \Omega)_x = \mathrm{Ext}^{r-q}_{\mathcal{O}_x}(M_x, \Omega_x).$$

Since M_x is isomorphic to $\mathfrak{F}_x$, and Ω_x is isomorphic to $\mathcal{O}(-r-1)_x$, hence to $\mathcal{O}_x$, this completes the proof.

To state Theorem 2, we shall need the notion of the *dimension* of an $\mathcal{O}_x$-module. Recall ([6], Chapter VI, §2) that a finitely generated $\mathcal{O}_x$-module P is said to have dimension $\leq p$ if there is an exact sequence of $\mathcal{O}_x$-modules

$$0 \longrightarrow L_p \longrightarrow L_{p-1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow P \longrightarrow 0,$$

where each L_i is free.[22] This definition is equivalent to that of [6], loc. cit., because every finitely generated projective $\mathcal{O}_x$-module is free (cf. [6], Chapter VIII, Theorem 6.1′).

Every finitely generated $\mathcal{O}_x$-module has dimension $\leq r$, by the syzygy theorem (cf. [6], Chapter VIII, Theorem 6.2′).

LEMMA 1. *Let P be a finitely generated $\mathcal{O}_x$-module, and let p be an integer ≥ 0. The following two conditions are equivalent:*

(i) P has dimension $\leq p$.

(ii) $\mathrm{Ext}^m_{\mathcal{O}_x}(P, \mathcal{O}_x) = 0$ for every $m > p$.

It is clear that (i) implies (ii). We prove that (ii) implies (i) by descending induction on p. For $p \geq r$, the lemma is trivial, since (i) always holds. Now pass from $p+1$ to p, and let N be an arbitrary finitely generated $\mathcal{O}_x$-module. There is an exact sequence

$$0 \longrightarrow R \longrightarrow L \longrightarrow N \longrightarrow 0$$

in which L is finitely generated and free (because $\mathcal{O}_x$ is noetherian). The exact sequence

$$\mathrm{Ext}^{p+1}_{\mathcal{O}_x}(P, L) \longrightarrow \mathrm{Ext}^{p+1}_{\mathcal{O}_x}(P, N) \longrightarrow \mathrm{Ext}^{p+2}_{\mathcal{O}_x}(P, R)$$

shows that $\mathrm{Ext}^{p+1}_{\mathcal{O}_x}(P, N) = 0$. Indeed, $\mathrm{Ext}^{p+1}_{\mathcal{O}_x}(P, L) = 0$ by condition (ii), and $\mathrm{Ext}^{p+2}_{\mathcal{O}_x}(P, R) = 0$ because $\dim P \leq p+1$ by the induction hypothesis. Since this property characterizes the modules of dimension $\leq p$, the lemma is proved.

Combining the lemma with Theorem 1 gives:

THEOREM 2. *Let $\mathfrak{F}$ be a coherent algebraic sheaf on X, and let p be an integer ≥ 0. The following two conditions are equivalent:*

(a) $H^q(X, \mathfrak{F}(-n)) = 0$ for all sufficiently large n and $0 \leq q < p$.

[22] The printed French says "where each L_p is free." The displayed resolution contains $L_p, L_{p-1}, \ldots, L_0$, so the definition requires every term L_i to be free. Diplomatic French preserves L_p; corrected French and English restore the quantified index i.

(b) For every $x \in X$, the $\mathcal{O}_x$-module $\mathfrak{F}_x$ has dimension $\leq r - p$.

75. Nonsingular varieties. The following result plays an essential role in extending the "duality theorem" of [15] to the abstract case:

Theorem 3. *Let V be a nonsingular subvariety of the projective space $P_r(K)$; suppose that all the irreducible components of V have the same dimension p. Let $\mathfrak{F}$ be a coherent algebraic sheaf on V such that, for every $x \in V$, $\mathfrak{F}_x$ is a free module over $\mathcal{O}_{x,V}$. Then $H^q(V, \mathfrak{F}(-n)) = 0$ for all sufficiently large n and $0 \leq q < p$.*

By Theorem 2, it suffices to show that $\mathcal{O}_{x,V}$, regarded as an $\mathcal{O}_x$-module, has dimension $\leq r-p$. Let $\mathcal{I}_x(V)$ denote the kernel of the canonical homomorphism $\varepsilon_x : \mathcal{O}_x \longrightarrow \mathcal{O}_{x,V}$. Since x is a simple point of V, this ideal is known (cf. [18], Theorem 1) to be generated by $r-p$ elements $f_1, \ldots, f_{r-p}$; and the Cohen–Macaulay theorem (cf. [13], p. 53, Proposition 2) shows that

$$(f_1, \ldots, f_{i-1}) : f_i = (f_1, \ldots, f_{i-1}) \qquad \text{for } 1 \leq i \leq r-p.$$

Let L_q denote the free $\mathcal{O}_x$-module with basis elements $e\langle i_1 \cdots i_q \rangle$ corresponding to the sequences $(i_1, \ldots, i_q)$ such that

$$1 \leq i_1 < i_2 < \cdots < i_q \leq r - p.$$

For $q = 0$, take $L_0 = \mathcal{O}_x$, and set

$$d\big(e\langle i_1 \cdots i_q \rangle\big) = \sum_{j=1}^{j=q} (-1)^j f_{i_j}\, e\langle i_1 \cdots \widehat{\imath}_j \cdots i_q \rangle$$

and

$$d\big(e\langle i \rangle\big) = f_i.$$

By [6], Chapter VIII, Proposition 4.3, the sequence

$$0 \longrightarrow L_{r-p} \xrightarrow{d} L_{r-p-1} \xrightarrow{d} \cdots \xrightarrow{d} L_0 \xrightarrow{\varepsilon_x} \mathcal{O}_{x,V} \longrightarrow 0$$

is exact. This proves that $\dim_{\mathcal{O}_x}(\mathcal{O}_{x,V}) \leq r - p$, as required.

Corollary. *One has $H^q(V, \mathcal{O}_V(-n)) = 0$ for all sufficiently large n and $0 \leq q < p$.*

Remark. The proof above applies more generally whenever the ideal $\mathcal{I}_x(V)$ admits a system of $r - p$ generators; in other words, whenever V is locally a complete intersection at every point.

76. Normal varieties. We shall need the following lemma.

Lemma 2. *Let M be a finitely generated $\mathcal{O}_x$-module, and let f be a noninvertible element of $\mathcal{O}_x$ such that $f \cdot m = 0$ implies $m = 0$ whenever $m \in M$. The dimension of the $\mathcal{O}_x$-module M/fM is then equal to the dimension of M increased by one.*

By hypothesis there is an exact sequence

$$0 \longrightarrow M \xrightarrow{\alpha} M \longrightarrow M/fM \longrightarrow 0,$$

where α is multiplication by f. If N is a finitely generated $\mathcal{O}_x$-module, there is therefore an exact sequence

$$\cdots \longrightarrow \operatorname{Ext}^q_{\mathcal{O}_x}(M, N) \xrightarrow{\alpha} \operatorname{Ext}^q_{\mathcal{O}_x}(M, N) \longrightarrow \operatorname{Ext}^{q+1}_{\mathcal{O}_x}(M/fM, N) \longrightarrow \operatorname{Ext}^{q+1}_{\mathcal{O}_x}(M, N) \longrightarrow \cdots.$$

Let p be the dimension of M. Taking $q = p + 1$ in the preceding exact sequence shows that $\operatorname{Ext}^{p+2}_{\mathcal{O}_x}(M/fM, N) = 0$; hence ([6], Chapter VI, §2) $\dim(M/fM) \leq p + 1$. On the other hand, since $\dim M = p$, one may choose an N such that $\operatorname{Ext}^p_{\mathcal{O}_x}(M, N) \neq 0$. Taking $q = p$ in the same exact sequence identifies $\operatorname{Ext}^{p+1}_{\mathcal{O}_x}(M/fM, N)$ with the cokernel of

$$\operatorname{Ext}^p_{\mathcal{O}_x}(M, N) \xrightarrow{\alpha} \operatorname{Ext}^p_{\mathcal{O}_x}(M, N).$$

This homomorphism is multiplication by f; since f is not invertible in the local ring $\mathcal{O}_x$, [6], Chapter VIII, Proposition 5.1′ shows that its cokernel is nonzero. Thus $\dim(M/fM) \geq p+1$, completing the proof.

We now prove a result closely related to the "Enriques–Severi lemma," due to Zariski [19].

THEOREM 4. *Let V be an irreducible normal subvariety of dimension ≥ 2 in the projective space $\mathbf{P}_r(K)$. Let $\mathfrak{F}$ be a coherent algebraic sheaf on V such that, for every $x \in V$, $\mathfrak{F}_x$ is a free module over $\mathcal{O}_{x,V}$. Then $H^1(V, \mathfrak{F}(-n)) = 0$ for all sufficiently large n.*

By Theorem 2, it suffices to show that $\mathcal{O}_{x,V}$, regarded as an $\mathcal{O}_x$-module, has dimension $\leq r-2$. First choose an element $f \in \mathcal{O}_x$ such that $f(x) = 0$ and the image of f in $\mathcal{O}_{x,V}$ is nonzero; this is possible because $\dim V > 0$. Since V is irreducible, $\mathcal{O}_{x,V}$ is an integral domain, and Lemma 2 applies to $(\mathcal{O}_{x,V}, f)$. Hence

$$\dim \mathcal{O}_{x,V} = \dim\bigl(\mathcal{O}_{x,V}/(f)\bigr) - 1, \qquad (f) = f \cdot \mathcal{O}_{x,V}.$$

Since $\mathcal{O}_{x,V}$ is integrally closed, all the prime ideals $\mathfrak{p}^\alpha$ of the principal ideal (f) are minimal (cf. [12], p. 136, or [9], no. 37), and none is therefore equal to the maximal ideal $\mathfrak{m}$ of $\mathcal{O}_{x,V}$ (otherwise $\dim V \leq 1$). We may thus choose an element $g \in \mathfrak{m}$ which belongs to none of the $\mathfrak{p}^\alpha$. This element g is not a zero divisor in $\mathcal{O}_{x,V}/(f)$. Calling $\bar{g}$ a representative of g in $\mathcal{O}_x$, Lemma 2 applies to $(\mathcal{O}_{x,V}/(f), \bar{g})$, and gives

$$\dim\bigl(\mathcal{O}_{x,V}/(f)\bigr) = \dim\bigl(\mathcal{O}_{x,V}/(f,g)\bigr) - 1.$$

But the syzygy theorem cited above gives $\dim\bigl(\mathcal{O}_{x,V}/(f,g)\bigr) \leq r$. Therefore $\dim\bigl(\mathcal{O}_{x,V}/(f)\bigr) \leq r-1$ and $\dim \mathcal{O}_{x,V} \leq r-2$, as required.

COROLLARY. *One has $H^1(V, \mathcal{O}_V(-n)) = 0$ for all sufficiently large n.*

REMARKS. (1) The argument above is classical in the theory of syzygies. See, for example, W. Gröbner, *Moderne Algebraische Geometrie*, 152.6 and 153.1.

(2) Even if the dimension of V is > 2, it is possible to have $\dim \mathcal{O}_{x,V} = r-2$. This is notably the case when V is a cone whose hyperplane section W is a projectively normal irregular variety (i.e., $H^1(W, \mathcal{O}_W) \neq 0$).

77. Homological characterization of varieties that are k-fold of the first kind.

Let M be a finitely generated graded S-module. By an argument identical to that of Lemma 1, one proves:

LEMMA 3. *In order that $\dim M \leq k$, it is necessary and sufficient that $\mathrm{Ext}^q_S(M,S) = 0$ for $q > k$.*

Since M is graded, one has

$$\mathrm{Ext}^q_S(M, \Omega) = \mathrm{Ext}^q_S(M,S)(-r-1),$$

so the preceding condition is equivalent to $\mathrm{Ext}^q_S(M,\Omega) = 0$ for $q > k$. Taking Theorem 1 of no. 72 into account, one concludes:

PROPOSITION 2.

(a) *In order that $\dim M \leq r$, it is necessary and sufficient that $\alpha : M_n \to H^0(M(n))$ be injective for every $n \in \mathbf{Z}$.*

(b) *If k is an integer ≥ 1, in order that $\dim M \leq r-k$, it is necessary and sufficient that $\alpha : M_n \to H^0(M(n))$ be bijective for every $n \in \mathbf{Z}$, and that $H^q(M(n)) = 0$ for $0 < q < k$ and every $n \in \mathbf{Z}$.*

Let V be a closed subvariety of $\mathbf{P}_r(K)$, and let $I(V)$ be the ideal of homogeneous polynomials vanishing on V. Put $S(V) = S/I(V)$; this is a graded S-module whose associated sheaf is precisely $\mathcal{O}_V$. We shall say[23] that V is a subvariety "k-fold of the first kind" in $\mathbf{P}_r(K)$ if the dimension of the S-module $S(V)$ is $\leq r-k$. It is immediate that $\alpha : S(V)_n \to H^0(V, \mathcal{O}_V(n))$ is injective for

[23]Cf. P. Dubreil, *Sur la dimension des idéaux de polynômes*, J. Math. Pures Appl., 15 (1936), pp. 271–283. See also W. Gröbner, *Moderne Algebraische Geometrie*, §5.

every $n \in \mathbf{Z}$, so every variety is 0-fold of the first kind. Applying the preceding proposition to $M = S(V)$ gives:

Proposition 3. *Let k be an integer ≥ 1. In order that the subvariety V be k-fold of the first kind, it is necessary and sufficient that the following conditions hold for every $n \in \mathbf{Z}$:*

(i) *$\alpha : S(V)_n \to H^0(V, \mathcal{O}_V(n))$ is bijective.*

(ii) *$H^q(V, \mathcal{O}_V(n)) = 0$ for $0 < q < k$.*

(Condition (i) can also be expressed by saying that the linear series cut out on V by the forms of degree n is complete, as is well known.)

Comparing this with Theorem 2 (or arguing directly), one obtains:

Corollary. *If V is k-fold of the first kind, then $H^q(V, \mathcal{O}_V) = 0$ for $0 < q < k$, and, for every $x \in V$, the dimension of the $\mathcal{O}_x$-module $\mathcal{O}_{x,V}$ is $\leq r - k$.*

If m is an integer ≥ 1, denote by φ_m the embedding of $\mathbf{P}_r(K)$ in a projective space of suitable dimension given by the monomials of degree m (cf. [8], Chapter XVI, §6, or no. 52, proof of Lemma 2). The preceding corollary then has the following converse:

Proposition 4. *Let k be an integer ≥ 1, and let V be a connected closed subvariety of $\mathbf{P}_r(K)$. Suppose that $H^q(V, \mathcal{O}_V) = 0$ for $0 < q < k$, and that, for every $x \in V$, the dimension of the $\mathcal{O}_x$-module $\mathcal{O}_{x,V}$ is $\leq r - k$.*

Then, for every sufficiently large m, $\varphi_m(V)$ is a subvariety k-fold of the first kind.

Because V is connected, one has $H^0(V, \mathcal{O}_V) = K$. Indeed, if V is irreducible, this is clear (otherwise $H^0(V, \mathcal{O}_V)$ would contain a polynomial algebra and would not be finite-dimensional over K); if V is reducible, every element $f \in H^0(V, \mathcal{O}_V)$ induces a constant on each irreducible component of V, and these constants are the same by the connectedness of V.

Since $\dim \mathcal{O}_{x,V} \leq r - 1$, the algebraic dimension of every irreducible component of V is at least 1. It follows that

$$H^0(V, \mathcal{O}_V(-n)) = 0$$

for $n > 0$ (for if $f \in H^0(V, \mathcal{O}_V(-n))$ and $f \neq 0$, the $f^k g$, with $g \in S(V)_{nk}$, would form a vector subspace of $H^0(V, \mathcal{O}_V)$ of dimension > 1).

With this established, denote by V_m the subvariety $\varphi_m(V)$; one plainly has

$$\mathcal{O}_{V_m}(n) = \mathcal{O}_V(nm).$$

For sufficiently large m, the following conditions hold:

(a) *$\alpha : S(V)_{nm} \to H^0(V, \mathcal{O}_V(nm))$ is bijective for every $n \geq 1$.*

This follows from Proposition 5 of no. 65.

(b) *$H^q(V, \mathcal{O}_V(nm)) = 0$ for $0 < q < k$ and every $n \geq 1$.*

This follows from Proposition 7 of no. 65.

(c) *$H^q(V, \mathcal{O}_V(nm)) = 0$ for $0 < q < k$ and every $n \leq -1$.*

This follows from Theorem 2 of no. 74 and the hypothesis on the $\mathcal{O}_{x,V}$.

On the other hand, one has $H^0(V, \mathcal{O}_V) = K$, $H^0(V, \mathcal{O}_V(nm)) = 0$ for every $n \leq -1$, and $H^q(V, \mathcal{O}_V) = 0$ for $0 < q < k$, by hypothesis. It follows that V_m satisfies all the hypotheses of Proposition 3, q.e.d.

Corollary. *Let k be an integer ≥ 1, and let V be a nonsingular projective variety of dimension $\geq k$. In order that V be biregularly isomorphic to a subvariety k-fold of the first kind in a suitable projective space, it is necessary and sufficient that V be connected and that $H^q(V, \mathcal{O}_V) = 0$ for $0 < q < k$.*

Necessity is clear from Proposition 3. To prove sufficiency, it is enough to observe that $\mathcal{O}_{x,V}$ then has dimension $\leq r - k$ (cf. no. 75) and to apply the preceding proposition.

78. Complete intersections.

A subvariety V of dimension p in the projective space $\mathbf{P}_r(K)$ is a *complete intersection* if the ideal $I(V)$ of polynomials vanishing on V has a system of $r - p$ generators $P_1, \dots, P_{r-p}$; in that case, all the irreducible components of V have dimension p, by Macaulay's theorem (cf. [9], no. 17). It is well known that such a variety is p-fold of the first kind, which already implies that

$H^q(V, \mathcal{O}_V(n)) = 0$ for $0 < q < p$, as we have just seen. We shall determine $H^p(V, \mathcal{O}_V(n))$ in terms of the degrees $m_1, \ldots, m_{r-p}$ of the homogeneous polynomials $P_1, \ldots, P_{r-p}$.

Let $S(V) = S/I(V)$ be the projective coordinate ring of V. By Theorem 1 of no. 72, everything reduces to determining the S-module $\operatorname{Ext}_S^{r-p}(S(V), \Omega)$. There is a resolution analogous to that of no. 75: take for L^q the free graded S-module having as basis the elements $e\langle i_1 \cdots i_q\rangle$ corresponding to the sequences $(i_1, \ldots, i_q)$ such that $1 \leq i_1 < i_2 < \cdots < i_q \leq r-p$, and having degrees $\sum_{j=1}^{j=q} m_{i_j}$;[24] for L^0, take S. Set

$$d(e\langle i_1 \cdots i_q\rangle) = \sum_{j=1}^{j=q} (-1)^j P_{i_j} \cdot e\langle i_1 \cdots \widehat{\imath_j} \cdots i_q\rangle, \qquad d(e\langle i\rangle) = P_i.$$

The sequence

$$0 \longrightarrow L^{r-p} \xrightarrow{d} \cdots \xrightarrow{d} L^0 \longrightarrow S(V) \longrightarrow 0$$

is exact ([6], Chapter VIII, Proposition 4.3). It follows that the $\operatorname{Ext}_S^q(S(V), \Omega)$ are the cohomology groups of the complex formed by the $\operatorname{Hom}_S(L^q, \Omega)$. An element of $\operatorname{Hom}_S(L^q, \Omega)_n$ may be identified with a system $f\langle i_1 \cdots i_q\rangle$, where the $f\langle i_1 \cdots i_q\rangle$ are homogeneous polynomials of degrees $m_{i_1} + \cdots + m_{i_q} + n - r - 1$. Under this identification, the coboundary operator is given by the usual formula

$$(df)\langle i_1 \cdots i_{q+1}\rangle = \sum_{j=1}^{j=q+1} (-1)^j P_{i_j} \cdot f\langle i_1 \cdots \widehat{\imath_j} \cdots i_{q+1}\rangle.$$

Macaulay's theorem cited above shows that the conditions of [11] are satisfied, and one recovers the fact that $\operatorname{Ext}_S^q(S(V), \Omega) = 0$ for $q \neq r-p$. On the other hand, $\operatorname{Ext}_S^{r-p}(S(V), \Omega)_n$ is isomorphic to the subspace of $S(V)$ formed by the homogeneous elements of degree $N + n$, where

$$N = \sum_{i=1}^{i=r-p} m_i - r - 1.$$

Taking Theorem 1 of no. 72 into account, one obtains:

Proposition 5. *Let V be a complete intersection defined by homogeneous polynomials $P_1, \ldots, P_{r-p}$ of degrees $m_1, \ldots, m_{r-p}$.*

(a) *The map $\alpha : S(V)_n \to H^0(V, \mathcal{O}_V(n))$ is bijective for every $n \in \mathbf{Z}$.*

(b) *$H^q(V, \mathcal{O}_V(n)) = 0$ for $0 < q < p$ and every $n \in \mathbf{Z}$.*

(c) *$H^p(V, \mathcal{O}_V(n))$ is isomorphic to the dual vector space of $H^0(V, \mathcal{O}_V(N-n))$, where $N = \sum_{i=1}^{i=r-p} m_i - r - 1$.*

In particular, one should note that $H^p(V, \mathcal{O}_V)$ can vanish only if $N < 0$.

§6. Characteristic function and arithmetic genus

79. Euler–Poincaré characteristic. Let V be a projective variety, and let $\mathfrak{F}$ be a coherent algebraic sheaf on V. Set

$$h^q(V, \mathfrak{F}) = \dim_K H^q(V, \mathfrak{F}).$$

We saw in Theorem 1 of no. 66 that the $h^q(V, \mathfrak{F})$ are finite for every integer q, and zero for $q > \dim V$. We may therefore define an integer

$$\chi(V, \mathfrak{F}) = \sum_{q=0}^{\infty} (-1)^q h^q(V, \mathfrak{F}).$$

[24] The printed French has $\sum_{j=1}^{j=q} m_j$. The basis element depends on the selected indices $i_1, \ldots, i_q$, the differential below uses P_{i_j}, and the next paragraph explicitly gives degree $m_{i_1} + \cdots + m_{i_q} + n - r - 1$. Corrected French and English therefore restore the missing i_j in the subscript; diplomatic French preserves the print.

This is the *Euler–Poincaré characteristic* of V with values in $\mathfrak{F}$.

Lemma 1. *Let $0 \longrightarrow L_1 \longrightarrow \cdots \longrightarrow L_p \longrightarrow 0$ be an exact sequence, where the L_i are finite-dimensional vector spaces over K and the homomorphisms $L_i \to L_{i+1}$ are K-linear. Then*

$$\sum_{q=1}^{q=p} (-1)^q \dim_K L_q = 0.$$

We argue by induction on p, the lemma being obvious if $p \leq 3$. If L'_{p-1} denotes the kernel of $L_{p-1} \to L_p$, we have the two exact sequences

$$0 \longrightarrow L_1 \longrightarrow \cdots \longrightarrow L'_{p-1} \longrightarrow 0$$

and

$$0 \longrightarrow L'_{p-1} \longrightarrow L_{p-1} \longrightarrow L_p \longrightarrow 0.$$

Applying the induction hypothesis to each of these sequences, we see that

$$\sum_{q=1}^{q=p-2} (-1)^q \dim L_q + (-1)^{p-1} \dim L'_{p-1} = 0,$$

and

$$\dim L'_{p-1} - \dim L_{p-1} + \dim L_p = 0,$$

which immediately proves the lemma.

Proposition 1. *Let $0 \longrightarrow \mathfrak{A} \longrightarrow \mathfrak{B} \longrightarrow \mathfrak{C} \longrightarrow 0$ be an exact sequence of coherent algebraic sheaves on a projective variety V, where the homomorphisms $\mathfrak{A} \to \mathfrak{B}$ and $\mathfrak{B} \to \mathfrak{C}$ are K-linear. Then*

$$\chi(V, \mathfrak{B}) = \chi(V, \mathfrak{A}) + \chi(V, \mathfrak{C}).$$

By Corollary 2 to Theorem 5 of no. 47, there is an exact cohomology sequence

$$\cdots \longrightarrow H^q(V, \mathfrak{B}) \longrightarrow H^q(V, \mathfrak{C}) \longrightarrow H^{q+1}(V, \mathfrak{A}) \longrightarrow H^{q+1}(V, \mathfrak{B}) \longrightarrow \cdots.$$

Applying Lemma 1 to this exact sequence of vector spaces gives the proposition.

Proposition 2. *Let $0 \longrightarrow \mathfrak{F}_1 \longrightarrow \cdots \longrightarrow \mathfrak{F}_p \longrightarrow 0$ be an exact sequence of coherent algebraic sheaves on a projective variety V, where the homomorphisms $\mathfrak{F}_i \to \mathfrak{F}_{i+1}$ are algebraic. Then*

$$\sum_{q=1}^{q=p} (-1)^q \chi(V, \mathfrak{F}_q) = 0.$$

We argue by induction on p, the proposition being a special case of Proposition 1 if $p \leq 3$. If $\mathfrak{F}'_{p-1}$ denotes the kernel of $\mathfrak{F}_{p-1} \to \mathfrak{F}_p$, then $\mathfrak{F}'_{p-1}$ is coherent algebraic because $\mathfrak{F}_{p-1} \to \mathfrak{F}_p$ is an algebraic homomorphism. We may therefore apply the induction hypothesis to the two exact sequences

$$0 \longrightarrow \mathfrak{F}_1 \longrightarrow \cdots \longrightarrow \mathfrak{F}'_{p-1} \longrightarrow 0$$

and

$$0 \longrightarrow \mathfrak{F}'_{p-1} \longrightarrow \mathfrak{F}_{p-1} \longrightarrow \mathfrak{F}_p \longrightarrow 0,$$

and the proposition follows immediately.